

# Neur 240 The Neuroscience of Optimal Performance and Well-Being

*Numbered Transcript*

NEUR 240 • Dr. Baland Jalal

## Contents

Lecture 1. The Brain in Balance

Lecture 2. Performing Under Pressure

Lecture 3. Fear & Flow

Lecture 4. The Architecture of Attention

Lecture 5. Dopamine

Lecture 6. Recovery

Lecture 7. Stress

Lecture 8. Decisions & Direction

## Lecture 1: The Brain in Balance

**At a glance** — Jalal introduces the brain in balance and optimal performance.

- 1.1 Have you seen the movie The Karate Kid? The one from the '80s? In one of the initial scenes—or actually, the middle of the movie—the kid, the skinny kid, he's on a boat. And on that boat, he's standing, trying to keep his balance. And then Mr. Miyagi, you know Mr. Miyagi? He rocks the boat and he falls into the water. And he's angry about that. But he gets up and he learns that there's something about balance that is important, that's very important for life.
- 1.2 And then in a later scene, I think it's Mr. Miyagi's birthday—maybe it's actually Daniel's birthday, I think—and they're sitting there and they're chatting and things like that. And then they're talking about karate: how to do karate, how to do various tricks, and how to prepare for that competition. And then Mr. Miyagi says, he says to Daniel, "Come, come here, I have a gift for you." He takes him out to the parking lot and he says, "These are one of my cars. Whatever you want, just take it." And then there's a nice yellow one. And Daniel points to that. He says, "I want that one."
- 1.3 And then he goes into the car and he's so happy about that. And Mr. Miyagi takes out this picture. There's a picture there of Daniel with the girl that he likes. And then he says, "Go to her." He says, "Everything in life is about balance." Balance. And then he goes. He takes his new car, he's happy. And then he drives. He drives over to that girl. And then the girl is angry with him—for whatever reason—she's very angry at him. But he manages just to calm her down and he hugs her and everything.
- 1.4 But I think the main lesson from this is that this course is all about balance. If there is any lesson we can take away from this course at the very end, it is that balance is the key. So we'll start with that balance. How do we maintain balance? How do we have a sense of balance in our lives? How do we keep everything in a coordinated manner? So that is something that we'll try to go through in this course. In fact, this is my own Achilles heel because I know nothing about balance myself, you see. So we have to learn together. I think that's the whole point of this course.
- 1.5 How do we create that balance and understand what's going on in the brain? How do we end up in burnout when we have no control anymore? We feel fatigued. We feel maybe even restless. Some people feel restless during burnout. How can we use sleep and dreams to improve our lives, improve our sense of direction, our well-being, how we perform? Can we make ourselves more creative, Juan? Is that possible? Are there techniques?
- 1.6 What about flow? What is flow state? Have you heard about flow before? Where you're completely absorbed in this activity. You love it, you lose a sense of time, space, you're just completely immersed. Everything is just, wow. Have you heard about flow before? Flow is interesting. The founder of flow—we'll get into that later, we have a

whole lecture on flow—but flow is so fascinating. We get into that. Creativity. Can we make ourselves more creative? We have all those things in this course.

- 1.7** Before we dive into this course and what it has to offer us, we have to understand the human brain. We have to understand the very machinery that allows us to experience optimal performance that leads to well-being. And that is the human brain. So let's take out our notebooks and jot down some notes.
- 1.8** First, we have the human brain—even you, Rachel—we have the human brain. We have the cortex, the very outer layer [is] the cortex. The prefrontal cortex, the top of your nose, forehead region, is the CEO of the brain. CEO, the head of the ship, steering the ship, decision-making, logic, precision, planning, future goals, time travel, mental time travel, all in the prefrontal lobes.
- 1.9** And the prefrontal lobes have the medial part and the sides, the lateral part. The medial is important—for my friends that were in the previous course, Alexander—medial is important for what? Emotional decision-making. I know you know, you know it, come on, man. Emotional decision-making, right? And the sides of the prefrontal cortex is important for what? Past students, come on. Exactly. Logic, calculating, cold cognition, like being precise, but in this non-emotional way. So the [lateral]—exactly, more analytic, exactly.
- 1.10** So the medial part is important for me making emotional decisions, right? It's the bridging point between emotion and cognition. That's the medial part. So if I have to make an emotional decision: should I fire this person, should I break up with this girl, should I do this or that? That's the medial part. That's the medial. So it has emotion and cognition bridging them. Exactly. Rosanna, right? [Yes.] Okay, Rosanna. Jot that down, Rosanna. Very good. And then the side: cold cognition; calculated.
- 1.11** As you will see, both of them are critically important for well-being. You want to be able to make sharp decisions. If you're a CEO, right? You want to make sharp decisions—working memory, calculate it fast. And then occasionally, you want to make more emotional decisions. Should I buy that stock? Should I fire that person over here? Should I do this or that? So that's more emotional. You want to have both. That's the prefrontal cortex.
- 1.12** Right behind it, in the cortex, you have the motor strip involved in [motion]: actions, movement. And just behind that, you have the premotor strip involved in things like planning movement. So whenever I plan, if I'm a soccer player—I'm Messi and I plan to dribble—then I use my premotor cortex to orchestrate that. I have an intention to move, that's the premotor strip. And then whenever I decided to kick that soccer ball, I involved the motor strip right in front of it. Does that make sense? Not too complex yet, right? Prefrontal: emotional decision-making [medial], calculated [lateral], motor, and then planning of action [premotor]; movement and planning of movement. Behind

that, you have the somatosensory cortex involved in sense of sensation and touch. Just behind it.

**1.13** And in the right parietal, in this region—we're mainly going from middle to sides—and then further back, you have the parietal lobe. Right here is a multimedia studio, metaphorically, of course, there's a multimedia studio over here. So whenever I move my body from A to B in space, that region of the brain is titillated. It's activated. It's dancing with activity, whenever I see myself in space, move around, and so forth.

**1.14** And then in the back of my brain, I have what? My previous students would know. The very back of the brain is important for what? Vision, exactly. Vision. So we have that—you can jot that down: vision. Excellent, Kelly. Thank you. Vision is in the very back of the brain. So that would be the cortex, six layers deep.

**1.15** Now, if you go further inside the brain, you have things like the ACC, which is involved in monitoring conflict in the world and assessing value together with the OFC. These are just—let's think about it, that's the ACC and OFC. Just put ACC and OFC. We get into why these are important later, but they're mainly important for assigning value to the world. Let's put it that way. Assigning value. Meaning, value, that's in the middle of the brain.

**1.16** And then you have the emotional core, the limbic structures behind your ears, further in, you have like a ring. Wow, this is so exciting, this part of the brain. My God, this is important for so many things. For any emotions that you feel, you have that limbic region becoming activated. Think about this. Put limbic ring, even draw the bloody ring, okay? Just like a drawing. In that ring, you have the amygdala, the fear part of the brain, Nikola—fear part of the brain. That's in that limbic [system].

**1.17** Within the middle of the brain, you have another very, very important system. The dopaminergic system: the reward, anticipation of reward system. Nucleus accumbens, striatal regions. Nucleus accumbens, striatal regions. Think about it, in the middle of the brain, I have things like reward and the anticipation of reward, and that's dopamine. So crucial, isn't it? Anything I do in this bloody world is tied to dopamine. What reward would I get from doing A, B, and Z? Why did I come to this course? Oh, I want to learn something. Reward. Everything is reward, right? That's what drives us. That's the very currency for existence, almost. Reward. Dopamine. Very good. Now we have emotion, limbic. We had assigning value: ACC and OFC. We had dopamine.

**1.18** Then you go a step further down in the brain, you have things like—this is more the brainstem: breathing, balance; sense of balance: cerebellum. So these are more raw survival—respiration—they're very critical for survival, but they're not important in the sense of— even animals have these more basic brainstem regions for respiration, breathing, things like that. So that would be the brainstem.

**1.19** Now, the chemicals. The chemicals for sustaining our well-being. Now, we often talk about neurochemicals in the brain. So we'll say we have dopamine, serotonin, and all

that. We'll list maybe five or six or seven. But really, you have more than maybe 100 in real life, right? But let's mention some of the most important ones. Which would be one? Dopamine would be very important. Dopamine is important for? Reward. Anticipation of reward.

- 1.20** So we have two reward systems in the brain. And my students would know, my hardcore students know, we have two systems for pleasure. One is anticipation of pleasure. Right? So let's say this weekend, I'm looking forward to going to that restaurant with my friends. Is that anticipation of reward or is it consummatory reward? Is it anticipation of reward? That's what I'm trying to say. Yeah, it's anticipation reward. That's dopamine, isn't it? Yes.
- 1.21** But you have another reward system which is involved in the actual reward that I'm experiencing. So let's say when I'm looking forward to meeting that girl, for example, I have a lot of dopamine; it's anticipation of reward. However, when I'm actually sitting there with that girl and I'm having that conversation, of course, there'll be some dopamine. But there's something else going on: endorphins, the actual reward. When I'm playing with my niece and holding her and playing with her, that would be more consummatory reward. It's endorphins. Different reward system.
- 1.22** In fact, people with depression illustrate this perfectly, as we talked about in my previous course. Their systems of reward are decoupled. So on the one hand, they lack dopaminergic reward. So if you say, "Come on, man, let's go and have fun. We'll go to the park. We'll go and do this exciting thing this weekend." And they'll say, "No, I don't want to go. I don't feel like it. I don't want to do anything." And that's because their dopaminergic system is flat. The neurons of dopamine [are] insensitive to reward. So they won't do it.
- 1.23** Now, if you drag this depressed person along, you say, "Come on, man, we have to go. Come on," and just drag him. And then he actually goes to that park, or he goes to that soccer match, or he goes to that restaurant, he'll actually enjoy it. That's because his endorphin system—the [ventral pallidum], and other regions, the striatum, and mainly [ventral pallidum] areas—would still be active. Those neurons that produce endorphins are very active. That's one issue with dopamine.
- 1.24** Now, dopamine is good. You want to have dopamine, you want to have—Giovanni, right?—you want to be able to do things and be excited and be excited for life. I want to do this thing over here because it drives my reward centers. So that's a good thing. But let's say I was to artificially ramp up your dopamine. How can we do that? What drugs can we—? I don't recommend this, of course, but cocaine, right? That could be, but let's take a drug, actual drug. "Let's do cocaine for the bloody sake, man." Let's say somebody takes cocaine—not you, of course, Giovanni—but somebody takes cocaine. What happens to them?
- 1.25** I don't know personally, but I think everything's elevated.

- 1.26 Absolutely.
- 1.27 There are all sorts of senses, sensations.
- 1.28 Absolutely. Everything is elevated. Everything is just madness, and everything is driven towards the external world. There's no sense of well-being. There's no sense of rest. Everything is just outward, external reward, one reward to the next. It's almost like an erratic CEO of some company. Everything is external.
- 1.29 But of course, the issue with that, Joe, is that you want to have a system that balances that out. You want to have inner rest. You want to have a sense of well-being and sense of calmness. You want to have a neurotransmitter system in the brain that's responsible for that sense of calmness. And what would that be, Nikola? What system seems to fight with the dopaminergic system? Serotonin, right? So you have to have serotonin in the system as well. That's what balances out that dopamine.
- 1.30 But yet, having so much serotonin that everything is still and everything is just calm, it's not good either. So you need to have an alertness chemical in the brain. You need to have alertness, right? You need to be alert occasionally. So for example, when you are reading a book, you have what? A little bit of alertness, right? Kind of. And then your mom comes to you and yells at you—like she does at me all the time, "Baland, do this or that." I have even more alertness. And then I'm on the freeway, I'm driving, and suddenly there's this car driving towards me. I have what? Tons of alertness. What neurochemical is important for that? Noradrenaline. Okay.
- 1.31 And let's go one step deeper. What inside the brain is responsible for creating noradrenaline? I know you know. I know you know. It's deep in the brain. It's important for producing noradrenaline. Locus coeruleus. Think of a gladiator; it's like a gladiator name. Locus coeruleus is important for producing noradrenaline. It's in the brainstem.
- 1.32 In fact, in the brainstem you have—this is how I think about it all the time—you have [the] dorsal raphe nucleus. Again, sounds like a Latin gladiator, right? Dorsal raphe nucleus: it's in the brainstem, the very low part of the brain. And next to it, you have locus coeruleus, another warrior. These two warriors, these two gladiators, produce maybe perhaps the most important neurochemicals we have in the brain.
- 1.33 One, locus coeruleus, produces the alertness chemical called noradrenaline. Dorsal raphe nucleus produces serotonin. And then they have wires projecting to the rest of the brain and flooding the brain with this neurochemical. Is that clear? This is how I think about it. I literally picture—I see these two gladiators deep inside the brain, and they're sending out these chemicals to the rest of the brain. Just very simple, but that's how it works. You have that. Two neurochemicals that are always antagonistic, always dueling it out, so to speak, wrestling it out. They seem to be involved in this dance we've been talking about.

- 1.34** There's this excitatory and inhibitory dance in the brain. Everything is either excited, excitatory, or inhibitory. Very simple, excitatory or inhibitory. There's two neurochemicals that always, in the circuits of the brain, in the various regions, they seem to always be in a conflict. Usually, if there's mental illness, if it's, let's say—it's funny, it's interesting, even with some of the things like depression, you have more leaning towards excitation, especially in fear circuits and anxiety circuits and things like that.
- 1.35** You have two neurochemicals involved in exciting the brain and calming it down, mainly two of them. What would they be? Glutamate and GABA, right? Glutamate and GABA. It seems like every disorder that I've talked about in my courses so far has this wrestling match between glutamate and GABA. Glutamate being the neurochemical that just fires things up, makes everything much more excited. Just putting fuel on the fire. And then you have GABA, slowing things down, making things calm, always trying to—and you could do that, of course, by taking a [benzodiazepine]. But the problem would be?
- 1.36** It doesn't really work long term.
- 1.37** The backlash. So when you take the benzo away, the system overcompensates and you have tons of excitation. That's why you can't really drug yourself out of these problems. But these chemicals are key.
- 1.38** And then, of course, we have acetylcholine, perhaps one of my best buddies in the brain. I love acetylcholine, man. I love it. You know why I love it? I'm 100% serious. I love this. This is one of my best friends, acetylcholine. I mean, I have friends outside science, outside the brain, a few, but I have some. Acetylcholine, okay? It's interesting because it produces what? What's it important for, Rachel? Come on, man, you know. Attentional focus, right? When you focus intensely on something, you have what? Acetylcholine. What process of the brain is acetylcholine important for? I'm going to get into that in a moment as well. But what is it? Look, when you focus intensely on something—I'm focused intensely on this horse over here; it's beautiful, it's running, it's just beautiful.
- 1.39** Salience.
- 1.40** Salience, but it leads to what, ultimately? Why would Baland love acetylcholine? It's important for focus, but why would Baland love it?
- 1.41** Plasticity.
- 1.42** Plasticity. Plasticity. So it's crucial for plasticity, okay? Acetylcholine, and it's produced by the nucleus—? Nucleus basalis. Close. Nucleus basalis. That is the acetylcholine-producing machine. That is the very attention system in the brain. You know why I love it? I want you to understand why I love it. It's not merely that I love it; I really love it. The reason why I really love it is the following. Think about this.

- 1.43 We all have a nucleus basalis system. We all have it. When we are kids, babies, infants, how is our attention system tuned into the world? Do we pay attention to everything when we are babies? Pretty much, yes, we do. Everything is just, wow. Everything is salient. Everything is just, "I have to pay attention to everything." And that's because the nucleus basalis system is perpetually active. Perpetually active.
- 1.44 And that's because a baby's brain, they don't know what's important in the world. Kelly, they don't know that this is important, this is not important. So their salience systems in the brain—we have something called the salience system, what we pay attention to— that system, it doesn't understand what I should pay attention to, so it pays attention to everything. And of course, the cost of that is their brain is hyperplastic, so it changes its shape all the time because acetylcholine is ultimately involved in changing your brain.
- 1.45 Now, as we age and become older, our acetylcholine system and nucleus basalis system shuts off. So it's not perpetually active, but it's only active when we pay attention to something. And that very paying attention to something, when we do it selectively, that's when plasticity occurs. Does this make sense? Do you see now why it's important? A baby, full acetylcholine, full nucleus basalis, everything is important, the brain is hyperplastic, always changing. Then you have somebody like you and I. We reach maturity— 12, 13, 14, 15—and the nucleus basalis system shuts off. So we can only turn it on when we actually selectively pay attention to something. That's why it's important.
- 1.46 The key of this course is understanding that for the human brain—we talk about well-being and optimal performance: how do I perform better and become just better on the soccer field or better as a CEO or better as this or that. We often talk about dopamine and how important dopamine is and things like that.
- 1.47 But really, as we'll see, it's balance, not only in life decisions and going and seeing that girl whenever you are overworked or going for that picnic with your family or using your Sundays with your niece, Sophia, playing, all this stuff. Not only is it important in terms of balance in our life decisions, but the very brain, our very brain, is about balance.
- 1.48 And so if we can keep this balance intact, GABA, glutamate, glycine, acetylcholine, dopamine, serotonin, so forth, and the various parts of the brain and have them operate in a balanced way, that's optimal performance. That's well-being. That's feeling good. That's being on track in life. That's how you can keep sustained activity in the future. Forget willpower, forget dopamine drive, forget, "I, me, I'm going to accomplish all this." It's simply balance inside the brain and outside the brain. I think that will be the key lesson of this course—that very balance I do not have myself, to be clear. Balance.

- 1.49 Stability is another one. You want to stabilize the system, and fundamental to all this is the fact that the brain is always in the business of predicting the world. Prediction is what we do. This morning, when I woke up, came here, I had mapped out this environment. I had seen the faces in front of me—Rachel's going to sit maybe over here, over there. I had some prediction model of how the faces would be. And I knew I had seen myself in this situation, maybe lecturing—this prediction of what will happen. So I'm running relatively smoothly. So relatively, right? It's running. Something is running and it's relatively okay, right?
- 1.50 But if I was to lecture and suddenly—and this is ridiculous—let's say an elephant came here. How would my lecture go? Terrible. It wouldn't be good. Imagine this big elephant coming here with the—? That'd be insane. But my model of the world would collapse. So that very model that I created in my mind of reality of me lecturing, just saying these things and looking at faces and all that, that would collapse completely because now this elephant came. My very predictive model of the world [would] shut down. So we all have models we create all the time, whether it be a soccer player, Messi, or maybe it's James, of Colombia, he's playing. Okay, James, he's playing. He has a model of that pitch of that soccer game. He has, like, "I'm going to go on that pitch and I'm going to dribble like this and that." And then he wouldn't expect somebody to come and tackle him suddenly in a very harsh way.
- 1.51 If something like that happened, if somebody tackled him, let's say, illegally, his very model would change. Maybe he'd hesitate a bit more. Maybe his dribbling would be a bit more stiff, rigid in his behaviors. That smoothness of operation would fall down because he didn't expect it. The point is that we all have these predictive processes. We predict the world, we model the world, we picture it in our brains: this will happen, that will happen, and so I'm going to do X, Y, and Z. How does prediction happen in the brain? This relates to my previous course as well, because in my—actually, I'd recommend, if you watch my courses, you should probably watch the neuropsychiatry course before this one. It'd be a good understanding of how the healthy brain completely collapses, and that would be an extreme example, then, of the healthy brain. Because in mental illness, you have things like this predictive modeling of the world completely flipping on its head. I'll give you an example. How do I predict the world? My eyes take information from the world. I view the world. Light hits the eye, the optic nerve comes in, to the thalamus. The thalamus is the bouncer of the brain. It filters information, says, "This is important, that's important, that's visual information. Please, buddy, come in." And then sends information to the very low part off the back of the brain. This is the visual cortex. The visual cortex [is] involved in lines and basic shapes and things of that nature. That's basic vision, basic visual percepts here.
- 1.52 Then information travels along this pathway up on the side, and then I give things motion. That's the motion part of the brain, the MT. I give things depth. I give them

color. So this is the V4 area. This is a fancy name for neurons over here, putting color on it: this is red, this is blue, this is green, adding to that whole visual perceptual line.

- 1.53** And then it goes further. It goes to Wernicke's area, another part of that up in the system. You notice the more I go up in the step, the more sophisticated it all becomes. Now it's very sophisticated. I give it meaning. "It's an apple. Oh, it's Newton." I think about an apple tree. I think about oranges. I think about whatever. I make semantic associations, even broader. Then I bring in the hippocampus, the memory part of the brain. How many trees have I seen in the past? How many apples have I seen in the past? How do apples look? And so forth.
- 1.54** Now, the brain is clever. Whenever I look at, let's say, an apple or let's say, even more simple, a table, it doesn't create the table from scratch, you see, because that would be stupid. It would be stupid because visual processing is very costly. Think about it: in nature, if I was to go through all those steps each time I saw a lion, I would be what? Its next meal very quickly, right?
- 1.55** So it says, how can I quickly create a lion without going through all these steps? Well, I'll do the following. Information comes in. Basic shape. "Oh, this might be a lion." Does this shape look like previous lions in the past? "Yes, it does." Information comes from top-down processing. Yes, match, fixed. "This is a lion." And then it creates a lion based on previous expectations of what a lion looks like. Does that make sense? So it uses top-down predictive modeling to then say, "Oh, make the lion look like this based on my expectations." Does that make sense?
- 1.56** Now, in fact, if I had some issues with my lower processing areas, let's say my visual cortex had a little scotoma, let's say, I had a brain injury to my visual cortex, or even my eye, let's say my eyes weren't working that well, I had an eye issue, the brain tends to fill in all the gaps in my environment with hallucinations. It's so dependent on predictive interpretation of the world that if there's some issues with the basic lower sensory areas, it'll bloody just interpret the thing; it will hallucinate for you. Does that make sense? So even if it lacks access, it'll just hallucinate the world.
- 1.57** Now, in mental illness, this balance sometimes shifts. So in schizophrenia, the entire world is based on top-down prediction. So I hallucinate everything almost. Everything is based on my expectations, and I ignore the sensory data completely. All the data in the world doesn't matter. What I think is what's out there is what's out there. Now, you don't want that, of course, but that's what schizophrenia is like. But it shows you that balance between what I expect of the world and what's out there. And it's often this quick comparison. I have some information over here. I expect this world to be like this, and boom, I create reality as quickly as possible. And if it's a lion, get out of here. If it's a mate, pursue that mate. If it's this, and so forth.
- 1.58** Your brain, in order to keep the model stable, to avoid any discrepancies to that model of the world that you are predicting, whether it be the football field or, let's say, the

lecture place where you're lecturing, or whatever it might be, you prefer familiarity. Familiarity is always good. Always good. Because things run more smoothly if you know that area.

- 1.59** Perhaps that's the reason why soccer players, they would go to the fields. Do they sometimes do this? If they have, let's say, they are going to play Inter Miami, let's say, Inter Miami, they're playing. We're not in Miami anymore. We're in Arizona. Any Arizona-relevant examples? I don't know. You have a soccer match here. Do you play soccer? Like soccer? You like, okay. So, there's a stadium you go to when you have this important match. Would it be important to go and practice on that match before you actually do the thing, the actual match? I mean, the very same field that the match is going to be on. Knowing how the stadium is and how the—okay, of course, the grass will be similar, but just having an idea of the setting is very important.
- 1.60** In fact, in your very lives, the way you want to arrange things is, as much as possible, let's say, the way you work, your offices, whatever you have, make the environment predictable as well. Predictable, like, where do I put my things? Is my schedule clear? So for example, me, I know exactly when I come here, I wake up at, let's say, 5:00. I go to the gym. Every day since I've been here, I go to the gym in the morning at the same time. It reduces friction. Any predictive uncertainties go away in that way.
- 1.61** At the same time, I go to the gym. At the same time, I go get my breakfast. The same waiter, in fact, I say the same words. Have you seen Groundhog Day? Exactly. I say the same words to her: "I need this, and then salt and pepper, and please give me that extra spoon and plate." Every time the same word, and she looks at me. She's like, "Weird. He says the same thing."
- 1.62** Wearing the same T-shirt: like Zuckerberg, wearing the same T-shirt every day, can actually be helpful sometimes. You reduce cognitive load, especially when you have to go do something, you have to give a lecture, you have to play a soccer match. You've seen Suárez, right? You know Suárez, the soccer player from Uruguay, right? He always has the same—is it yerba mate he drinks? Everything. Same ritual. Rituals in this regard [are] important because they cue the brain for new—
- 1.63** The match itself is overwhelming. A lot of cognitive load, a lot of friction. So I keep habits. Everything's smooth, everything's simple. Reduce friction, reduce any possibilities of my model, like being in friction and expectations have to be on track. So I have that. OOkay, so we had that: smooth sschedule, no friction, do the same thing. I check my email at the same time. Everything iis running smoothly now. And then I wanted to say on the learning tthing, that's also important, just for contextual learning, the iimportance of contextual learning. So the brain, how your brain best works is often in terms of contextual cues.
- 1.64** So if I really wanted to give the best lecture of my life, I would come here maybe Sunday. Let's say Sunday. I say, "Josiah or Nancy, give me the key to the studio." I come

here. The very same time that I'm supposed to give this lecture, let's say 10:00 a.m., I come here, I open the door, go inside, wear the suit, tie, everything, get my makeup done, everything, come here.

- 1.65** I can't have people come here—you wouldn't come here on a Sunday. You have to do your family stuff. Maybe, I don't know, go to church, whatever you're doing, you have all your things. So Baland would be alone. So maybe I hire an audience, another audience I bring in. Come in, come here, they sit here. And then I give the exact same lecture to this audience, this mock audience. Now, that would be great for my circuitry. Why? Because I did contextual learning. I did the exact same lecture as contextually close to this lecture that I'm giving today. That would be contextual.
- 1.66** In fact, when I was in—I talked about this in my previous lectures, courses and stuff. But as a student, I was a very poor student in school. Oh, my God, I got so bad grades. Couldn't do nothing. I was so bad. I was so bad in school. And once I went to—I'd moved to Cairo at some point in my life—I got really good at the university in Cairo. And the reason is I started to love the information, knowledge, and all that. It just became much more meaningful.
- 1.67** And one of the ways I practiced for exams is I would go to the very room of the exam. And that was weird, like on the weekend, but I would go. And I would sit in that very room where the exam was supposed to be. And I had a recording of all—I was so OCD that I recorded all my lectures for all classes. I had this recorder that I brought from Copenhagen. I brought it with me to Egypt. And every single class I had, I recorded the whole class. It's weird. That's why I'm saying I have only a few friends, not many friends.
- 1.68** And then I sat on the very place where I knew I was going to sit during the exam, and I listened to the whole lecture, and I took notes, and I just learned. Can you imagine this? It's a bit weird, right? But it worked. It did work. So that's contextual learning. Your brain learns much more in that environment. So the exam came, and then it was like the information just came to my mind.
- 1.69** I want to make this clear, too. So I talked about stability, Joe, routines, predictability. That's all good for an event. Why? Because that event is already high friction. It's high cognitive load, high emotional load. So you want to keep things smooth, keep things stable.
- 1.70** But generally in life, though, in order to grow, in order to expand your sense of self, even your brain, you have to move from the low-friction area into the middle-friction area. You have the low-friction area: this is boredom, comfort, routine, schedule, sameness. This is up here. Then you have what we might call the growth zone: some uncertainty, some friction, some discomfort. This would be the middle zone. And then you have overwhelm: too much, I'm overexcited, I can't regulate. The middle is the arena where growth happens, the middle, because, Giovanni, that's where you can

grow maximally. And neurochemically, of course, that is important. But why is it important neurochemically? Okay, based on everything you know from this course, my previous courses or anything I've said, why is the middle zone the key zone for maximal growth?

- 1.71** Because if I'm going above that, we'll overwhelm all of the brain systems.
- 1.72** Balance, right? Mr. Miyagi, right? So the middle is— Think about it neurochemically. Middle. Balanced. Noradrenaline, right? Dopamine, balanced. GABA, glutamate, balanced. Acetylcholine, focus. TThe first, boredom—do you have focus? Not really. If I'm doing the same tthing over and over, if I'm going to the same— II don't know. Give me an example. If I'm doing the same soccer practice rroutine every day, boring. Do I have a lot of attention? No. Acetylcholine system [is] down. WWhat about do I have dopamine? If everything is just routine, bbasal ganglia movements and me just passing through life—basal ganglia, by the way, is a region of the brain iimportant for more automatic habitual movements. Dopamine is important there, but key reward circuitry of dopamine is what's called the nucleus accumbens.
- 1.73** I don't mean to be overtechnical on you here, but my point is middle zone, middle zone is what creates an anticipation of reward in dopamine. Acetylcholine, right? Noradrenaline, up as well. Prefrontal, on. If everything is the same, if everything is just routine, if everything is the sameness, my prefrontal cortex shuts down, right? So you want to have novelty.
- 1.74** Let me give you a concrete example. Let's think about a concrete example we can think about that's really good in terms of middle and not overwhelmed. What would be middle? Low is obvious, right? So it's, let's say, me going to university and doing the same things. Let's say I'm a student, I'm back as a student in school, I go to university, take the same bus, then I get out and then go to the class, sit in the front row, go home. Not much acetylcholine and dopamine. Not very interesting.
- 1.75** What would be a middle example of something that's middle-challenging? I know, I know, I know. I go to the gym, okay? Instead of just doing my routine things, now I add a few more weights and a few new machines. And you know what? I'm going to do the squat for the first time. I won't take tons of kilos there and overwhelm myself. I just do a little bit extra, just a little bit of improvement, right? And then I do all this, and then I have what? Moderate growth zone, okay?
- 1.76** Now, if I was to go and say, the hell with it, I'm just going to put tons of weight over here, and then instead of doing four sets, I'll do seven sets, and then I'll do bench press, but I'll put tons of kilos on it for the first time. And then I'll do, I don't know—what else do I do that's extreme? I do a Zumba class with my eyes closed. I don't know. I do something crazy. This would overwhelm the system.
- 1.77** Now, noradrenaline, alertness would go up. Acetylcholine focus would be all over the place. In fact, my prefrontal cortex, important for regulation, would probably shut

down. It would just go, "My God, I can't handle this anymore. Too much." My amygdala, fear part of the brain, so the limbic portion, you have a fear department that would fire with activity. And when that fires with activity, and glutamate circuits in that circuit put fire and everything, the prefrontal cortex will shut down.

- 1.78** Now, when the CEO shuts down, then what? Can you be regulated? Can you be calm? Can you be peaceful? No, right? So everything collapses. Everything collapses. That's the prefrontal cortex. You want that to stay stable. Dopamine probably just completely collapses. There's no reason to feel reward. What reward? I'm dysregulated. I'm out of balance. I'm out of a sense of stability.
- 1.79** And this is also where plasticity comes in, neuroplasticity, the brain's ability to change its shape. When I have attention, that's the scenario, too, with the squats and everything, and I'm just keeping things middle; I'm having acetylcholine, nucleus basalis system. The attention system is on point. I have attention. The first ingredient for neuroplasticity.
- 1.80** And I have reward. My reward system of dopamine is not overwhelmed by all the activities, so I have a sense of accomplishment. I do this new seven kilos, 10 kilos. This is manageable for me. I do 10 kilos. I feel reward. This is so rewarding. I'm going to have all these new muscles, and I'll impress that girl, for God's sake. Take a little bit, one more. Dopamine gets encoded in the system. I want to do more of it. I have attention on the bloody thing, and I have reward as well, the ingredient for neuroplasticity.
- 1.81** Now, if that was 30 kilos and 30 kilos, would I have reward now when I couldn't lift it, and everything is overwhelming? Would my attention become totally fragmented? Yes. Same goes for that job interview, right? Or anything in life. If you overwhelm the system, you become dysregulated. You have no neuroplasticity. Neuroplasticity, by the way, assuming you all know what it is, it's basically the brain changing its shape. So whenever you're doing a new activity, you want your brain to adapt. You want the neurons to say, "Oh, my God, this is so cool. Let's build these new connections with other neurons and build new synapses. So next time, something like this comes, my brain is already prepared. I already strengthened the circuits." Think about how Messi, when he plays soccer, everything is effortless, everything is smooth, everything is beautiful. Poetry. Why? Neuroplasticity. His brain is already wired; the experience is already wired into the system. It's like a dance, you see? It's beautiful, like ballet, you see? That's neuroplasticity. Neuroplasticity. So you want that, and you have that in the right balance where you're doing things in the middle zone.
- 1.82** One of my favorite examples, I think—and then we'll go over how you can train for regulation, how you can really train for better balance in regulation. I'll have some steps I've written down. We'll go over that. But before we go to that, let's think about—I have actually one movie that I really love. I think it encapsulates what regulation is all about and the very essence of this course, which is not willpower. It's not feeling unafraid or feeling like you're on top of the world, but it's rather regulation, being able

to regulate when things are hectic. That's Gladiator. Have you seen Gladiator? Maximus, right? Maximus. How is Maximus inside the arena?

- 1.83** He's strong.
- 1.84** He's strong, right? Now, imagine, I want you to transport yourself back to ancient Rome. Right now, Nikola, you're a gladiator. You're going into that arena as a slave. Okay, your hands are chained; you're going into that arena. You know, everything is—people are shouting, the crowd. You're thinking, "My God, these people out there with swords," saying, "I'm going to die now."
- 1.85** Have you seen the scene where there's this guy? He pees his pants. He's so scared. What is that an example of? Lack of regulation. In other words, his amygdala, the fear part of the brain, overrides his prefrontal cortex. He's going, "My God, I'm going to die now." What probably happened to that guy? He probably died, right? He couldn't regulate. I mean, this is not something to be ashamed of. We all experience this, right? On that first date or whenever you're going to do something completely new, sometimes we just can't regulate. Or that new boss or that job interview, I collapse, I can't regulate. I'm so scared. My heart is beating. My noradrenaline, it's all over the place. Lack of regulation.
- 1.86** But Maximus, different, different. He goes into the arena and looks at that guy who pees his pants and he's like, "Okay, I know this is going to be hectic. This is serious stuff." But he goes in. He's able to shut down his noradrenaline. His amygdala is screaming, probably, "We're going to die now. High chance of me dying." But me listening to that emotional core of the brain and the fear parts of the brain will, in fact, not help me survive, but will probably lead to my death.
- 1.87** So he has prefrontal override, meaning the prefrontal logical part of the brain overwhelms the emotional part and says, "Look, buddy, you stay silent now. Be quiet. We have to use strategic, logical, long-term planning. We have to think about how to get out of this scenario alive. We have to think strategically." And mind you, the prefrontal cortex—this part of the brain, the CEO—and the emotional part of the brain speak different languages.
- 1.88** That's why you can shout at somebody and be angry at him. You don't think rationally when you're angry, for example. Have you noticed this? Or when you're sad, whoever did that thing to you, that man, you just hate him. He's just the worst person in the world. You can't stand him. You never want to see him again. You're done with him, okay? But then two hours later, you think about it, you go, "Maybe it wasn't that bad," right? That's when the prefrontal comes online again, you see, and it says, "Oh, it's fine. Maybe it wasn't that bad." You do a reinterpretation.
- 1.89** Logic part of the brain, emotional part of the brain, speak two different languages. They came about for different purposes in the brain, so they often are in conflict. What you want is these two to speak together in a nice way to be friends, to have good

communication, but you always want the logic to be overriding that one. So fear and other emotions, sadness, anger, jealousy, don't overwhelm the prefrontal.

- 1.90** That's what Maximus is doing. He's overriding that. He says, "Okay, I'm going to override that, and I'm going to stay focused within the arena. I'm going to stay calm. And you know what? I'm going to recruit a couple of these guys over here and make them friends." So he has these big friends. He makes friends with them. He said, "You go over there, and you go over there." He strategically knows what to do. He keeps [dysregulation] down. He keeps noradrenaline down. He keeps cortisol down. He keeps prefrontal up. He keeps amygdala down, fear part of the brain down, and he strategically maneuvers.
- 1.91** Now, when he can do that, when he's keeping these systems down, his basal ganglia, the movement part of the brain, motor cortex, can operate in a frictionless manner. He can smoothly operate. The sword is coming there. This guy is coming with this brutal thing here to kill me—spear—he can work effortlessly. Almost like Messi, almost like a soccer player. LeBron James. Anybody, think of anybody. Why? Regulation. He was able to keep it all down. For somebody like Maximus, the key is: what made it so that he could survive in that arena? What made it so that he was different? So we want to look at that. Okay, before I go to him, though, I want to have one example.
- 1.92** Who would be an example of somebody who was not able to regulate himself under these circumstances? Zinedine Zidane, the poet of all poets, okay? In the World Cup 2006, I was watching this. I was a young boy, okay? I was watching this on screen, and somebody came and cursed at his sister and mother in the final, okay? "Because your mother is like this, your sister is like this." What did he do? He headbutted his chest. He was thrown out. Example of what? Lack of regulation. See, the guy was very smart who did this. He knew that if I get the best player in the world to bite my bait here, we'd win the World Cup. And in fact, Italy did win.
- 1.93** But this is how your enemies—you will have enemies in business and whatever. They will sometimes poke at you. Competitors, rivals. Don't call them enemies, but rivals, competitors, right? They will poke at you. They'll come at you like the devil, and make you angry like Zidane, right? So you have to stay calm in these situations.
- 1.94** In fact, in Gladiator, have you seen Commodus when he goes to him in the very final battle they have? Have you seen Gladiator? Giovanni, you should watch Gladiator, man. My brother saw it 10 times in a row. One day. It's on the curriculum. You have to watch it, okay?
- 1.95** Now, in Gladiator, you'll see Commodus, that's the evil emperor. He goes to Maximus in the final match. First of all, he does something very evil. He takes his knife and stabs his back like this. So now he can't fight. But nobody can see that he's hurt; he's injured. But then he says something to him. He says—I forgot the exact wording—but his son

and his wife were killed by the Roman soldiers. Very smart, this Commodus, Caesar, very smart. What is he doing? He's trying to overwhelm his system.

- 1.96 Have you seen Zorro? What happens in Zorro? In the initial scene, one of the initial scenes, the old Zorro, this would be— What's his name? I forgot the actor. Not Sean Connery, the actor. Antonio Banderas is the young Zorro, but then there's the old Zorro. He's Anthony Hopkins. Anthony Hopkins is the old Zorro. He's training the new Zorro, okay? And then he says, "Come, show me how you—" And then he takes out this [sword], the young Zorro, Antonio Banderas, takes out the sword, and he attacks, attacks. And then he does— And Hopkins, he's about 80 years old, he does like this. And then Banderas is on the floor. And then he says, Hopkins says, "Never attack in anger." Never attack in anger. Why? You're using the overwhelmed system. You're in a lack of regulation; amygdala, fear parts of the brain overriding the prefrontal areas. What Commodus was trying to do to Maximus before that fight was to make him more emotional in his fighting, less strategic. He was trying to tell him, "Don't use your strategic prefrontal cortex. Use your fear parts of the brain. Use your emotions. Be angry at me. I did all this to your wife and [son] and kid and all that. Be angry." But he was able to keep himself calm. In fact, he won the fight. If you've seen the movie, he won it.

## Lecture 2: Performing Under Pressure

At a glance — Performing under pressure.

- 2.1 Now, as I was sitting there in my room, usually before I come out here and give a lecture one night, I just turn off the light and sit there and meditate, just let my mind wander and relax. And this is very good. It makes my brain much, much sharper. You see, it makes it less crowded, less fragmented.
- 2.2 But as I was trying to meditate and I was trying to focus on nothing there, some thoughts came to me, and I thought, "Okay, hold on, these thoughts might actually be important. Perhaps I could use these thoughts in the next lecture." So I didn't know if this was a trap or whether this was actually good. But I said, "Okay, hold on, let me just get the thoughts in." And when the thoughts came to me, I thought, "Okay, here's something important."
- 2.3 See, I was talking about preparations. So you have a job interview, you have a soccer match, you're going to see that girl, that date, important. You've been waiting for one month. Okay, you're so excited. Now, you prepare for it, right? You have your mind and brain. Everything is just prepared. You've trained in an environment similar. You've done things that could make you optimally able to deliver the job that's supposed to happen.
- 2.4 But I think a key component that I left out is the following, and this works for me always, always, always. If I go to give a lecture, I go for a meeting, I go for anything important, and I have the attitude of preparing and saying this or that, and just putting on a performance, it never works. It never works.
- 2.5 You have to switch off, you see? You have to go, "I'm going to go out to that interview, or I'm going to go see that girl, or I'm going to do this or that, that soccer match, and I'm just going to be myself completely." Let go of any inhibitions, any fears you might have, and just be yourself completely. And I know this sounds a little bit mushy. It sounds a little bit like, what is this authenticity business, right? But actually, there's something to it. There's something to it. And I was just trying to chew on this in the room over there. What exactly is it about?
- 2.6 And I think there are several components to this. I think one key component is the following. When you are authentic, and you go in there with no fear and say, "The hell with it. If the girl is going to not like me after this, it doesn't matter." I know this is hard because you've been waiting for a very long time. You're very excited, right? This is important. There's high stakes—that interview, that job, that career. When we go in with the attitude of no fear, that person across from you can sense it; they can smell that you come here with no fear. "I'm going to be myself no matter what." They can just sense your emotional expression.

- 2.7** You see, in my previous lectures, I've talked about when you smile, you have something called a Duchenne smile. Like an authentic smile—different muscles than an inauthentic smile. So you use this, use your eyes like this. Can you make me laugh? Say something funny. You see my eyes are moving? You see my eyes are moving? Okay, now I'm going to do an inauthentic smile. Not very charming, is it? Completely different muscles. You'd use your cortex, the premotor regions, the motor regions for the inauthentic smile. Now, when you do an automatic smile, the same muscles, but you also use the basal ganglia, the automatic movement part of the brain.
- 2.8** Now, our brains are wired to detect liars because when we were in the savanna back in the days and we were negotiating with our fellow peers for food and all that, we were looking at their faces. "Is this guy lying when he says that all the animals are over there, we can go hunt them? Is he trying to get me away from the actual place where the animals are so he can go get the animals himself?" So we have this.
- 2.9** Being able to detect lying faces is good for survival. To detect authenticity, to detect sincerity. The voice. How is the voice? The tonality, all of that. You see? And that is completely detectable. And I think that is the key. That's why going in with authenticity—it's hard to say, you see. Being authentic in these scenarios is key for surviving that interview, coming out as a winner.
- 2.10** Also your brain, right? When you do that, then you train yourself to go, "You know what? I'm not going to care about the outcome either. I'm interested in that. I don't love that girl yet, but I'm interested in that girl. But it doesn't matter if she's not into me that much. I'm going to be okay with it." You train yourself to deal with it later on, and the whole circuitry becomes different.
- 2.11** So I think being authentic, being fearless, letting go of the prefrontal cortex a little bit, and letting the emotional part also just be authentically you, I think it's key. So when I have that, let's say, part of the whole thing. So the prefrontal cortex is important for inhibiting fear, but you want to have authenticity there and letting go as well. In fact, in this lecture we'll get into more of that, about letting go and just flowing and being yourself.
- 2.12** So I did mention some points. How do we become more like Maximus? How do we become more robust, resilient in these high-stakes situations? What is it that he was able to do in that arena when he was fighting off all these people trying to kill him? If you know the movie, Maximus was a general, I think the Northern Army. He was a general, right? So he had experience on the war field, on the battlefield.
- 2.13** So that experience, he could translate. Remember we talked about translation of scenarios, how the brain generalizes from context A to B and C? Same thing. Whatever you are doing, high-stakes, go and practice in that situation or a very similar situation, and that will give you a very robust sense of—it can make things easier when you go and do the actual thing. So that's first.

- 2.14** Exposure. Now, of course, sometimes it's not possible to do the actual exposure. So we do something that simulates it, or even using virtual reality in some situations is often used. But of course, this would be somebody who has a fear of public speaking. They might do a virtual reality version of that. But I think that's gone a little bit too far. It's not very practical.
- 2.15** What else can you do in the actual situation to be like Maximus? What can you do? And I would say, if the anterior cingulate is monitoring the landscape for importance, and the OFC—we have something called the OFC in the brain—it's the police officer. The police officer is always detecting threat, Kelly. It says, "Is this threatening? Is this dangerous? Will this kill me?" This is the OFC, orbitofrontal cortex. It's hyperactive in OCD. I call it the police officer of the brain, detecting fear and assigning value. With the ACC, the anterior cingulate cortex, detecting conflict, danger. These are highly active.
- 2.16** These will—if there is danger in the landscape, they will signal to the amygdala, "Oh, danger. Somebody coming towards me with a sword." This is terrifying. I'm on the soccer field, and somebody's trying to tackle me. Danger. It sends information to the amygdala. The amygdala is in the emotional part of the brain. The amygdala has circuits there talking to something called the hypothalamus, which is the hormone factory of the brain, producing hormones.
- 2.17** In other words, the police officer detects a threat. "Threat present." Sending signal to the emotional, the fear part of the brain. The fear part of the brain lights up, sending signals to the hormone-factory part of the brain. Hypothalamus becomes activated, now sends out signals. Pituitary, release, send down signals to the adrenals. Release cortisol, noradrenaline. Noradrenaline: you're agitated. Cortisol: you become focused. Actually, cortisol in mild doses is good. It makes you focused. It makes your attention sharp as well. It's very important. Now you have this.
- 2.18** Now, Debbie, there's information going to the movement part of the brain as well. Signals are being sent. The cascade is continuing. Movement part of the brain: move, act out, move your hand like this, move your body like this. As long as the threat is not too overwhelming, it's frictionless, it's smooth. Of course, in some situations, it can become overly excited—I mean, overly terrifying. The danger is just too overwhelming. I can't deal with this.
- 2.19** The way to do this is the following. Shift your attention. Attention is crucial here. Attention: we have two types of attention, endogenous and exogenous attention. Two types of attention. Endogenous attention is me looking at the screen over there. Endogenous, coming from inside me, choosing to pay attention to that over there.
- 2.20** Then you have exogenous attention. That's when I'm looking at the screen, or I'm looking at Max over here, and then suddenly there's a cat, and I look at the cat. This is a bottom-up process. My attention shifts from endogenous to exogenous—exogenous means external. These are two types of attention. When you are in a high-stakes

situation, in something that's highly arousing, do you shift your attention inwards or outwards?

- 2.21** In other words, I'm Baland, giving a lecture. I'm highly aroused, so much so I'm overwhelmed. In fact, I can't give this lecture anymore. It's just too much. I don't know what to do. Should I focus, Giovanni? Should I focus on myself? Start self-monitoring, becoming self-critical. "Oh, my God, maybe my tie is not right. Maybe the makeup, maybe the hat. They don't like the hat. God, for God's sake. They don't like that. Or maybe it's the belt. Maybe I said something wrong." I become self-critical.
- 2.22** When I become self-critical, what part of the brain lights up, by the way? Max, you know. Come on. Default mode network. Self-criticism network, right? Frontal, parietal, medial. You have this, right? This is a little network we have in the brain, a little system that's important for whenever we think about the self. When Juan is thinking about himself, his life, his place in the world, this self-criticism and mind-wandering network goes ballistic. Starts flaring with activity.
- 2.23** Do you want that on that day? You meet that girl, is it going to be that? This is the one. This is the girl I'm going to be with. You like her. Should you become self-conscious, Joe? Self-conscious is good? It's very bad. On that soccer field, Messi, Ronaldo, should they become self-aware, self-conscious, or should they become outer-focused? This is the key.
- 2.24** Am I boring you guys? I'm becoming—see, this is what's tricky. I'm becoming self-conscious now. You see? Not good. You have to be external. You have to look at faces. Is Debbie enjoying this? Maybe she's enjoying this. You see? You become externally focused. Like the gladiator, Maximus, what he does on the field, he focuses outward. The hell with the default mode network, okay? I'm going to focus all my attention on this warrior over here, on this horse, on this sword, right?
- 2.25** Or the girl. What is she saying? Is she smiling? You're focusing outward. Forget my inner state. Okay, maybe I shouldn't talk too much right now. I'm overwhelming the girl. Stay calm. Look at the waiter. Be nice with the waiter. You have to give a good impression. You focus outward on the environment and you shut down the default mode network, the self-criticism network.
- 2.26** Is that clear? It's important. This is now an example. Shift attention outwards in high-stakes situations. That's what Messi does. That's what Ronaldo does. That's what a good CEO does in a board meeting instead of focusing on himself. Self-criticism, the default mode network, the medial prefrontal cortex. Shut that whole cascade thing down and focus on the environment.
- 2.27** How can I improve the situation? What's the next actionable step? How do I overcome this? How do I include the group to overcome these other warriors on the field? Of course, your nervous system—can your nervous system be hacked? Can you hack your

own nervous system? So we have two branches of the nervous system: the sympathetic and the parasympathetic.

- 2.28** Sympathetic means—Debbie, what's a sympathetic nervous system?
- 2.29** In charge of that.
- 2.30** Is it—
- 2.31** We can guide it.
- 2.32** We can guide the nervous system, yeah. When you are aroused, you're giving a lecture, you're playing that soccer match, you are running from a tiger, is that a sympathetic nervous system that's active? Or is it a parasympathetic nervous system that's active?
- 2.33** Sympathetic.
- 2.34** Sympathetic. It's sympathetic, right? So that's when you are agitated, you have high cortisol, you have high noradrenaline, right? So that is the sympathetic nervous system. Then you have the parasympathetic nervous system. That's when you're calm, you're rested, you're just chilling on the couch, maybe watching a, let's say, boring movie. That's reading a boring book. That would be the parasympathetic nervous system. Things are calm, you're just resting, you're chilling.
- 2.35** When you're getting a massage, what is that? Parasympathetic or sympathetic?
- 2.36** Parasympathetic.
- 2.37** Exactly. When you're listening to this lecture, parasympathetic or sympathetic?
- 2.38** Parasympathetic right now.
- 2.39** No comments, okay? All right. Can you hack the system? That's the question. That'd be a good thing, right? See, the thing is, when you're facing these gladiators or you are in the soccer match, obviously, you will have a sympathetic activation. You're activated. You're looking at various other players. You're focusing on the goal. The whole thing is in front of you, right? Now, being too overwhelmed by the sympathetic nervous system and being too high amygdala, too much noradrenaline, cortisol, high arousal all the time, bad thing, especially as a gladiator, for God's sake. That's a fatal mistake. And can you hack the system? What can you do?
- 2.40** Breathing.
- 2.41** Breathing. You can do breathing. So maybe you can do—and this is a practical step, right? In those high-stakes situations, you can make the system tilt a little bit more towards parasympathetic nervous activity. You do maybe some deep breathing. It could be two long inhales and then out, okay? This is known as the physiological sigh, very well known. [Inhales twice]. Keep your breath in, maybe hold it in. Five seconds. [Exhales]. You feel it? [Inhales twice, holds it for five seconds, exhales]. Very simple, but you can apply this, okay, directly on the battlefield.

- 2.42 Meditating is difficult in this scenario, but maybe you can—and this is me thinking aloud—visualize as well. That'd be a good visualization. Let's say you actually go to a different country, or a different state, and you have this unknown event. You don't know how—it's unknown in the sense that it's a lecture or it's a job interview or it's this thing you're doing that's important.
- 2.43 What you can do is you can try to visualize the event as much as possible. Just try to sit down and visualize and see how it would actually play out in real life. Is there anything that I can do to manage it? It's just like looking at various scenarios. A, B, C. This might happen: outcome A, B, C. This actually helped me once in a situation.
- 2.44 Let me give you an example. Don't be too frivolous, okay? But there was this girl, and I wanted to make an impression. I liked her. I don't do this every day. I'm not like this—I'm a pretty modest guy, okay? But I actually liked her, and I still like her anyway. And I wanted to make an impression. So I had these scenarios. If I do A, based on my past experience, if I go up and talk to her, I know she's been mad at me because some stuff I did. So probably if I go and talk to her initially, she will lash out at me and be harsh.
- 2.45 I knew, based on personality theory, these charming girls, though. But they're very fiery. That's part of the whole charm. You never know when it's going to be a ninja kick on you. But it's just part of it, right? Like psycho chicks. Anyway. I thought in my brain, I knew a scenario here. If I go to her and I talk to her now, she'll be angry at me. I did foolish things. But I know already she's going to be angry at me and she's going to be very mean, say something stupid, say something like, "I don't want to talk to you."
- 2.46 I already modeled that. It was part of my modeling. So I said, "My plan is the following. I take her on the rebound. I go to her now, let her shout at me and reject me so she can feel good"—I modeled that. "Then I will go away and wait for her to come to me." You see, that's part of the modeling.
- 2.47 And it worked. See, I already knew how prediction would work. And then I went to her. Instead of becoming agitated because of what she said to me, I didn't. I already knew. The first time she became angry at me and all that, I was flustered. How dare she say this to me. What chick is this? But I modeled it this time. I said, "Okay, I know her. I predicted it." I had a scenario, went away, let her come to me slowly. She came over, and A, B, C. Is that clear? Predict the event, have a good modeling. So whenever you are in that situation, you predict the boss might say something stupid here, or somebody might shout at you here, or somebody might tackle you very harshly.
- 2.48 I know when I go on the field as Messi or Maradona in his 1986 days, I'm going to have five players around me trying to tackle me all the time. It's just how it is. I'll put that in my prediction. I'll visualize that. They'll come left, right, tackle me. I'm not going to get flustered. I'm not going to be like Zidane. I'm going to expect that somebody's going to come and curse me out, try to make me agitated. I'll build that into the prediction now. And so when I go on the field, you swear at me, curse at me, tackle me, reject me, it

doesn't matter. I already predicted it. Now I'm going to wait, wait, wait, and then do my thing. So put that in. I think that's a good personal tip.

- 2.49** And I think the next one gets to what Juan said, really. So perhaps I'm repeating, but this is the reason why practice under heavy load and pressure is the key way. So do anything that resembles the actual event. Go to a soccer match where they'll pressure you. If you're military, do something that mimics the military situation. CEO, go to another meeting. Be around people that will pressure you. So this goes with the territory.
- 2.50** Number six, I think, is crucially important. We might have an entire lecture to do this, but it's really about the whole idea of—I think we think about somebody like Maximus or Messi or Maradona or Zidane as just people that are greatly talented; they do those things on the field, but they do it nonstop. It's just a false impression.
- 2.51** Even the great CEOs, the great ones, I think they know when to rest. They know when to recalibrate. Actually, even in the situation itself, even during the fight itself, as I talked about before, there's this dance between high arousal, like the Lamborghini, right? Lamborghini example. Lamborghini is like the [makes revving sound]. It accelerates, it's up, and then it goes down, comes down. [Makes revving sound]. And then it goes up and comes down, like a Tesla, right?
- 2.52** I think this is crucial, not only in terms of when the event is over, you want to have sleep to consolidate, have neuroplasticity, have rest, recouping the whole system. Forget all that. Even on the field itself, even during the high-stakes situation, if there are moments where you can do micro-pauses, that's a good thing. Doing micro-pauses is actually—sometimes I go up and I drink some water. Have you noticed this? What is that about? Coming down a little bit, yeah.
- 2.53** Sometimes I even pause in my speech. But you can do that, right? And then I think it does allow, even if it's momentarily, you just calm this whole thing down a little bit. Your brain goes a little bit down. It goes a little bit down. I think going down like that, even in a high-stakes situation, can really help.
- 2.54** So it has this balance between going—then whenever it goes up, it just becomes better, like Messi on the field. Some young soccer players just run around. Have you seen young soccer players or other sports? Basket? Maybe basketball. You like basketball? Alexander, you like basketball? What do you like? What sport? You like sports, Rachel? I know you're from Canada. Come on. Ice hockey, right? Not really.
- 2.55** Maybe basketball.
- 2.56** All right, basketball.
- 2.57** Young players, how are they? Very agitated, high arousal, hot-headed a little bit, right? They don't have that—hyper; they don't know how to calm down and keep the system down. I think that balance is important.

- 2.58** Let's go beyond sports. What other example? Yeah, but beyond sports, though, in your practical life, how can we apply this? How can we apply this? In what high-stakes situations would you fire?
- 2.59** New hires.
- 2.60** New hires?
- 2.61** Disciplining your children.
- 2.62** Yes. Yes, disciplining your children could be. But let's say—
- 2.63** Who said something here? Public speaking. Do you have a public speaking event soon? But then we already went over. Another example, Joe, what do you think? Other practical example you think of?
- 2.64** Presenting in front of a boardroom.
- 2.65** Being asked a question by you.
- 2.66** Yes. My interest in that would be, while I'm not speaking, you calm things down, and then your mind is ready. But yeah, board meetings, lectures, it's all about that balance between going down and going up. This whole idea of being erratically all over the place all the time, it's hyper dopamine. It's crackhead-like. You don't want to do that. Don't be a headless chicken. The whole thing is that you want to calm the system down and then have optimal performance. It's interesting. Optimal performance. There's so much we can cover, right? And depression, of course, and anxiety and all those things—we had a lot of notes for this.
- 2.67** I'm thinking, I'm wondering the reason that I have all these notes could also be an expression of my own life lately. I'm trying to think out of the box. My own life lately, there's so much going on, so many new things, so much stuff.
- 2.68** I recently did an interview with this guy. He wanted to interview me. He's a cool guy from the hood. He wanted to interview me. I have a little book that came out recently, not to brag, a Danish book. It's about my life in the ghetto as well as the brain. But he sent a DM to me. He said, "You want to come on my show?" I said, "Okay."
- 2.69** I said, normally, if something like this happened—I'm not saying it happens a lot, but let's say it was to happen—I normally maybe say, "No, I don't want to do it. I just want to do my science, have fun a little bit." I saw this guy, same neighborhood, same thing, the same origin, Kurdish, from Iraq. Okay, let me do it, man. Represent the culture and all. Let me do this thing, podcast.
- 2.70** Anyway, I went over there. He had all these—I hope he doesn't see this. Anyway. He's a great guy, and I love him. I love him. He had all these questions on this paper for me. "I want to talk about anxiety, then stress, then sleep, then dreams, then we'll talk about your life in the hood." The list was longer and longer. I said, "How long do we have for this?" I was expecting four hours, Rogan style. He said, "About half an hour." I said,

"Goodness me. Goodness me." Then, as the conversation flowed a little bit, the conversation was good. I talked about my life, growing up in the neighborhood. It was fun. I enjoyed that.

- 2.71** But then I asked him at the very end. I said, "We talked about stress." I looked at him and I said, "Are you stressed these days?" He said, "I'm very stressed. I'm hyper stressed." I said, "Oh, okay, I understand." The reason I knew, it's instinct. Being in the field, you get to my age, you're in the field a long time, you know that people, when they are stressed, they become overcrowded in their thinking. Do you know this? You've noticed this. You prepare for an exam, you over-prepare almost. You have so many ideas; things become a little bit more cluttered, right? You can't think straight.
- 2.72** It's not that old simplicity you always had, like when it's just more simple. It's like this idea, "Oh, let me go with this idea as well. Ah, this? Okay, we've got to have this as well." And you just keep building on and piling on. And I wonder—I'm just trying to have self-insight here as well as teaching a key lesson—I wonder if my own stressful days these days are contributing to this.
- 2.73** Stress has many forms. Stress can be from work, it could also be from romance, it could also be that I have a lot of things going on, you see? So it could be that it's a combination of all that that's really contributing to my—there's a form of OCD, a symptom of OCD. It's called excessive list-making. Do you know? Did you know that? You put everything in a list.
- 2.74** Let's talk a little bit about plasticity, how plasticity works, how we can use the principles of neuroplasticity to improve our brains and get optimal performance in our lives. What is neuroplasticity?
- 2.75** It's the ability of your brain to change its shape.
- 2.76** It's the ability of a brain to change its shape, right? Now, as kids, we have tons of neuroplasticity. Why do we have tons of neuroplasticity? What's one factor?
- 2.77** Acetylcholine.
- 2.78** Acetylcholine, right? Nucleus basalis system. Is it perpetually active when we are children? Yes. You pay attention to everything?
- 2.79** We said dopamine is important for neuroplasticity. As babies, we have a lot of REM sleep as well. REM sleep is important for neuroplasticity. As children, our brain is hyperconnected. Everything flirts with everything. You have a hyperconnected brain, 50% more synapses. Synapses are the junctions between neurons. You have neuron A, neuron B, you have the space between them. It's called a synapse. The synapse is where the communication happens between neuron A and neuron B. This is called a synapse.
- 2.80** In babies, infants, you have 50% more of these synapses, but you have a gene then. It's called a synaptic pruning gene. It shapes away excess synapses. And so when you reach

your teenage years, you already have a perfectly shaped brain, let's say. In other words, you have a brain that's better shaped for the environment. I didn't use these circuits over here. They were redundant. Put this away. These circuits over here were not useful. Or they were useful—let's keep them. So you have that. That's part of neuroplasticity. Synaptic pruning.

- 2.81** Now, for dopamine, when you are repeating the same behaviors, let's say, you are now that CEO. You're now that surgeon who's doing the same action over and over again. Here's a surgeon. He's suturing something, right? I have attention, right? I have reward. It feels good. I have that. What else do I have? You have behavioral repetition. Behavioral repetition leads to myelination. That's how efficiently—on the neuron itself, on the axon, you have something called the myelin sheath. It allows it to send signals efficiently.
- 2.82** As you repeat actions—this is very basal ganglia territory, automatic movement—the more myelination you get in the system, things become more efficient and faster over time. In other words, the first time I juggle a soccer ball. First time you juggle a soccer ball, is it difficult? Or play a piano for the first time. Very difficult. You recruit many centers of the brain. You have prefrontal, you have parietal, you have auditory, you have all these centers being activated.
- 2.83** Or the first time you're doing—give me an example—your personal life.
- 2.84** Sewing.
- 2.85** Sewing. Joe, something? Cooking. Anything. This is the first time, right? Demanding, challenging, right? But as you do it, what happens? More myelination, okay? You have dopamine, you have attention, you have repetition. The circuit is laid down. As the circuit is laying down, we have a principle in the brain, Kelly, it's called use it or lose it, okay? This is like a principle. It's very interesting, actually.
- 2.86** It's like whenever I'm repeating something over and over and again, neurons dedicated for that task will expand. You have like monkeys in experiments, they will use their fingertips for spinning this disc for like six weeks. They do it, do it, do it, do it, do it, do it over and over. Eventually, the finger area in the brain—I mentioned, here you have motor cortex. Right behind it, you have the somatosensory area. That region is important for bodily representation of the entire body. Now, that part of the brain, the fingertip area of the brain will expand, will expand the more I use my fingers.
- 2.87** So if I was to juggle a soccer ball, if I wanted to be the next world star soccer player, what would I do? I'd play soccer all the time, right? Especially as a kid. You know Messi as a kid? They say that he was in—my sister-in-law is from Peru. And she said their family, they're actually here. They're actually not here. They're in Copenhagen, visiting us right now.

- 2.88** And their dad, I'm not sure if this is a lie, but he told me once that over there—he meant well, of course, even if it's not true, but I think it's true. He said that one of their friends' family had Messi come visit him as a child. Messi came to their house as a kid, five or six. And this kid always had the soccer ball on his feet all the time. He's playing soccer, soccer, soccer all the time, obsessively. They couldn't remove the ball from his feet.
- 2.89** Now, as we know of neuroplasticity, especially in children, how do you think he'll have control of that ball for the rest of his life? Incredible. All right? So all of us, if you want to develop a new skill, even as an adult, whatever you want to do, something novel you want to do, Rachel, something completely different. You want to change careers. You want to do something completely different as a sport. Let's say you want to go into basketball, let's say. By continuously doing that thing, over and over and over and again, you will grow that region of the brain. Those neurons will expand, take over more cortical real estate, more expansion in that part of the brain, and then other parts, let's say, that have to do with other tasks, let's say you have another sport you'd like, let's say, ice hockey or football or something, those regions will become less.
- 2.90** Of course, you can't have big expansion for everything. You don't have an unlimited brain, so you will have to pay a price. So you become bad at that thing. Actually, that's funny. That's how, when you learn a new language—do you speak different languages? Spanish, probably? English? French?
- 2.91** And Spanish.
- 2.92** Wow, impressive. Very impressive.
- 2.93** Now, okay. I learned a few languages in my life, not because I was smart, but because I had to. Pure survival. But I noticed something. When I learned a new language, I was speaking Spanish in school. I learned it in school. And then, see, I learned Italian at some point. I had to learn Italian, mainly because I had—it's a bit of an embarrassing story. But I told a girl I was half Italian. I was going to impress her. Then I had to learn it; otherwise I'd be in trouble. I was a teenager, mind you. I wouldn't do that now. I was 17, 18.
- 2.94** Anyway, I had to learn it very quickly. So I rented all these Italian movies and all that, got them in, watched them. Cinema Paradiso, all these movies. Interesting. I watched them, and I looked in the dictionary. I had a few words, enough to get by. But I learned my Spanish got worse. And each time I tried to speak Spanish, I was like, "Anche, and yo hablo." You mix it all up. I think—not think, "I know that's because there's this cortical competition in your brain. The regions are always competing for real estate. Is that clear, Alexander?
- 2.95** But of course, what happened, let's say you become like that soccer player, right? You become like Messi, you become like Ronaldo, you become like Zidane, and you become extremely good at that task. What happens in the brain then? Would it keep

expanding? Would the neurons there and that tissue keep expanding? What do you think?

- 2.96** What would you do if you had to design a system and you depended on plasticity, you depend on being able to take away from other tasks and other regions and expand that, do like a military invasion, take over these regions, and then use them for that particular task. And eventually, Nicola, you run out of space. What would you do?
- 2.97** I'd want to stop it at a certain point so I can do other things as well.
- 2.98** That's the more balanced answer. But the brain, you know what it does? It makes those neurons more efficient. It drives up efficiency. That's where myelination comes in. That's where neurons start to fire faster even. It's quite incredible. In fact, your brain is built for survival. It's a survival machine.
- 2.99** I know balance is how we achieve the best in all our brain health and performance, but ultimately, it's a survival machine. So it will keep increasing those neural capabilities, improving those neurons. So that shows us how repetition, doing these tasks over and over, can plastically rewire the brain. Of course, then, when you have—
- 2.100** The equation for plasticity is not full. Something is missing. And you guys took my previous courses? Dopamine, acetylcholine, repetition, myelination, all good. Synaptic plasticity, if you're a kid, all good. But what else is missing for plasticity? I want to become that new soccer player, or I want to become that new person that can deliver this board meeting speech.
- 2.101** When you are anxious and depressed, when you have high levels of stress, when you're highly stressed, there is a protein that's missing in your brain. We call it a growth factor. Growth factor. When you put it in a dish with neurons, they grow more synapses, these neurons, in a dish. In fact, that's how it was discovered by some of these experiments with tissues.
- 2.102** But what is a growth factor? What is this growth protein? I sometimes call it a neural fertilizer. It's like a fertilizer in the brain. You sprinkle it on the brain and you will have more growth. You have more synapses, you have more myelination, you have more overall new connections. What is that? You know it, come on. It starts with a B. BDNF, brain-derived neurotrophic factor, and something called GDNF, glia [cell line]-derived neurotrophic factor. Those would be examples. So these are growth factors.
- 2.103** It's almost like any system, any car, anything—you always want to have fuel. Humans, food, right? Something that could fuel the system, Kelly. And this would be the BDNF, GDNF. We all need this. When you are hyper stressed, then, when you don't get enough sleep, then, when you have high cortisol elevated all the time, you have a decline in the BDNF and GDNF growth factors.
- 2.104** This is why having breaks is so important, calming the system down, making sure you don't have chronically elevated cortisol, because that will really inhibit the BDNF, the

growth factors. Growth factors we all need. Can we do anything in our own lives just to boost BDNF? What about fish oil? Can I go buy fish oil and get some BDNF? In fact, it can boost. So you write: fish oil, omega-3.

**2.105** What else? What if I go exercise? Really? Exercise? Can I produce it? Aerobic exercise? Really? It can. So I'll put that too. So you can do that as well. You can do some exercise. All these things will boost BDNF.

**2.106** Okay, long-term potentiation is what? What's long-term potentiation? Okay. LTP, LTD. Long-term potentiation is whenever I then keep doing the same behaviors over and over, I get more synapses, get more neural circuits. Do I get more neurons as well? If I keep doing these actions over and over again, I go to that meeting, I prepare for that meeting, do novel things, do the soccer preparation, all that. I do it in the right way. I have a lot of low stress. I make sure that I'm rested, all that. I get long-term potentiation, new synapses. Long-term potentiation is also strengthening of these synapses as well. Long-term potentiation, new synapses.

**2.107** Do I get new neurons as well? So I got new synapses. Do I get new neurons as well? Yes, no.

**2.108** Yes.

**2.109** Yes?

**2.110** No.

**2.111** No? Do we have neurogenesis in the human brain? Do we grow new neurons? It's a trick question, isn't it? You can sense it, right? The answer would be we have some neurogenesis in some part of the brain, something like the hippocampus a little bit, the memory part of the brain, maybe the insula part of the brain, which has to do with mapping bodily states. But really, you don't have a lot of neurogenesis. In other words, you don't have a lot of birth of new neurons.

**2.112** That's why when you have a stroke, for example, you're relying on old tissue and other synapses and other circuits. You don't really grow new. Have you seen somebody who had a stroke and became completely normal? Not really. There's always something off, slightly off. Even though they can function, I've heard of cases of people that had a stroke, and then they could barely move, they could barely do anything, and then they climb a mountain. But still, you can sense it's not completely the same. You don't have neurogenesis, birth of neurons, but you have new synapses, new connections, typically.

**2.113** When the brain then wants to—let's say you have a new skill that's no longer useful for you. What could be an example of a skill that's no longer useful?

**2.114** You finish your sports career.

**2.115** You finish your sports career. That could be good. What else?

**2.116** You retire.

- 2.117** You retire. Or let's say you have a bad habit. How's that? Isn't that better? You tend to become impulsive at interviews or when you play soccer, right? Or when you do whatever thing you're doing. You have a bad habit. How do you get rid of that habit in the system? LTD, long-term depression. Put that down. LTD. Here it's the opposite of LTP. Now the scenario is, how do I get rid of those behaviors? How do I get rid of maladaptive behaviors?
- 2.118** When I'm at that boardroom as a CEO, if I'm a surgeon, I tend to zone out during surgery. It's not very good. I focus on other stuff. My attentional system is not very focused. I have this bad habit of going and checking my phone each time I'm doing surgery or something. Some bad habit. How do I get rid of this habit? This is called LTD.
- 2.119** The way that LTD works is the following. You do actions that would—first of all, you pause that behavior as much as possible and reduce it. So you reduce doing it. That could be telling yourself, "I'm not going to go and check my phone under surgery."
- 2.120** Or "As a CEO, I'm not going to be impulsive. I'm going to hold myself back from talking." Let's say there's some impulsive CEO. Have you seen The Godfather? Sonny was very impulsive, right? Was it good for the family business? No, he got killed very quickly, right? So if you want to prevent something like that, you can learn to control yourself. The more you can regulate and try not to get angry, over time, what happens is that you have long-term depression. It's called LTD—not depression, depression, like the mental illness, but in the brain, the circuits for that action become less under those circumstances. So you're less likely to become impulsive, flustered, angry, the more you keep away from it.
- 2.121** And then you can do a habit replacement. That's when you start doing a different habit that's beneficial. So some habit replacement. So each time that I want to check my phone during a surgery, or I get angry and frustrated during a board meeting, or a soccer player, each time I want to lash out, each time somebody pulls my shirt, I get angry and I do like this and I get the red card, for example. You know this? Some players are like that. They're hot-headed. Messi is very good. Never hot-headed. He's very humble on the field, right? But some players are not like that. They get angry, very flustered.
- 2.122** Try to, especially initially, replace that habit with a different habit, almost like a good habit. So each time that I want to go check for my phone, I'm going to consciously do something else. I'm going to consciously take a breath and stop and focus on something positive, let's say. This is a random example. Change it, okay?
- 2.123** Another example could be anything, any habit you have, replace it with something else that removes—it competes. It's always competition. That's why I'm saying. Even with languages, that's the whole point: when you have competition of circuitry, it's better than having absence because with the absence, the old circuit might come back online.

- 2.124** So always replace it with a different habit if you can. What would be a good example? I'm running out of examples for how can you do a habit change that would compete with the old bad habit, the habit of becoming impulsive at a meeting or becoming angry during a meeting or lashing out at another soccer player or so forth. What could be something you can do?
- 2.125** I've seen some people count to 10 instead of getting angry.
- 2.126** Count to 10. I love it. Something like that. Initially, it's not very practical, but I think like that. Count to 10. Maybe focus your attention on something positive. Maybe take a deep breath. I'm going to take a deep breath at this point. Anything that will teach you to replace the habit. Fill that gap. That's very good.
- 2.127** Whenever you do an action, whenever you are planning to do something, let's say, go to the gym, do you always feel like going to the gym? Do you always feel good about going to the gym every single day? No. There's a cost calculation in your brain. There's a cost calculation because the brain has limited attentional capabilities. The glucose consumption is very high.
- 2.128** For any demanding task, anything in life you're planning to do, whether it be exercise, whether it be reading a newspaper—reading the newspaper is pretty fairly simple. But anything you do that requires effort, you have this rationale for doing it. Should I do it or should I not do it? And I think what is crucial to point out here is that that system, whether you decide to go to the gym or you decide to look for that new job, or you decide to go and do something productive in society, whatever it might be, that cost analysis you're doing in your brain—should I do it? Is it worth it? Is it worth going to the gym? Is it worth going out and trying to get a job? Is it worth doing A, B, C?
- 2.129** Of course, it relies on various parts of the brain. One would be the OFC, assigning value to things. It is the ACC that's involved in conflict. Is it worth doing it or should I just rest? It's reliant on the dopaminergic centers, anticipation of reward. If I go out there to the gym, will it be worth it? If I go pursue a job, would it be worth it? If I go and play that soccer match right now, would it be worth it? If I go do A, B, C, D, is it worth it?
- 2.130** So you always have this cost calculation inside your brain. But this calculation is biased by the system that's already running, the baseline of the system. Depending on where you are on this whole—remember I mentioned all the chemicals in the brain, like acetylcholine, GABA, glycine, dopamine, all that? Depending on how that system is balanced, that will bias your decision.
- 2.131** If you are extremely tired, let's say you have chronic stress for months, would you be more likely to go to the gym or not? Why?
- 2.132** Too overwhelmed.
- 2.133** Too overwhelmed. What about if you're chronically sleep-deprived? Are you more likely to go to the gym or not? So your background situation, your background mood,

the background noise in your brain decides whether things are worth it or not. And the reason is the following.

- 2.134** Let's take stress as an example. So you have chronic elevation of stress, and then you are sleep-deprived as well. And when you are sleep-deprived and when you have chronic sleep deprivation and chronic high stress, what happens to the prefrontal cortex?
- 2.135** It shrinks.
- 2.136** It shrinks. Literally, it will shrink. So in the prefrontal cortex, you have receptors for cortisol. So cortisol will latch on to the prefrontal cortex and thin it, make it thinner. What's the prefrontal cortex, the CEO of the brain? What is it important for? Is it important for goals and planning the future and all that, looking into the future?
- 2.137** Yes.
- 2.138** Right. When we think about going to the gym, that is a future investment. Do you like going to the gym? Let's say there seem to be two types of people: person A, who loves going to the gym and being in the gym. They actually like working out. They feel like, "Oh, my God"—this is probably endorphins as well, and not only dopamine; they just like going to the gym. Other people completely dread going to the gym. They just hate going to the gym. They hate working out, too, because it just doesn't give them a reward. But they do it because they like to look good. They want to have muscles and be in shape and all that. There tends to be person A and B.
- 2.139** Where would you put yourself? Who's A? They love the gym. Actually, they love going to the gym. They actually feel that. It's a neurochemical thing. It's like chemistry. They like working out. Who's B? Maybe the rest, right? I assume the rest is probably B. They don't like that feeling. Okay?
- 2.140** Actually, knowing where you stand on this, whether you're A or B is important, right? Then you can decide whether— it's easier for you to make the decision if you're A. But most people, maybe they're B, right? I don't know for sure, but let's say most people are B. The point of all this is the following. It's an investment for the future.
- 2.141** If the prefrontal cortex, involved in future planning, is shrunk, it's thinning because of the high level of stress and low level of sleep, what happens? Is it harder to make that decision? Yes. It's harder to make that decision.
- 2.142** What happens with chronic stress as well, it will create desensitization of reward pathways within the nucleus accumbens and striatal regions, important for a sense of anticipation of reward. So you can put that down. Reward sensitivity goes down. You're less sensitive to reward. And so for that reason, it doesn't feel rewarding. That whole calculation in your brain, like, should I go and do that working out? You go, "Probably not. I won't go for it because it's not worth it right now." And then the vicious cycle happens because then you [don't] go to the gym, of course, going to the gym and

working out, doing exercise reduces cortisol, right? And so you stay in that high-cortisol loop. You don't improve your sleep and so forth.

**2.143** So the background noise in that way can affect a lot of your decisions. I think that would be my concluding remark on that, the idea of how your brain is always making predictions, predicting the world, predicting how costly the world will be, and depending on your environment, and depending on your background noise, let's say, the last few weeks of experiences, that will taint your decisions. And in somebody with chronic stress, it will be heavily tainted towards the not doing. Now, if you have depression, it's even more towards not doing. That would be chronic stress, but even more chronic stress. So I think that would be my thoughts on that.

**2.144** Next lecture, we go into something more fun. How's that? Creativity and flow and things like that. No more stress and burnout. You don't want to hear that. There's actually a whole lecture on burnout. I'll get to that later. All right.

## Lecture 3: Fear & Flow

**At a glance** — Fear and flow, and the role of meaning.

- 3.1 Your question made me think, Giovanni, the whole idea of meaning and purpose and I think really positive affect and having positive emotion. And I was trying to weave in some of those thoughts with my previous idea of being in the arena and being overwhelmed and having a threat all over the place. And how can you deal with that? And so I have some thoughts about visualization and also how changing attention onto something positive can be very helpful in any given threat scenario.
- 3.2 So you have a scenario. Let's say I've talked about this in previous lectures, but there's a condition called sleep paralysis. You may have heard of it. You're lying in your bed, paralyzed, you can't move, you can't speak. It affects about 20% of people. And then during that condition, they may see a ghost. Ghost comes to them, threatening, and they're paralyzed. So given the paralysis, this is highly overwhelming. Now, use it as a metaphor for our daily life, right? This could be analogous to somebody in a threatening board meeting or a soccer match or anything that's overwhelming. It's similar. You feel paralyzed and something is overwhelming you.
- 3.3 Now, my treatment for sleep paralysis, the way that I tell the sleeping person to get out of that state is by using four steps. And I actually developed these four steps based on my own experience with sleep paralysis.
- 3.4 So I was a teenager in Copenhagen, and one day I was lying there sleeping, and then suddenly I felt there was a ghost in my room. And I realized I'm paralyzed. I can't move. I can't speak. What do I do? And in those moments, the ghost came closer to me and tried to strangle me and kill me. Quite overwhelming. And this kept happening many times, not only that time, but several times. And I would have all kinds of ghost hallucinations. I would see my ghostly friends. I actually had a friend who passed away many years before that, and I saw him in my hallucination. It was quite terrifying.
- 3.5 So I thought to myself, how can I develop a therapy that can be useful for this condition? And this very therapy for this condition, as we'll see, is important for a lot of these overwhelming states we can be in. Four steps.
- 3.6 First, when terror strikes, whether it be sleep paralysis or anything else, you do an attentional—first, you do cognitive reappraisal. Let's say that. Cognitive reappraisal. This is something you can all do in any threatening scenarios. That's reinterpreting the experience and saying, "Look, given that this scenario is temporary, this is not something that will kill me. Okay, maybe perhaps I'll lose this soccer match, or perhaps I'll lose this girl, or I'll lose this, or I'll lose that; I won't die from it. And perhaps I won't even lose her for just having a bad day." So this is cognitive reappraisal. This is reappraisal where you're cognitively, mentally changing things. And by doing that, we know there's so many bidirectional connections between the prefrontal cortex and the

emotional core of the brain. So this is a cognitive correction of whatever situation I'm in. So this is the first step.

- 3.7 Then there's a step you can use in sleep paralysis or any threatening situation. It's called emotional distancing. In that board meeting, in that soccer match, during that interview, you emotionally distance yourself from the event as much as possible, meaning you're trying to look at the event from the outside. You realize you have emotions and you have fear and terror, but you don't identify too much with those emotions.
- 3.8 So you take a step back here. You acknowledge the feelings, but you don't become overwhelmed by them. It's an emotional distancing. Some people, a therapist might say, not a therapist, but a therapist might say, "Oh, imagine your emotions as a cloud that will just pass. They came by and they will just float away."
- 3.9 It's very practical, but in certain situations, it might be good to realize, at least in a very high-stakes situation, very hectic, you might realize, "Okay, I know noradrenaline is pumping through my system. Oh, I know I have high cortisol. Oh, I know I have maybe elevated activated dopamine or I have terror right now, amygdala is firing away."
- 3.10 This is what you might call—I always tell my friends when they ask me about neuroscience, I say it's very enriching to know about the brain because you know what's happening under the hood, under a stressful situation. And it's soothing, isn't it? What do you think, Alejandra? You know that this is happening under the hood. This is happening inside my brain right now. Why do I feel agitated right now? Why do I feel so bloody anxious right now? There's noradrenaline.
- 3.11 And we know it's something interesting about anxiety, for example. It could be situational anxiety, but we know that interpretation is so important for changing anxiety. What is anxiety and excitement? They look very similar, don't they? Almost indistinguishable. You have somebody who's anxious and you have somebody who's excited. How do you know what is anxiety, what is excitement? If they look the same, they're both shaking, both sweating. How do you know?
- 3.12 Inside the brain, there's only one key difference. In the amygdala, in the bed nucleus of the stria terminalis—it's a fancy name for the fear part of the brain and the anxiety part of the brain that anticipates future threat is highly active. There's a little tuning in the brain. Otherwise, it's the same.
- 3.13 But the person doesn't really know himself, right? How do you change that interpretation inside his brain so he goes, "Oh, my God, I'm actually not anxious. I'm excited." You see what I'm saying? It's just so close. It's just tuning down one thing, one button you press, and then you become more, "Oh, this is excitement." This is cognitively by telling yourself, this is just excitement.

- 3.14** Fool your brain, for God's sake. You can fool it. Say, "This is excitement. I'm not really anxious. I'm going to see that girl. Well, it makes sense that I'm excited. This is not anxiety. I don't really fear the future. I'm just excited for the event." That really calms things down. So that's a way you can trick the system.
- 3.15** And this is the key step of the whole therapy or this treatment approach for threatening a looming danger. It's called intense emotional focusing. And the way it works is the following. Inside our brains, we talked about the visual system, and we talked about the police officer in the brain. What's the police officer? OFC, right. You have good memory, Alejandra. Very good. You have good memory. Alejandra, what is it doing? What is it doing, the OFC?
- 3.16** It's checking to see what's important.
- 3.17** Threat detection, assigning value. Very good. I'm very impressed. Great job. You guys are smart. You see, I have a detection. I've developed the detection for smart people. You see? This is my work, what I do. I have to detect it so I can, for students, hire the best one, recruit the best one for projects. You're very smart. Good.
- 3.18** Now, here, you do the following. The OFC has connections to the amygdala. The amygdala is the?
- 3.19** The fear center.
- 3.20** The fear center of the brain, right? Good. Connections. And the OFC also has connections to the? The back of the brain. The back of the brain is what?
- 3.21** Vision.
- 3.22** Vision. Now you're being too fast, okay? Very fast. Okay. Bragging, you see, we don't want that.
- 3.23** Okay, the visual cortex, that's true. OFC, amygdala, there's this connection, there's a loop. That's why when you have fear, you have threat detection in the environment that can send signals to the OFC saying, "Oh, this is terrifying. This is dangerous." And then this will spill over into the amygdala. This will spill over into the visual cortex eventually and bias your visual perception.
- 3.24** In sleep paralysis, for example, let's give that example, you're lying there paralyzed, you can't move and speak, you're in this state between dreaming and reality. By having amygdala activation and fear centers firing on all cylinders, what happens then? Your visual percept of the world will change depending on your emotional state. If you have fear, you have anxiety, you're feeling all these emotions, Juan, what happens is that what you see out there, these hallucinations, dreams spilling into reality becomes monsters, ghosts, demons, evil things, based on your OFC biasing the visual cortex.
- 3.25** And in my previous course, I talked about how in depression, for example, you may have a depressed person, and he's looking at somebody sitting over there in a chair,

and I'm looking, and he may just be leaning and resting, but he's seeing somebody who's crying. And then he looks again, oh, he wasn't crying. His visual, because his amygdala is always tainting the OFC, he's always hallucinating, almost, these scenarios of fear and dread.

- 3.26 We might get an SMS, and the SMS says, what? Oh, you are terrific. And then he looks at it, he says, "You are terrible. Oh, I knew." He's getting a little bit sad. And then he looks again, "You're terrific. Oh, I didn't know." This actually happens, these [misperceptions], because their OFC is always sending these signals to the visual cortex because they are biased by the amygdala.
- 3.27 Now, how can we all use this information in our daily lives? When you are in a threatening scenario, you are on that date, you are in that soccer match, any place you are, you're in, right? Focus your attention intensely on something positive. Intensely on something positive, right? And this could—I mean, in a soccer match, I don't know if the soccer players are doing this, but it could help. It could definitely help in any situation where you're highly frightened.
- 3.28 Now, what would be a good example of that? Would you do this in an overwhelming board meeting? You could do it before an interview, right? In the interview, before you do the interview, during the interview, you start paying attention, paying attention on something positive, your mom's face. But you can do it before, right? You do it before.
- 3.29 And this is where things like—this is where Giovanni's point came in before about things like prayer and meaning and all that that I mentioned before. I think something like— and this is a whole new field of neuroscience, but things like spirituality and meaning and God and prayer and things like that [are] super important for mental health, well-being, and we'll get to that, of course, in one of the lectures later.
- 3.30 But also for any situation right now where I'm in terror. Have you seen the soccer players? They're always doing the crosses and praying to God and doing all. It is very soothing for the nervous system. Extremely soothing. This would be an example of how you can soothe your nervous system. Is that clear? And by doing that, you dampened the activity of the amygdala as well. So I wanted to add that as another point. So those are my steps for the therapy and how that could be applied.
- 3.31 And then I wanted to also add a point on visualization because I think we did cover that, but I want you to know the importance of visualization, what it actually can do. So visualization is that before the actual event, I imagine scenarios that could unfold. I predict the environment, and I even predict the unpredictable. I make it so bloody unpredictable. I take into account all kinds of scenarios that could happen. The broader, the better, right? The more outcomes I imagine and the worse, the better.
- 3.32 Like Ramachandran, my mentor always used to say, whenever you have an event or you have something, imagine the worst possible outcome. Imagine the worst possible outcome because then you have that flexibility built into that system. And so if it

happens, you can deal with it. So during that interview, you imagine perhaps that it could happen that the interviewer might get a rage and start yelling at you, let's say. Highly implausible, but you say, "Okay, what if that actually happened? How would I deal with it?" So that could help.

- 3.33** But on visualization, I wanted to add that it's extremely powerful for a very, very key reason, and that is the brain. In our brain, our circuits, for example, whenever I perform an action, I have neuroplasticity. Let's say I'm kicking a soccer ball, whatever I'm doing, I have plasticity, right? But whenever I close my eyes and imagine myself kicking that soccer ball or giving that lecture, whatever I'm doing, I'm also using my brain and the same circuitry. Just a subset of the neurons, and they're not firing as vigorously, but there is activation there when I'm merely imagining it. Does that make sense?
- 3.34** When I'm dreaming, what happens when I'm dreaming? Do the neurons in my motor area become activated when I'm dreaming? For example, I see myself giving a speech. Are my hands moving in my dream, for example? Yes, the motor neurons are represented; the same neurons are firing as if I was awake. And this is the key.
- 3.35** They've done studies showing that people—and I've talked about this before, so I'm just going to summarize—but people that play the piano, they're trained for about five days, play the piano for five days. Another group imagined the same playing for about a couple of hours a day or so. By the end of that week, those who imagined the event was nearly as good as the people that for a whole week actually trained the piano for a whole bloody five days or so. Just as good. It only took one hour of extra lesson for the one who imagined the whole thing, the fantasy group, to become just as good as the first group.
- 3.36** I've talked about before how people that are trained—so one group was trained to use their finger muscles by lifting some weights at the gym. They lifted, lifted, lifted, by 30% improvement in muscle strength. One group just imagined the whole thing for the whole month. Twenty-two percent improvement. Just from imagining the whole thing.
- 3.37** So what I'm saying is, visualization is not only predicting the environment, but you're actually literally reinforcing the circuitry. You are changing your brain. You're plastically changing your brain by imagining yourself in that soccer field over and over, by matching yourself and seeing yourself in the board meeting over and over, on that exam, seeing yourself in that room and imagining things.
- 3.38** I think as a society, we're not really using this enough. What do you think, Kelly? Do they teach this in school? Not really. Not much, right? Like the fact that we should visualize and how the neurocircuitry. I never heard it for sure. I think it's helpful.
- 3.39** Have you heard about the concept of flow before? What is flow? It's a peak state where you lose yourself, a sense of self. You lose your sense of—the fact that you're doing a goal-directed behavior as playing tennis or playing soccer or what examples could be?

You're playing soccer, you're playing tennis, you are writing, painting, playing video games, playing the guitar. Absolutely.

- 3.40** The first time I heard about flow, I hadn't heard about this. I was an undergraduate student in Cairo. I had moved to Egypt. In Egypt, I happened to run into this professor. He was a very nice guy. His name was Samy. Very cool guy. He was American originally. Or originally, he was from the Middle East, like his roots. That's why his name was Samy. He was in Egypt. But he was from America. And I was from Denmark. So this was American. For me, he was from a foreign country, right? So I noticed, right? He was an American professor.
- 3.41** And this professor, he told me about various experiments he was doing with another professor in University of Chicago. He was from the University of Chicago. This professor, he did his PhD there. He had a mentor called Csikszentmihalyi. They were doing experiments where they were having all these young guys, students, play these video games and looking how immersed they were in the video game.
- 3.42** He told me about something called the experience sampling method where people would wear—in some of these experiments that Csikszentmihalyi had done, people would wear these pagers on them the whole day, and then they would measure their states, their emotions, how they felt, and so I found it intriguing.
- 3.43** And then he asked me, he said, "Do you want to do an experiment?" I said, "Yeah, that sounds fun. I haven't really done experiments before, but that'd be cool, man. I love it." We did an experiment together, and the experiment was the following. You had students play video games. They play video games. As they were playing these video games, they thought they were playing against an actual, like another person. But they weren't. It was just a machine. It was all precalculated by a machine that was running the whole thing. In one version of the video game, they won with a small margin. Okay? In another version, they won with a big margin. The question is, which led to more flow and enjoyment? A or B? Winning by 4–3 or winning 10–0?
- 3.44** And before you answer, I want you to think about how winning by 10–0 is a good thing. Don't you feel more competent? Don't you feel more like, "Oh, my God, I have so many skills." You feel very good about yourself?
- 3.45** Maybe the first time.
- 3.46** First time. Then it gets boring. So you like challenge more, perhaps. So you think A, right? That's correct. That's what we found. Okay. So version A, where you win by a small margin, leads to much more enjoyment than the second one. The version A.
- 3.47** I was lucky because this experiment was done with Csikszentmihalyi. Mihaly Csikszentmihalyi was also involved in the project. And as a young student there in Cairo, having no research experience and having a past of really poor grades in school,

it was great to work with somebody like Csikszentmihalyi and do a study like this, right? So it was great. I was happy about it.

- 3.48** The flow state, losing your sense of self. So what happens during the flow state? Let's look inside the brain and think about the various centers that quiet down, the various centers that become activated during flow.
- 3.49** You lose your sense of self. Things are running smoothly. No friction. You just lose your sense of time and space, and it feels very good. The prefrontal cortex: does it shut down or go up? Shut down. There's a shutting down of the prefrontal cortex, at least partially. Default mode network involved in self-referential thinking, self-monitoring, self-criticism. Does it go up or does it go down? Go down. Very good.
- 3.50** What about the amygdala? Does it shut down or does it go up? Fear part of the brain. Shut down. It shuts down.
- 3.51** What about the sensory motor circuits, the basal ganglia and the cerebellum? This is a mouthful. But what about these regions? Sensory motor regions—that has to do with sensory experiences of the world. I'm painting; I'm watching this sensory scenario. My motor regions, let's say, if I'm using my hands to paint, the basal ganglia, automatic movement, and reward circuitry in that loop, and the cerebellum involved in balance, but also timing of reward. Do they go up or go down? They go up. Sotemporarily, these automatic regions within the brain shut down.
- 3.52** Of course, in the ventral striatum, in the nucleus accumbens—a fancy name for the reward parts of the brain, dopamine parts of the brain—those circuits go up. But also that makes sense, right? When you think about it, the flow state is a state of no self-doubt. The default mode network, out of the picture. There's no self-doubt.
- 3.53** It's very rhythmic. It's very rhythmic. Have you seen Batman when he fights? Is it The Dark Knight or is it—the version where he has a very interesting karate style? I think it's The Dark Knight and Batman Begins. Forget it if you haven't seen it, all right? But it's very rhythmic. There's something rhythmic about the whole flow state. It's almost like a mind-body connection. There's something going on that's not merely mind, but it's more mind-sensory-motor dopamine. It's a flow. It's a dance. It's Messi when he's dribbling past five players and scoring, or Maradona in '86 when he dribbles past the whole bloody English team and scores. It's like that, right? That sensory motor, like automatic movements of the basal ganglia.
- 3.54** So what's the basal ganglia? We have in the upper regions of the—you have reward circuitry in the nucleus accumbens, right? And whenever you do an action, initially, it's not automatic. But when the action becomes automatic, it goes to the basal ganglia. Is that right? Basal ganglia is involved in automatic behaviors, things you don't really have to think about, things that are just floating.

- 3.55** Somebody who's an extremely competent basketball player, an extremely competent CEO in a board meeting, or an extremely competent tennis player, surgeon. What is happening? There's this unison, this harmony between the prefrontal cortex shutting down, the default mode network shutting down. There's no self-criticism. He's not steering anything. He's being steered. It's this dance. It's this floating. It's this being where there's this rhythmic dance to the whole thing, and that's the whole sensory-motor, dopamine reward centers going on. You lose your sense of self at that time, which makes sense, right? It's a beautiful state to be in.
- 3.56** What else happens in the brain of the flow person? Well, specifically in the prefrontal cortex, you have the—first of all, you have the dorsolateral prefrontal cortex, the sides as well, involved in inhibition, logical serial thinking, shutting down. This is based on a very interesting study that I'll get to now. It's quite fascinating. Then you have the medial prefrontal cortex as well, shutting down, regions of it.
- 3.57** Okay, so we looked at the brain and what happens during the flow state in the brain. We had an idea of the various centers that shut down, shut up, go up. What centers were they? The amygdala goes down. Default mode network goes down. Medial goes down. Lateral prefrontal cortex goes down.
- 3.58** What about the neurochemicals of flow? What neurochemicals are involved in flow? Do you have attentional focus in flow? Noradrenaline, does it go up or down? It goes up. What about dopamine? Does it go up or down? Dopamine goes up. Dopamine goes up. Okay. Dopamine goes up in flow, which makes sense.
- 3.59** And that's why often I link the flow state to dreaming. It's almost like a dreamlike state. It has some overlap with the dreaming world. That whole dopaminergic aspect of the dreaming world is very much similar to the flow state. Wouldn't you say? When you are in flow and things are running smoothly, it reminds you of almost like a dream, being inside a dream. It's very interesting.
- 3.60** Okay, so the noradrenaline goes up, but only to the extent that it becomes more precise, you see? So it's not like it goes too far up. What happens if it goes too far up? You overwhelm the system; you lose control. But it's more like the signal-to-noise ratio is just more sharpened.
- 3.61** Okay, we talked about the endorphins. Endorphins are involved in what system? It's the consummatory pleasure, right? Is when you're having that meal, when you're feeling good about that meal. In the flow state, do they go up or go down?
- 3.62** They go up.
- 3.63** They go up. Okay. Endocannabinoid system as well, feel-good, endorphin-like system as well, feel-good hormones also go up. So you have these systems that rise during this peak state.

- 3.64** It's interesting with flow. In my own life, before going to Cairo, I had this change in my life. So when I was a teenager in Copenhagen, I did very poorly in school, very badly in school, Juan. I got really bad grades. I was always late to class. In fact, Monday was the—I told my mom, "We don't have any classes." It was my day off. It was an extra weekend every Monday. Pretty much. I never went on Monday. It's not a good thing, but this is what I did. Twenty-five percent absent rate, no homework. Never had any enjoyment of reading books. School—never gave a presentation in my entire life in school. Nothing. I was 18, 19 at this stage. And mind you, it wasn't like I was so young. I was like, really poor in school. The sense of being immersed in books, nothing.
- 3.65** Around that age, I had a change in my way of thinking. I had like an epiphany. I had an opening, a vista, a change in how I see the world. And there was a lot of things happening in my neighborhood. People were shot, people were killed. I mean, this was everyday life for me. Every day life, I'd go, maybe my neighbor would kill, or my best friend was stabbed 13 times, or violent things would happen. Eric, these things change you. They change you. You want to change yourself at some point. And I had this feeling around that age: if I don't change right now, this might happen to me. I might experience one of those things.
- 3.66** I went to the library for the first time, perhaps in my life, and I looked up a book and I started to read books. What's happening in the brain, let's say, during sleep paralysis? I had a few sleep paralysis episodes. Or what happens in the brain when you dream? Or is there a way to—how can you understand love better? I would start reading books.
- 3.67** And at this point, a change happened in me. There's two modes of approaching the world. One is you're intrinsically driven, so intrinsic reward versus extrinsic reward. My entire life was pretty much extrinsic reward. It was just like, you go to class so you get the reward of getting the grade. I didn't even get the grade, so it was more running from punishment, really. But I had no internal reward system at all for any knowledge.
- 3.68** But that changed with me reading, and I got immersed in these books. And it was so fascinating. Initially, obviously, I couldn't read it. It was very difficult to read. It was English books, many of these books. But I tried, I made the effort, and each time I actually learned something, the brain does this, this, this.
- 3.69** Oh, and then I get a spike of dopamine. When I have a spike of dopamine, Alejandro, you know what happens? The brain says, "Oh, this was important. Wire that into the circuitry now for future." You have attention, of course, I talked about before, and you have reward. So you learn a new thing about the brain. Oh, another spike of dopamine. Aha. We call it the aha moments. I had a few of these aha moments more and more, and I would learn and I would immerse myself. Completely flow state, completely immerse myself in these books.
- 3.70** And by the time I knew it, in six months, my room went from just having the Tupac picture and all this stuff, to suddenly—it was like, have you seen the Library of

Alexandria? It's all books, piles of books. My family thought, what's happened to this guy? He's gone crazy. He's insane now. My hair was long. It became long. It was an interesting change. And suddenly, I was sitting in class and answering all these questions, and they thought, "This guy, what happened to him?" In six months. When I passed the exam now, I barely made the cut. So if I hadn't had this period of six months, I would have completely gotten out of high school. They would have thrown me out. But I was able to pass through high school.

- 3.71** So I went to Cairo after that. And in Cairo, I kept the same routine, but even more severe, more obsessed, more excessive, more extreme, reading books, complete change. But this was how I went from extrinsic motivation to intrinsic motivation. I went from being driven by customs and norms to being driven by internal drive, motivation that was driven by a flow state. And lo and behold, I hadn't imagined that I would end up meeting, working with the godfather of flow over there. I hadn't pictured that. This is how life can be serendipitous and things can change, right?
- 3.72** And this is what I'm saying. Sometimes when you have a good intention, you work hard, and you just do the right thing, things will happen. I later had another mentor in California that also changed my career. But I'm saying, for me, that intrinsic motivation was the factor. If anybody asked me, how did I end up here, where I am now, right now in my life? I think it was intrinsic motivation. It was this joy of learning, of being completely immersed.
- 3.73** We know just play, being playful, tons of neuroplasticity. Like being rigid, a personality, Giovanni and all that, very low plasticity, in fact. Play, fun. Ramachandran, my mentor in California, used to say, and he used to have this habit of, he said, "Let's do work and let's do play." So we would work, and our work would almost also be play. We'd talk about science, about ideas, the brain, about all these fascinating things. And then after half an hour, "Oh, let's go for a cookie. Have some milk. Let's go to the beach for a walk. Talk about science. Let's go to the cafeteria. Oh, let's go hang out with these students." It was fascinating. But this change from this oscillation from work and play, work and play. I think this is very important.
- 3.74** So what are these endocannabinoids I mentioned before? They're very important. They're very interesting. They contribute to this flow state. They help you regulate your emotions, keep stress down, and just be less rigid. And they dampen, of course, the prefrontal cortex, keep things down, and they dampen the amygdala. How can we have more of these endocannabinoids in the system? What actual practical steps can we take to have more of these, to eventually have more flow-like state or being able to reach flow more easily?
- 3.75** Let's think about it. First thing, aerobic exercise, 20 to 40 minutes, in fact, running, cycling, brisk walking. I think, as you would notice with all of these, they would have an element of something to do with reward and rhythm. There's something about reward and rhythm that's crucial for these endocannabinoids.

- 3.76** Learning new skills, novelty, problem solving, creative works, painting, taking up tennis for the first time or some other sports you haven't played before. Slightly challenging but not overwhelming. That's the key. They come with social connection, close conversation, bonding. This is where the oxytocin and vasopressin interact with the endocannabinoid system. You feel calmer, safer, you feel better.
- 3.77** I would say actually as a side note here, before—and this could actually work, too—before any important event, let's say you're going on that date with that girl or you were having that important meeting or whatever you're doing, if you can tap into the oxytocin in the cannabinoid system, oxytocin and vasopressin system, for example, that would be a great thing, too. So what actually I would do sometimes if I have to go see this girl that I like, and I feel nervous and anxious about it, hang out with your mom, hug your mom, kiss your niece, play with your nephew.
- 3.78** Touching like that—I'm not going to touch you. I don't want to get sued. But if you touch somebody, somebody you care about, you have a reduction of the amygdala and vasopressin systems come online. That's why I say if you want to go and see somebody or you have an important meeting and you want to connect with that future boss or future colleague, don't be all stressed and agitated. Be calm. First of all, reduce stress so you have less of prefrontal activation. Prefrontal activation, you want to have such that it's online, but you want to have the vasopressin-oxytocin systems flooding your system, keeping you calm, already priming you for a warm social interaction. That's my point. Is that clear? So do that for God's sake. I've done it. It works.
- 3.79** Social connection, touch, music, and rhythm. Rhythmic music. That's very helpful. Anything that's rhythmic, chanting, prayer, anything that feels soothing. But again, notice that reward rhythm for endocannabinoids. Reduces internal noise, internal criticism, self-criticism. And here's a potential one. It could work. I'm not sure if it does, but heat and cold exposure. Have you tried this before? Do sauna, go to the ice bath, right? An ice bath, sauna. Sauna, ice bath. Potentially, it could work.
- 3.80** Mindfulness and breathwork, lying there, slowly breathing in and out and doing meditation. Omega-3 fatty acids, dark chocolate. I have to mention that, but it seems to be linked to the cannabinoid system. So I said that those would be the main things I would do.
- 3.81** On the whole music business, what is that? Can music be helpful? What happens during music? I sometimes think about it. There's actually this movie. It's not even a movie. It's a book. It's a book that I read many, many years ago. I'm not exaggerating. I'm 80 years old. It was about five years ago. I was reading this book. It's Norman Doidge's book. It's called *The [Brain's Way of Healing]*. It's a very interesting book. I recommend it, in fact.
- 3.82** But there's this chapter about these monks in this monastery. Initially, they're very happy. They're doing their job. They're going around in the monastery. I don't know what they do, but they have chores they have to do every day. But at some point, they

develop some depression. They feel very sad. They're very low-mood. They think actually it may be physical, so they bring in a doctor, a physician. He looks at them. He looks at them; he looks at their vitals. They're okay. They can't really figure out what it is. The next doctor tries again. Can't figure out what it is. Fourth doctor, they keep searching. Is this depression? Is this physical? Maybe it's a virus that's spreading. But it's, for some reason, only affecting these monks. What is it about? What's going on?

**3.83** Eventually, a very smart [physician] noticed something. Actually, he's the one who pinpoints that, in fact, this may be depression. It seems like they're all depressed. They have a collective depression. What might be going on? And he was trying to figure out what's happening. He was smart. He was like Sherlock Holmes. What has changed in their routine right now? What has happened to them collectively? Have you given them some food all of them are eating now that's making them depressed? Maybe they have less visitation. I don't know. What could it be?

**3.84** It turned out that they had this singing they had every morning where they were singing, like humming. I don't know what they were exactly doing, but some singing ritual that they were doing every single day for a couple of hours. They had been told not to do this for some reason. Imagine if you're a monk; you don't have much dopamine in your life. It can be boring, I guess. And so this whole circuitry and the whole rhythmic singing that they were experiencing every single day and engaging in every single day was affecting how they felt. And eventually, because of the whole ritualized dopamine that they were getting every single day from this singing—and then we talked about how rhythmic and rhythm and reward leads to endocannabinoids—[not singing] actually led to things like depression in this group. They had depression.

**3.85** And how would you do the experiment to figure out whether this was actually the case? You let them sing again, and suddenly they felt happy, and their positivity came back from singing. There's something there. There's something there that's very interesting.

**3.86** In fact, some people with depression, they will tell you, "I can no longer feel pleasure from music, from singing. I don't feel any pleasure." Then, Rachel, you might ask, does it matter which music you listen to? Let's say it's music. Maybe. When you're listening to music, have you noticed—

**3.87** On the weekend in the hotel I was in here, I was trying to go over my notes, you see, and review. They were playing techno outside my room, very loud. I couldn't figure out, is there something wrong with my brain that I can't inhibit that information? I was trying to analyze the situation, or is it the music is too loud? You know the situation? Is it just me, or is it the music? But I did open the door; it was like, boom, boom, boom. It was just blasting, right? They had a pool party, basically.

**3.88** What happens during techno music, and any music, in fact, you have entrainment, meaning your neural circuits will try to mimic that music. So if it's techno, if it's fast,

and you want to meditate, what happens? Is it hard? What about if you're agitated and everything is aroused, you feel aroused, and then you're listening to something calm, like a lullaby, something. How does it feel? Soothing, entrainment. So the neurons, again, they try to mimic and mirror the music itself. So this is very important. And I think that's why for stressful situations, when you feel stressed, it does make sense to listen to something soothing.

- 3.89** That's why, especially if the lyrics—and this is what I would add at the prefrontal layer—if the music itself you're listening to has a positive message, even better. If it has a soothing message, even better. So I'd say that'd be even better.
- 3.90** I think that's why the lullaby is so universally used. The mom singing to the child, soothing. Tons of neural entrainment, synchronization of brain waves all over the place. So the eyes will move like the eyes of the mom. The neurons will fire just like the mom's neurons. So much soothing, so much oxytocin and bonding going on there. It's quite something.
- 3.91** In what kinds of scenarios is the flow state useful? We talked about painting. If you're an athlete on a soccer field, let's say you're a basketball player. You're a basketball player. What parts of the brain are involved in the flow state? For that moment where you're dribbling past all the players and you're throwing the ball, what regions of the brain are involved? Basal ganglia, what else?
- 3.92** Coordination of movement is where? And time, coordination, the part of the brain that goes down and goes wacky in ADHD. Remember we talked about this before in previous courses? Rhythmic dopamine coordination of time and motor movement and reward, the cerebellum, highly active. Visual cortex as well. Default mode network goes down.
- 3.93** Combat sports. Again, you have literal reduction in the amygdala and the fear parts of the brain. It's just something that is highly useful in situations like that. But what makes that state then so—it feels good, too. I mean, it's not only shutting down on various regions and amplifying others. What is it about the state that feels good? And why does it feel good? What makes it feel good? Dopamine and endorphins, that's part of it, absolutely.
- 3.94** But more than that, going back to the whole entrainment and all that, it has to do with the fact that your brain and your systems, as we talked about before, you have the emotional part of the brain talking a different language than the prefrontal part of the brain. These are antagonistic. The brain is in the business of always wrestling it out. You have this part of the brain fighting with this part of the brain. The septum, the pleasure part of the brain, fights with the fear part of the brain, the amygdala. There are always these combating forces inside the brain.
- 3.95** During flow, what you have is really you have a sense of alignment. Is that alignment where the prefrontal cortex, when it shuts down, the sensory motor balance system

works—neurally, if you look at it in terms of the neural activation, it looks very frictionless. So the behavior looks frictionless on the outside. When you look at it, it looks beautiful, poetic, magisterial. But inside the brain, there's also this neural harmony with the neurons firing without friction, with no sense of agitation and disparities. It's just floating. I think that's a very huge part of it. That's why it feels meaningful. It feels like something that's beautiful to look at, but it feels good from the inside.

**3.96** So how can we enter this flow state quicker, better? How can we harness it and get into it? One thing would be the action you're performing should be clear. This is the first part. Meaning the action is clear. Now, I'm sitting down, Baland, in my neighborhood. I'm not just vaguely googling stuff. I'm actually taking up a book. I'm actually reading something. It's a very clear goal in front of me. Or I'm playing soccer, or I'm playing tennis, or I'm a surgeon doing surgery. It's very clear. The goal in front of you is clear. It's not vague. It's the first point.

**3.97** And then you want to have the second pillar would be immediate feedback. This could be a video game, right? You're playing the video game and you're scoring that goal on the video game. You're playing soccer on the video game, you're scoring that goal. Or in case of a soccer player, he's scoring that goal, or he's dribbling that player. Or if it's Baland reading a book, "Oh, I'm learning something new. How great." I'm getting immediate feedback from what I'm doing. And that will titillate the dopaminergic center. So having that immediate feedback is great. It's good.

**3.98** And somebody mentioned raising their children. This could also make sense for when you want your kid to get into flow, right? You make sure that you give them a clear goal: read this book, play soccer. And you give them—maybe if the feedback is not clear from the actual event, you give them the feedback, the positive feedback, the reward in that sense.

**3.99** And then I would have the third pillar, and this one you already know. You can guess this. The challenge should be too low, too high, or middle? Middle. If Baland went to the library and got books in Latin, would that be good? No, I wouldn't get anything out of it. If it was children's cartoons, not good either. Middle book, right? Middle book. Maybe I start with a slightly easy book on the brain. It could be even a magazine, and then I move up. That would engage my dopaminergic centers.

**3.100** Also, I think generally having an environment that cues your brain for the flow state is good. This is good. I know when I go to the library and I enter that bloody library door and I go inside and I see all these books in front of me, I know it's focus time. I know it's flow time. I won't get distracted. I won't mess things up for myself by going into an environment that's going to kill my distraction and my attention to start with. And this could also be how you design your room. Design your room in a way that you are putting the desk in the right place and the desk is not too messy, and you have a clear way of making everything, cueing your brain for optimal flow.

- 3.101** And then again, as I mentioned before, you have a gradual cognitive ramp-up. So you slowly add more and more challenge to the experience, making it more challenging. This is where other CEOs have various techniques. Other CEOs—I'm not a CEO, but certain CEOs have things that they would do. They will talk about, for example, uninterrupted blocks of time. Does that make sense? Every morning, let's say, you have time for deep work, or time for tennis, or whatever it is. But you have that block where you remove notifications on your phone, or you remove your phone completely, perhaps, and you design the environment for an uninterrupted block of deep work, let's say, 90 minutes or so. That's a very important part of it.
- 3.102** Creativity and flow. The flow state is this state that makes you immersed. It makes you forget the sense of self. You're doing this activity. You have all this neuroplasticity as a result. But it's not really creativity, is it? In the broadest sense of it, where is it? Are you becoming more creative from flow? Not really, right? Not necessarily. Thinking outside the box, that's part of it.
- 3.103** Now I'm trying to think with you guys. I don't have a clear answer. Is it connecting things in the world that are usually unconnected, seemingly unrelated things? Is that what creativity is about?
- 3.104** Yes. It's like establishing connections between words or maybe like that dream state that you were referring to. Like connecting things that might be like experimenting.
- 3.105** The answer is we don't really know what creativity is. We don't really know exactly what creativity is, but it's somehow flexibly looking at the world in a novel way and being able to link seemingly unrelated things in the world. But the way that I'm linking it here with the flow state, and we'll get into creativity later, the way that I want to link it to the flow state is that there's a similarity there of letting go. You see? Get your damn self out of the picture scenario. Don't be so stuck in the prefrontal lobe and executive thinking and dorsolateral inhibition of your thoughts, your impulses, and all that. You want to let that system become more free. And that's what creativity is about. So there's some overlap between creativity and flow. What else is there of an overlap?
- 3.106** During the creative state, it seems like various different parts of the brain are activated. You have many parts of the brain that normally will not work together. You see this in brain imaging studies, brain imaging. You have many different parts of the brain being activated all at once, and that leads to the new aha, the out-of-the-box thinking. So you have that. That's also similar to creativity. That's always similar to flow.
- 3.107** When you get out of your own way, when you go for a walk, some of the best ideas—Einstein used to talk about this. He used to go for a walk, and then ideas would come to him. There's something about that. It's not, in this case, flow, but it's just letting go of the prefrontal cortex, letting go of the self-criticism, letting go of rigidity. That's highly critical. Steve Jobs was like this. Ramachandran, my mentor, always talked about this,

this idea of cross-fertilization of ideas, letting your mind wander so much so that you make connections in novel ways, in ways that are not usually made, connections that are not usually made.

- 3.108** You described this epiphany that you had when you were 18. From the previous course, around 18 is when people become schizophrenic.
- 3.109** Yes.
- 3.110** What's the difference in the brain of someone like you who becomes focused on more learning versus someone who becomes schizophrenic?
- 3.111** That's a great question. I think that for somebody like me, when I was not going to class, always very close to getting thrown out of school and all that, my mom, my dad, too—my dad was working a lot, trying to provide for us. But my mom was very good. I mean, since I was a kid, she would always go to these meetings where you meet the teacher. You know these meetings? What do you call them here in America?
- 3.112** Parent-teacher conferences.
- 3.113** Yes. My mom always went and came back crying until 18, 19. This broke my heart, too. I got sad. But what can I do? And always two weeks before those meetings, I'd try to suck up to the teachers. Never worked. Never worked. But I had this loving mom that was just very, very warm and always soothing and understanding. Obviously, sometimes she gets angry. She's Kurdish, so she could be a bit hotheaded, too. But that's part of it. I felt scared from her, too, so I wouldn't go out in too much trouble. It's a good thing.
- 3.114** But I think that's what it is. I felt extremely loved by my parents; my mom, my dad was always there for me. I think that's a key, crucial aspect of it. Obviously, there's genetics, but I think that even had I had the genetics for, let's say, schizophrenia, I wouldn't have had a schizophrenic break.
- 3.115** Also, honestly, I want to tell you something. This is just my heart. And I hope—in life, so many things happen to us that are out of our control. For example, when I was 18, 19, one day, I was with my gangster friends in the hood, and I didn't go to high school for three days. I just went back to this friend's criminal house and slept at his house.
- 3.116** Then one evening, we were planning to go to this party. At that point, just before the party, I got a phone call. My mom and dad said, "Come home. We need you. Don't leave us. What are you doing?" Something in my heart, I felt so sorry for them, and I went home that night. All my friends went to this party: bloodbath; they were stabbing. Everything happened, you can imagine. They would kick this guy, one of my friends, kick his head. Everything you can imagine would have happened.
- 3.117** Had I gone, maybe I would have been stabbed, maybe a brain injury. Maybe that would have been enough to tilt me into mental illness. But I felt like the universe—whatever

you believe in; I believe in God, for example. I felt like something was protecting me through my parents, my loving parents, and I was saved at that moment.

**3.118** I have 10 stories like this where I could have ended up in prison. One day, we were about to do something. My friend was about to do something very horrible, horrible like a robbery. And I managed to get away from them, but again, instances where [I was] seconds away from being involved in all that. So I think that's where you have to think that you're being blessed as well. That's what I feel like. There's something more than myself. I've been fortunate. Let me put it that way.

**3.119** Home

**3.120** Quad

**3.121** Messages

**3.122** Map

**3.123** My Courses

**3.124** Curriculum

**3.125** Store

**3.126** Search

**3.127** Create Post

**3.128** Notifications

**3.129** Profile

**3.130** 10

**3.131** Seek backward

**3.132** Pause

**3.133** 10

**3.134** Seek forward

**3.135** Pause

**3.136** 0:10 / 56:12

**3.137** Mute

**3.138** Quality

**3.139** Playback rate

**3.140** Captions

**3.141** Enter fullscreen mode

## Lecture 4: The Architecture of Attention

At a glance — The architecture of attention.

- 4.1 Today, we'll be talking about various topics. One of them will be attention. What is attention? Why is it important? How does it work in the brain? What regions are involved in attention?
- 4.2 Yeah, attention is interesting, isn't it? We all have limited attention, and we can only focus on so many things in a given moment. Why is this important? How can this help us? Can we train our attention? Are there any techniques we can employ in our daily lives to make attention better? And so this lecture will get into that. Where is attention located in the brain, Nikola? Is it here or here, or is it in the medial part? Is it in the parietal part? It's a rhetorical question you don't have to answer, okay? I'll cover that.
- 4.3 And so one study that I always like to talk about—have you seen this YouTube video? These guys are playing with this basketball. They're throwing this basketball around. One team has white T-shirts; the other team has black T-shirts. And you're told you have to focus on the ball and count how many times the ball is being thrown, right? So boom, boom, boom. And you're counting there, one, two, four, seven. And as you are counting, there's this gorilla, this man in a costume, a gorilla costume. He walks in. He may even do a little dance, and then he goes out. And about 50% of people will not notice this gorilla. Have you seen this, Debbie? It's quite intriguing.
- 4.4 What does it tell you about the human brain and attention? It tells us that we have limited working memory, that you have regions of the brain; the DLPFC up here is involved in working memory and attentional control on an object. And when you're engaging that, you can't attend to multiple things in the environment. Your brain and the neurons are very much focused on that single task. So this is very important for all of us to know.
- 4.5 So you have different types of attention. You have what's called top-down attention. This is when you focus on something intentionally. This is when I'm focusing on, let's say, I'm looking for a camera or a pen, or I'm looking for—I'm looking at Rachel. I want to look at Rachel right now, for example. I'm intentionally looking at her. That's top-down. And there's a cat right here. That's bottom-up attention. So you have these two types of attention.
- 4.6 And I think it's important to keep in mind that attention is really that. It is a currency that you can train. It's something that you have a limited capacity of. It's something that you can't—I mean, it's scarce. It's scarce in the brain.
- 4.7 You have various parts of the brain important for various aspects of attention. So you have Kelly here. In this part of the brain, the parietal lobes, what is that important for,

Alexander? Is this part of the brain the multimedia studio or is it the calculating studio, or is it the long-term memory storage studio, the parietal lobes? Multimedia, right?

- 4.8 So in the attentional—when you pay attention, you have various aspects. So first of all, you have to hold an object, right? You have to hold an object and you have to focus on it. This is top-down attention. Would this be perhaps the cognitive control areas, the DLPFC? It would be. So you put DLPFC, sides of the brain. That's important. And holding an object, focusing in on the object and holding it, and even suppressing alternative sources of information coming at you.
- 4.9 So we mentioned you have a salience network in the brain. It's the ACC, the cingulate cortex, and the insula. These parts of the brain are called the salience network. So if anything is salient in the environment, if I'm walking here and there's a cat, there's something interesting over there that pulls my attention, my salience network is recruited, and the anterior cingulate, the cingulate cortex, the insula gets activated. So that's that part of attention.
- 4.10 And then the parietal lobes that I mentioned, the multimedia studio, put that here, that's involved in orienting attention. So when I'm orienting my attention, that would be the parietal lobes. When I'm holding the information, that is the lateral parts of attention. That's the DLPFC, the sides.
- 4.11 I don't want this stuff to be me mentioning brain regions and you just jotting it down. I want to really understand because there's always an underlying logic of the brain. So when we're dealing with the sides of the prefrontal cortex, we're talking, typically, about what? Calculated cognition, cold cognition, math, working memory, me remembering 7 plus 47 plus 6 minus 19 doing calculation, that's always the sides, right? That's the DLPFC. DLPFC, it's right here, right?
- 4.12 Anything to do with spatial orientation, a sense of self, of me moving my attention or my spatial awareness of the landscape, that's always the parietal lobes, mainly the right side. So that's why it makes sense in the attentional loop, that that would be the parietal lobes.
- 4.13 Then you have, when you pay attention, the salience is there. The salience landscape is there, pulling your attention in various directions. But you also need to have an incentive to keep looking at things. You need an incentive. So if I'm watching this soccer match, let's say, I'm watching—what soccer do you like? Do you like soccer? What sports do you like? You like sports, right? American football? Yeah, sure. What kind of American? Which team? Tom Brady? Well, let's say Tom Brady. Okay, you're watching Tom Brady. You're watching Tom Brady, right? So you're watching him, and you're watching his game. When you're watching his game, you have orienting attention of the parietal lobes. You have the DLPFC holding that attention.
- 4.14 In fact, the DLPFC is also important for when the salience networks come in and say, "Look, buddy, you have a notification. Look, buddy, your mom is saying you have to

cook." Then there's this thing over here. There's that. There's this. That's the salience network being pulled in all these directions.

- 4.15** The DLPFC is also important in suppressing that information, you see? That's also the DLPFC. Suppressing alternative sources of attentional pull. That's also the DLPFC: suppression. But then, of course, that has to be a motivational pull. Something has to pull you. It can't just be me looking, right, Sharon? It has to be something that's just pulling me. And that's when the medial part of the prefrontal cortex comes in. That's involved in what?
- 4.16** Now, I've mentioned this about—okay, if you were in my last course and you can't answer this, there's something either wrong with you or me. I've mentioned this maybe 10 times, but maybe I'm not a good communicator.
- 4.17** Emotional decision making.
- 4.18** Emotional decision making, my friend, that's right. Emotional decision making, okay? So when you make emotional decisions, the part of the brain that has to do with cognition blended with emotion, you're always talking about the medial prefrontal cortex. So if I have to make a decision about whether I want to marry this girl or take that new job or move to that new place, I use what part of my brain? The medial, right? It's that bridging point between emotion and cognition.
- 4.19** In the attentional circuit, when I'm then paying attention to that Tom Brady football match, I'm using the medial prefrontal cortex a lot because I have focused attention, but there's an emotional overlay of motivation. And the circuitry going to the dopaminergic centers of my brain in goal-directed action, which would be what center? I know you know this. Now, what part of the brain is it, though? Max, come on, Max. You know it. You know it.
- 4.20** Okay, so you have two centers of dopamine. One is automatic. I'm saying it now—but when I'm doing automatic, non-goal-directed dopaminergic actions, movements: basal ganglia, right? That's when I'm walking up the stairs or I'm just doing something that's—it's not goal-directed; it's habitual. It's like an OCD person washing his hands 100 times. That's the basal ganglia.
- 4.21** When it is goal-directed, like me pursuing a mate or pursuing a meal or pursuing this movie over, watching this movie, it's what?
- 4.22** Nucleus accumbens.
- 4.23** Nucleus accumbens. You're learning now. Okay, Juan, you have to put it down. Nucleus accumbens: goal-directed motivation. Dopamine centers [are] there. Striatal circuits of the nucleus accumbens. These names will become familiar to you over time. They sound boring. They sound like gladiator names. But in time, you'll understand that these have personalities, these brain structures. You see, one is rigid, one is flexible,

one is fun, one is boring, one is emotional, one is rigid, one is angry. You learn over time.

- 4.24** Now, in the attentional circuit, the medial prefrontal cortex is important, and its circuitry going down to the nucleus accumbens and dopaminergic centers. When you pay attention, you have to have some brain neighborhoods, neural neighborhoods, you have to have them calm down. For example, you need to be less self-critical and less rumination on the self and me and I. And what part of the brain is that? Yes, exactly. Default mode network, Max. Exactly. Yes. Put that there. Default mode network. That's involved in self-critical thinking, self-rumination that has to go down.
- 4.25** Amygdala, fear-processing part of the brain, the fear departments of the brain, do they go up or down? I'm paying attention to this very interesting movie over here. It's so good. It's interesting. I'm being pulled. It's good. Is that—it goes down, right? Yeah, exactly. It goes down. So amygdala and all that goes down. Very good.
- 4.26** Okay, before I get to this, I wanted to—there's memories. I just wanted to give you a short summary of memory, because you have different types. There's working memory. Working memory is important for holding that number in your head. Having like 469372441, for example. That's working memory, DLPFC.
- 4.27** But you have different types of memories, and they have different functions. For example, you have episodic memory involved in—that would be long-term memory, but specifically the type that has to do with your life. It's episodic memory. So in your entire life, you have memories of your life that are episodic. And then you have another type, which is called semantic memory. That's knowledge of places, people, and so forth. That's me knowing a fact about a certain thing. So not autobiographical, but just me knowing facts. So that's another type of memory. Then you have procedural memory: sense of balance, riding a bike and all that.
- 4.28** Let's pull our attention back to attention. Okay, so what about—now, I think this is important to note. So attention— we have this circuit. There's a goal-directed aspect to it. You're focusing in on something. Of course, when you're doing that, you have to suppress any other information coming from any other source, right? So you have to do that. But you also have to generally filter out sensory information from the world. So we have a place in the brain that's important for censoring information that comes in. That's a filtering gate, a bouncer in the brain. That's also important for attention. I know you know it, right?
- 4.29** The thalamus.
- 4.30** The thalamus. So that's also involved. The thalamus. This is also important for attention.
- 4.31** Okay, so I'm mentioning these brain parts. Now you know. And as we are going through any aspect of the human condition, human nature, well-being, whatever, you will find

that these structures of the brain, as those who were in my previous courses, they overlap. It's similar. Whether you are a schizophrenic with everything coming in through that sensory gate, well, then the thalamus is involved. Or if you're an autistic child, that's also getting everything in, but in a different way. Everything from the sensory world is overwhelming you, literally, in terms of overwhelming light, auditory stimulation, also the thalamus. Or if you're in dreamland, if you're dreaming, the thalamus is also closed up for the world, but open for anything internally driven. Does that make sense?

- 4.32** What I'm saying is the thalamus is involved in all these various conditions and states of the human existence. But now, why is this important? Why is attention important? What is it about attention? There are several aspects to this. First of all, you can train it and use it, and you can also avoid being hijacked by it— very relevant for current society. I mean, your attention is being pulled in all these directions.
- 4.33** I mean, you sit down on your computer, laptop, you want to just prepare for this lecture, right? Because you're sitting there two seconds, and ding. And you say, "Well, should I look?" Your DLPFC is saying, "Don't look." Then your emotional circuits, nucleus accumbens, and those circuits, the reward circuits are saying, "Look, for God's sake, check that damn email. It's something important. Come on. It's so important now. You're getting that promotion. Oh, it's the girl, man. Check, please. Go, check." The DLPFC, "No, don't check."
- 4.34** Then you have the ACC. You know what the ACC is doing? It's involved in conflict monitoring. So the ACC is like, it says, "Oh, you have this big conflict. Are you doing it? Are you checking? What are you going to do?" It's monitoring all the conflict. It's the threat detection of the world. So that's involved, too.
- 4.35** And then eventually, I choose either to look or not look. Personally, I always look. It's a bad thing, right? But of course, you can train that. And that's where neuroplasticity comes in. You can train that. You can train your attention, right? One, you can train it.
- 4.36** So instead of being this person, have you had—okay, so let me ask you. I have this all the time. There are periods of my life where I am completely driven by attentional sources all the time. My salience network is always pulled. In fact, I'm getting SMS, email. I'm getting this here, beep. Mom is calling, my friend over here, everybody's pulling me, and it seems like I can't say no to anything. I mean, my attention is fragmented. And then I'm trying to sit down and read a book or listen to a lecture or read a paragraph or do anything, and I can't. Like deep work seems to be deteriorating. Do you know this feeling?
- 4.37** It's like almost your brain can be in two modes. Either you can train it in certain times of your life or you're able to be less distracted. And so other times you're completely distracted and you just can't find reward. It's almost like your reward circuitry in the

brain has now been trained to gain reward only for these short, immediate gratification type stuff. Do you know this feeling or is it only me? You know it, right?

- 4.38 But then other times when we actually sit down for, let's say, one week, two weeks, three weeks, and really try to pay attention and sit down and have these blocks of time where we are engaged heavily in this information, it seems like we're able to—it's almost like the salience network, over time, quiets down.
- 4.39 It's like that friend that keeps annoying you by sending you emails, and then he realizes you're not going to reply immediately. And then over time, you learn to train him to expect delayed replies. Your brain is like that, too, right? It expects. If you take out that salience charge by not always replying to things, you tone down the system. You train it. "Oh, things are not that important." You can sit for an hour, two hours, four hours at work, and it's going to all be fine. You're going to survive. So it has that component to it. It's really very interesting.
- 4.40 So we can train our attention by sitting down in these blocks of time. So how long would you sit? What's the ultimate? From a brain neuroscience perspective, Alexander wants to know; he comes to Baland. He says, "Baland, tell me, okay, what's the optimal time frame for deep work? Is it 10 minutes? Is it 20 minutes? Is it 30 minutes?" From a neuroscience perspective, for intense deep work, how long do you sit? What do you think, Alejandra?
- 4.41 Me, I need a lot.
- 4.42 A lot of time. I mean, in terms of like, okay, so you're now sitting down. Alejandra has to prepare for this presentation she's giving, and she has to pull in so much data, so many sources, so much information. So it leads to many hours of work. But she has this morning block. By the way, in the morning, attention is optimal. So in the morning, you do deep work in the morning. Now, Alejandra is in the morning. Should she have breakfast or no breakfast? What do you think?
- 4.43 No breakfast.
- 4.44 No breakfast? We'll get to that by the end of the lecture because I have tips for that, too. Breakfast, ketones or no ketones, glucose, all that's important. People want to know.
- 4.45 But now you're sitting there. You're sitting there. For how long should you sit? One hour, 60 minutes? That's the ideal? Completely without any distraction. It's actually about 90 minutes. Can you believe this? We're not trained to do this, but ideally, it's about 60 to 90, more like 90 minutes, in fact. But it can vary from person to person. It can, but it's about 90 minutes. Almost like the sleep cycle, how the sleep cycle is. You have sleep; every night you go through 90-minute blocks. Same for attention, for concentration. Hormonally, everything, attentional focus, neural synchronization, everything is about 90 minutes. It's ideal for deep work.

- 4.46 And then, ideally, you have a break after that. How long is the break? One hour? What do you think? Two hours?
- 4.47 Thirty minutes.
- 4.48 Ten to 20 minutes. Can you believe this? This is the ideal scenario. But most of us, including myself, maybe I do 20, 30 minutes of work, then I watch a little bit of YouTube, then I do a little bit of this, then I go out. I have to reward myself. You can't just work all the time. You're a workaholic, my friend. What is this, right? This is how we live. But ideally, this is how the brain is designed.
- 4.49 So you want to have these 90-minute cycles. Okay, so you have this. Now you're sitting there, but then Max is going, "I can't seem to focus. Something's up. What do I do?"
- 4.50 Maybe put a reward at the end of the 90-minute period. Get yourself motivated for it.
- 4.51 Okay, now we're getting into something interesting. Intrinsic versus extrinsic reward. Okay? So you want to reward yourself by giving yourself a reward. Now, I'm going to be diabolical here and say there was this study where they had children who loved painting. Okay, like Max, you like painting? You like painting, right?
- 4.52 Sure.
- 4.53 Sure. So they like painting and drawing. And that was so rewarding. They loved it. It was fascinating. But then they decided to give them some reward. Could be candy, could be cookies, could be anything they like at the end of drawing. Okay? Then they took the reward away. They took the reward away. Guess what happened? They stopped painting. They stopped painting. You see? You see the point here?
- 4.54 I see, yeah.
- 4.55 The point here is that, Juan, your medial prefrontal cortex should not be, ideally, so much tied to the external rewards. If it is, it's not good. You have extrinsic motivation, extrinsic outside reward. And those external rewards in the world can go; one day it's here, the next day it's gone. But if you tie your dopaminergic system to the actual action itself, you go, "My God, learning is so good."
- 4.56 Like Peterson Academy, you come here and learn, right? You don't get diplomas right after this. I don't give you money to come here and listen to me, right? But if I did that, it would be different. Now you're here for knowledge for its own sake. It's something different. Right, Giovanni? It has a different taste. You tie your dopamine system to the actual action itself. That makes it much more engaging and makes attention much more sticky. It sticks better. So you can do that.
- 4.57 How do we tie ourselves to that?
- 4.58 How do we tie ourselves to that? This requires the prefrontal cortex to think contextually and say, in my life, what is important? What do I enjoy? Do I enjoy

learning? What's fun about learning? How can I employ this in my life? How can I use this? "Oh, this is fascinating because it's intriguing."

**4.59** For example, let me tell you, these days, I've been listening to a podcast on CIA agents talking about the stuff they did in Iraq and this country. It has no relevance to my life, really. Maybe it's taking things away, not giving me anything. But I like the storytelling. I like the intrigue. It's like a movie. Like a movie, they talk about how this terrorist did this and then that happened. And it's just fascinating. I'm sitting there and I'm listening. Wow, one hour went by. It was so fascinating. Like watching a movie. So it's that, the human brain. And you can ask, what is interesting about that? I think we like stories. I think we like intrigue. I think we like anything that makes us think about the world in a new way.

**4.60** But when it comes to learning and something perhaps boring, you have to prepare for a presentation, I think the reward could be thinking about how this can help your life, how this can tie into your personal experiences. I think anything we can tie to our own personal experiences is actually very good.

**4.61** And I'm going to be blunt. I'm very blunt, right? Sometimes it's good, sometimes it's bad. But let's say you want to impress a girl. You say to yourself, "Okay, if I know this, it's going to be impressive to her." I'm not saying you should always be in that performative, impressive mode, but it could be something like that, right? I'm going to know this knowledge. It's interesting because I can use it in this context. It's relevant. It's going to make my life richer. So you have all these examples, okay? Does that make sense?

**4.62** Yes.

**4.63** Neurons, Juan, that fire together, wire together. Have you heard this expression? So whenever, for example, you are doing a certain action over and over again, what happens then? Do the same action, whether it be riding the bicycle, playing soccer, giving a speech, paying attention to this football match, or paying attention to your next presentation. What are you doing? Repetition, right? Repetition involves a lot of what? What part of the brain is repetition? Basal ganglia. So you do have goal-directed behavior, of course, but there's also a basal ganglia component to the repetitive types of attention as well.

**4.64** But you are paying attention fully to this thing, to this presentation. I've given it context. It's important for my life. It's intriguing. I can use this information for all these various aspects. So Juan is paying a lot of attention. You have dopamine because it's rewarding. The outcome of that is what? Plasticity. Repetition. The circuits, the more I pay attention, the longer I pay attention, the more I train my brain to become plastically wired to being able to hold long-term information and have long-term focus.

**4.65** Like each time I grab for that phone or grab for that cookie or decide to check my—I don't know what it is, the pager, whatever—the more I do that, the more I fragment

my attention. So next time you have this urge to go check your phone, you have this urge, just hold it for a few more seconds. Give it five seconds more. Give it 10 more. Give 20 seconds more. You'll see with time, it becomes easier. It's not willpower. Remember? We talked about it in the first lecture. It's not willpower; it's regulation. It's how you decide to—with each step, it becomes easier.

- 4.66 And in fact, you might say, you want to train yourself for attentional pull in an environment that has attentional sources pulling you a little bit, especially in the training phase. If you could go to a cafeteria and do your work there, it's actually very good. You can move attentional pulls and stay focused.
- 4.67 This is where neuroplasticity comes in. You keep focusing; more plasticity, the circuit becomes stronger. Myelination happens, meaning the wires from one neuron to the other—you have axons and dendrites. There's a myelin sheath here. It becomes more efficient over time. And with that efficiency, you have more and more long-term attention. So you keep the attention more.
- 4.68 But attention has, as Nikola knows, many manifestations. For example, I can have—and this is very practical. Let's say I have attentional focus. So you have a social circuit—I put it that way—social cognition and the social brain, right? If I have depressive tendencies, let's say, I have depression, I'm on the depression spectrum, my attention, default mode network attention, it's inwards, right? Is it positive or negative? Negative. Is it self-critical or not? Self-hatred. Always criticizing myself: I'm a loser. I'm this, I'm that.
- 4.69 Then what if you train your attention to be more external? If attention can be trained and I train it to be more externally directed, is that helpful, for example? It is. So you can train your attention circuits, if they are more internal, to be more external. And how do you do that? It would be something like whenever you have the attentional pull to engage in self-rumination, it's somehow to break that loop. It sounds easy, doesn't it? Maybe it's difficult, but the more you do it, the better it becomes. So you quiet that circuitry by—I mean, there's so many suggestions.
- 4.70 I remember once I had this guy; he sent me an email. He said, "Oh, I have this is new—" There's a fine line between genius and crackpot. So I get an email, you don't know, is this the next genius or what? He says, "I have this new idea. We're going to make a breakthrough in science." I said, "This is exciting." And then he says, "People with mental disorders, they tend to focus all their attention inwards." And I said, "Yeah, this is true. I read the paper." He said, "And as a means of psychopathology, each time they—to treat psychopathology, one way would be to simply have them look at something else when they have their inner rumination." And I went and thought about it. I said, "This is all true, but we already know this." I mean, this is how a lot of therapy works. So it wasn't really revolutionary in this case.

- 4.71 My point is, in the way we treat mental illness, anybody that has excessive self-rumination and inner talk, self-critical inner talk, we do tell them to put their attention outward, to watch movies, to watch a comedy. The person in the camera is smiling. What happens to you? Mirror neurons, you start smiling as well. You smile, and then they're laughing, and you find yourself also laughing. It makes no sense, right? Here you are, one second, ruminating about how you're the greatest loser. You want to kill yourself. And then you're watching something over here; it's funny, and you start laughing. You start laughing. Completely shifting the system. Training attention to go outwards.
- 4.72 In fact, that is attention training. It's one of the—I started my career in attention training. Can you believe this? I came all the way from—this is how it started. I was in Cairo doing my research over there. And then I decided, okay, I'm going to go to America now. Take the big steps. So I got on the boat. Of course, flying. The Godfather, a different movie, right? I took the flight, came to America.
- 4.73 And then I go to this laboratory and they tell me—I said, "I'm going to study the brain, how you get mental illness, how you get depression and anxiety." And the guy, he looks at me and said, "We're going to look at how attention works in the brain." I couldn't believe my own ears. I came all the way from Egypt to America to look at attention. What is attention? What is attention? I mean, I have attention, you have attention. There's nothing interesting about attention, right?
- 4.74 But apparently, it was very—this whole lab, a whole lab, 24 hours, literally, they were there from 9:00 a.m. to 9:00 p.m., studying attention only to treat mental illness. I was like, "What is this?" I was disappointed. I was very disappointed. I'm telling you, so disappointed. But when we got into it, and I realized, my God, this is actually pretty revolutionary.
- 4.75 The treatment was like this. He showed me this guy. He said, "Okay, you have anxiety. You look at two faces. You have social anxiety. You look at two faces. One is sad, one is happy. When you look at these, focus your attention on the positive one." So you go like this and you go on the positive one. Okay? Again, face, angry, sad. Look at the happy one. And you keep training your attention like this. And over time, the person becomes less socially anxious. When he goes into the crowd, instead of focusing on the sad face or the angry face, he looks at the positive face. He's reoriented his attention. His parietal lobe and all that. He goes towards the happy faces now. Incredible, right?
- 4.76 Or you have somebody with OCD and you show him a face, you show him something contaminated over here and something that's less contaminated, something that's clean. And you say, "Focus on the clean one." He looks at that and it becomes better. You train your attention. It's quite interesting. And so it made the effort. Now I started to see the point and it became more interesting. But eventually, I left all this attention business. It was still too boring. But nonetheless, they did have a good point. They did have a good point. Attention training, it's called.

- 4.77 Also, do you know there's a type of OCD— I don't have the answer to this, so we have to, in a group, come up with the answer. But I knew a girl once, and she told me—we were very young. I mean, I must have been, what, 14, 15? She was the same age, maybe a couple of years younger, and she said, "I tend to break up with guys very quickly." I said, "Why? Why do you break up with guys?" She's a very sweet girl, very sweet, long, brown hair, your deer eyes, these big brown eyes. What can you do? Okay.
- 4.78 And then she said to me, "I break up with them because initially I'm with them. It's exciting. But then after a few weeks or maybe a month, I start seeing their flaws. In fact, I can only attend to their flaws." So if they have some deformity—we all have. Nobody's perfect, right? Even me. Okay, this is a joke. That's a joke. You can clip that out. These jokes are not good. They're just to keep you laughing.
- 4.79 Anyway, we all have our flaws, right? So she may look at my bushy eyebrows. The eyebrows over here, she started looking at them. She said, suddenly, they become huge. Normally they don't become like this. And when nose becomes like—they have this thing. Or she had this thing, and she was like, "Wow, that's really bizarre." She actually started focusing on the flaws, like OCD-like focus. And then she said, "The moment the guy breaks up with me, then the flaws go away." It's a really bizarre thing, but she was OCD; that's obsessive-compulsive disorder. That rumination circuit—we have a rumination circuit, but it's more like an obsessive rumination circuit that became overactive in her. So she looked at all this.
- 4.80 But I think all of us, to some extent, have that circuit, not necessarily to outward flaws, but personality quirks. So do you have this uncle that's irritating because he's doing something, or your boss maybe has some weird habit? Do you know anything? You know it. Come on, don't hide it. Come on. You know this, right? Do you know this? You know this feeling of something that annoys you about somebody's demeanor or it could be your niece or anybody, right?
- 4.81 Can we train this, too? How do we train this?
- 4.82 You focus on the good things.
- 4.83 Focus on the good things, yeah. How exactly, though?
- 4.84 Ready for the acceptance?
- 4.85 Acceptance, true. We accept. Absolutely. I would say, yes, exactly, except in OCD, it's difficult because it's very sticky, the thought. But in most of us, if attention can be trained, and whether it be self-criticism, outward criticism, it's a matter of changing that view. It's almost like accepting it, but in a way, you accept it; it's focusing your attention away from it. That's what I'm trying to get at. Focus your attention away from it.
- 4.86 So let's say the thing that annoys you about somebody, let's say—well, give me an example. It's very hard to pick an example. What annoys you? Is it like when your dad

drinks, has a soup, he goes like [slurps], for example. Could be annoying, right? So how do you get your attention away from this?

4.87 Just focus on my food instead.

4.88 Focus on your food. But it keeps coming to your mind. It's so annoying. You can't get rid of it. So what do you do?

4.89 I don't like people chewing. That can drive me crazy. So one thing I do is I pretend it's me chewing.

4.90 Me chewing.

4.91 And it suddenly switches it—

4.92 It's interesting. So you change it. That's very interesting. So you go—I told you this is open-ended. I don't know the answer to this, but I think it has to do with the fact that the person is chewing over there. They're doing this annoying thing. You have critical behavior up there. Then you do an empathy mirroring. You say, "Okay, if it's actually me, I can do the same thing." Yeah, I think so. So you do a cognitive correction as well. So attentional pull-away from that. But then you say, "Oh, I sometimes do the same thing. Other people may also find me annoying. Maybe I should—" So you have a cognitive override. Maybe that could be. I think that is true.

4.93 At the very least, what I can say is that pulling your attention away from it over time will make it less annoying. I think that would be the critical clue regardless of whether it is inner or outer criticism. The more you put your attention away, the better, and it will become less over time. The question is how you will find a way to do it. But the more you focus on that and find it annoying, the more the circuitry going to the amygdala between the amygdala, the prefrontal cortex, and that neural representation will become stronger over time. So put your attention away.

4.94 Okay, so I talked about how in the flow state, there's a thing where different parts of the brain start to work together in this rhythmic dance, almost, right? And that's the whole—the brain hates incongruities. In the flow state, for example, when the parts of the brain start working together, it feels good. The prefrontal goes down. We mentioned it. What else? Default mode network goes down. What else?

4.95 Sense of balance.

4.96 Sense of balance, cerebellum goes up, and so forth.

4.97 And there's a level of synchronization of the network which feels good. It just fits. It feels good. That is exactly the same thing when you hold attention for a long time: it starts feeling good. Does that make sense? When the prefrontal goes up in attention, this part of the brain—lateral—goes up, medial goes down, goes up. Default mode network goes down. Thalamus, what happens with the thalamus? Goes up. Filters more information, right? What else? Dopamine goes up, nucleus accumbens. Basal ganglia

goes up. What else? Parietal—orienting goes up. And that whole cascade of neural activity that's becoming activated as I move my attention over to an object and I keep it there, it feels good. It feels good. There's a feel-good to it in the brain. So whenever the brain regions work together, it always produces this subjective feeling of feel-good.

**4.98** But not only does it feel good—this is counterintuitive in some sense—but it's also, let's say, less fatiguing. Can you believe this? Because it sounds like if I'm sitting there and doing math or preparation for work, say you have some preparation, you're doing Juan, you have to prepare, it feels like a lot of energy.

**4.99** Okay, what is more energy draining: sitting there and focusing for 90 minutes, or sitting there for 10 minutes and then checking an email and coming back? And then sitting for another 10 minutes and then checking a YouTube video and coming back. Which is more?

**4.100** The latter.

**4.101** The latter, right? But why?

**4.102** You're fragmenting.

**4.103** You're fragmenting, but that's not enough. You're doing more than fragmenting, but you are fragmenting, but you're doing more. You're multitasking, yes, and you shouldn't multitask when you're doing deep work. But it's very costly because you're starting over. You're starting over each time. Each time you have a new—it's like almost starting the cycle.

**4.104** Did you know that for attention, the neurons need time to get into deep work and deep attention? You can't just go from complete fragmentation and waking up and brushing your teeth and speaking to your mom and drinking that coffee and looking at the news and looking at the newspaper to sit for two hours. This is impossible. You can't. You have to have that 10, 15 minutes of resting a little bit into it. Resting into it, getting your brain in. You have to have a warm-up, for God's sake. That's what I'm saying. You have to warm up for attentional focus. That takes at least five, 10 minutes. So you have that. So that's why you can't really fragment it.

**4.105** But all of us can decide to live our lives in various attentional modes. And I think this is key. You can be inner-focused, self-critical, self-deprecating, or you can choose to pull your attention inwards whenever you have that thought. This is much easier for somebody without clinical depression. The balance of the circuitry is just easier to play around with and change. But through neuroplasticity, you can change your attention to be less self-critical. This is one mode. You can be outward critical and be critical of everything, people, but even perfectionism towards ideas and things in the world, job, careers. You're critical of everything. You can pull your attention away from that and focus on—you can be less—put your attention away from it.

- 4.106** So let's say you have two opportunities. Let's say you have two things you're buying, like you have a suit over here that costs \$2,000, \$5,000. It's beautiful, it's gorgeous, it's completely perfect. That's how you want it. This one is \$1,000 over here. In my pocket, go for the \$1,000. You can't do the \$5,000, okay? Might even go for the \$400 one, right? What you do over here is you train your attention to accept things that are not completely perfect as you'd like. Does that make sense? It's just a mode of thinking. You can become less self-critical and less critical about things in the outside world. So that's another mode.
- 4.107** Another mode of being is—have you seen *Beauty and the Beast*? Have you seen that? Belle and the people in the town, how are they different? How are the people in the town? Are they more driven by the external world? Attention. And where does attention flow in that cartoon? All the girls in the village, they like who? Gaston. How is Gaston? Big muscle, 12 eggs a day, right? Big neck. At least, right? Attention in that whole town is completely superficial. Completely. Always looking at the outside. Okay? Belle. How's Belle? Inner-driven. She doesn't like Gaston. She likes the Beast, for example. That would be an example of how you change your attention.
- 4.108** Okay, now the cartoon. We're in cartoon land now, okay? In *The Lion King*, you have two modes of attention. When Simba, when his dad dies, the scene, Scar makes him feel like it's his fault. When he falls down, the father, he's on that cliff, he's falling down, right? And then Scar gives him the impression that it's actually Simba's fault. He says, "Leave this whole land. Go to this faraway land now. Leave. Never come back."
- 4.109** When he goes to that faraway land, Simba meets these two animals. What is it? A pig and what? And another animal.
- 4.110** A meerkat.
- 4.111** Exactly, right? And their motto becomes? Hakuna Matata, right? Hakuna Matata. What's Hakuna Matata about? No worries. Attentional focus now is shifted. There's another mode of attention. You can either be like Hakuna Matata mode. This is the guy, sits in his basement, drinks beer all day, sits on social media, doomscrolling all day, looking at this video and that video. He has a lot of bottom-up attention, very little top-down attention, and he has very little prefrontal, future-directed attention. You see that? This is what? Hakuna Matata.
- 4.112** But then you have Juan over here, and you have all the students that come and have a long-term goal, long-term vision, long-term plan. This is when Simba, Rafiki comes to him and says, "You have to go back. Remember who you are." You know the scene?
- 4.113** Yes.
- 4.114** Okay, and he goes back. What is that attention mode that's switching to? He's switching to a different mode now. He adopts responsibility and directs to the future. He becomes future-oriented now. His attentional pull goes away from bottom-up to

top-down to prefrontal long-term vision. See what I'm saying? So you put that here. That's the second mode.

- 4.115** When you are stressed, do you have better or less attention? Why do you have less attention? Why?
- 4.116** Because your default is ignoring the emotional part.
- 4.117** The emotional part, the amygdala becomes hypersensitized. The prefrontal lobes are highly sensitive to cortisol. They will shrink over time, and you will find yourself always hijacked by your salience networks. Exactly. Great answer. Rohana, I like it. Thank you.
- 4.118** Okay, how do we train our attention? First of all, you want to have single-tasking. Do single-tasking, do not do multitasking. This will make your ACC, involved in conflict detection, happy. It will say there is no conflict. And whenever there is a conflict, let's say something comes up, the dorsolateral prefrontal cortex, the DLPFC, will be able to reduce that.
- 4.119** If you have difficulties with single-tasking and you find yourself pulled by a lot of attention, train yourself, okay? Train yourself. Delay checking that phone. Go to that, I don't know, bakery or whatever you like. Let's say you like sweets, so that coffee or chocolate. Go near that coffee store or to a chocolate store and walk by it, but don't go in. I think this is key for the brain. You want to cue it; give it a little bit of, like, you want to go in, but then you don't. You keep yourself outside of that bloody store. So I think that's key. Expose yourself in certain scenarios, but be able to delay gratification and delay impulse.
- 4.120** You could do some ways to train working memory. It would make it better, technically. I wouldn't do it myself. I would just find it too boring, to be honest. But you could do math. You see, ChatGPT. ChatGPT helped me with these notes. Not a good thing, man. In the past, I used to have this in my head. Go listen to my first lecture. I'm not looking at notes. I'm just talking. But our brain with ChatGPT is becoming lazy. They give us beautiful notes, organize everything for us. What is this? No thinking, no logic. It shows us we're using less brain capacity and fewer areas of the brain, especially working memory as we're using AI.
- 4.121** So try to maybe do some math, if you like, or at least use your brain more, if I could say. Do mental arithmetic, if you like. A trick that we used to use in neurology was, you can count, subtract seven from, like, 100, and then you go 93. Can you do this right now? This is actually a test. Can you subtract seven from 100? Keep going. I'm not going to put you on the spot, but let's do it in your head right now. I'm not going to expose you. Okay? Ninety-three and so forth. I don't even know it myself right now.
- 4.122** It's actually a neurological test. If we bring in a patient, we actually, Ramachandran and I did it so many times. Sit with a patient, and if he has a brain injury, this is the first thing we do. We say, "Subtract, do the subtraction." And if you can't do it, definitely

this guy needs a brain scan; he's out of it. But most of us today have difficulty with things like that. But this will actually expand the working memory in the DLPFC. So if you have the patience— I don't—but if you have the patience to do things like this, please do it. It does help.

- 4.123 Language switching. This is especially pertinent over here. Rohana, right, you guys, you know a lot of languages. French, perhaps Spanish. Okay, so you speak one language and try to switch over to another one. This is very difficult because now you're even using competing circuitry. So do this. Speak a little bit of Spanish.
- 4.124 When I was in Egypt, I spoke four languages in one day, every single day. Can you believe it? I spoke Kurdish with my mom and dad on the phone. I spoke Arabic with the street kids. I spoke Arabic [speaks Arabic]. I spoke Kurdish with my parents. And then I spoke Danish in my head, Danish, and then with my siblings, perhaps. But Danish was my main mode. This is my main language, right? My mother tongue, I dream in Danish, for example. What else? English at the university as well. Four languages every day. Can you believe this? Very good cognitive switching.
- 4.125 So you do that. You switch languages. You're Canadian, maybe you speak French.
- 4.126 You're supposed to.
- 4.127 You're supposed to, right? So you jump around. This is a very good training of the executive network.
- 4.128 Stress regulation, sleep optimization, aerobic exercise, social bonding. This is me repeating myself, but it actually is very helpful.
- 4.129 Meditation. Lie down, focus your attention on your, maybe, forehead or your breath, and you keep going 10 minutes, five minutes. And whenever thoughts come to you, either you push them away or you let them float over you like a cloud. You go, "Okay, they're coming, but I'm not going to be reactive to them. I will not respond to them. Let them just flow." So this is one way of meditating. Very helpful. This will make your prefrontal cortex grow and become thicker. So do that. Initially, understand that it will be very hard to sit and pay attention to that presentation for 90 minutes. Very difficult. Even for me, 30 minutes, very hard. Forty minutes, very hard. But with time, it becomes part of your identity. It's a shift. So you have to be patient.
- 4.130 Environmental engineering. Okay, so true, you do want to put yourself in scenarios where you're challenging yourself, especially when you want to train your attention. This could be walking past that store that pulls your attention in or sitting in a cafeteria and there's a lot of noises and sounds. But occasionally, when you're actually doing the actual work that's important and you now need 100% focus, that's when you actually remove a lot of these notifications, move anything that can pull your attention.
- 4.131 Now be strategic about it, Alexander. Be strategic. Notifications, remove. Phone, remove. WhatsApp, remove. No beeping and none of this, you have to go. None of this,

okay? You have to focus on this. And then maybe you tell your family members, you tell your mom, dad, your brothers, your spouse, whatever; you tell them, "Look, I'm working. Give me a break. Give me a half an hour."

**4.132** You prepare the environment. You remove visual clutter, even if that's a thing. Is visual clutter annoying, Kelly? What do you think? For attention? See, this is—what do you think? What do you personally think? In your office, for example, did you find it annoying?

**4.133** It can be.

**4.134** It can be. We're trying to full attention. This is where we get into this whole thing of you can be—one can be very black and white about it. The question is, are you optimizing for focus or are you optimizing for creativity? Being in a room, for example, me, I like to have a creative room. I do like to have some maybe paintings, maybe have a big window so I can look out. Occasionally, I look out and there's somebody walking by. He's wearing a weird hat or something. I get inspiration. So I can put that in the stuff I'm doing. This is the dilemma you're facing. Is it creativity you're optimizing for or is it attention? This is the point: it depends on the outcome, but I would say at least the notifications of the technology, you want to dampen that.

## Lecture 5: Dopamine

At a glance — Dopamine and motivation.

- 5.1 Let's finish attention, and then we go to dopamine. So this lecture will cover attention, and we're going to dopamine. Dopamine is so popular. All these health gurus talking about dopamine all the time, right? Dopamine this, dopamine that. Get more dopamine. Eat more chocolate. Do you get more dopamine, Juan? You do, right? A little bit? We'll get into that.
- 5.2 But there were actually some points I wanted to cover on the whole attention stuff. One is regarding internal salience and internal distractions when you have worries. It could be self-critical, but it could just be general worries, like generalized anxiety disorder. It's also the default mode network. Are there any techniques we can use, Max, like actual steps? This is what I'm going to give you now, okay?
- 5.3 When a thought arises: could be worry, could be an unpleasant memory, could be an urge. What do you do? The first thing you do? Mentally, you label it. You say, "This is a worry." Why is this important? It gives control to the prefrontal cortex. So it's a good way of contextualizing it. And being aware, look, this worry right now is my enemy, right? And Max is even more sophisticated. So you go, "This is my default mode network screaming at me right now." So you have one layer of sophistication extra, which is good. It's called psychoeducation, and it's very good to have meta-awareness.
- 5.4 Knowing what's going on in your brain and how you can correct it, and you say, "Now the default mode network is screaming, and then I need to orient my attention to the parietal lobes, and I have to have my prefrontal cortex engaged." Having that knowledge is good. So you want to label it clearly, engaging the prefrontal cortex.
- 5.5 Second, you gently bring your attention back to whatever you're focusing on in the outer world, but gently, okay? Not forcefully. That would be step two.
- 5.6 Now, if you have an annoying loop that keeps coming into your mind, could be about that, I don't know, meeting you have with a person, your friend, a girl, whatever, you're doing something, and it keeps coming to you, and it's a bloody obsessive thought. What do you do then? You write it down. You write it down. If you know you can't beat the thought, you actually just put it on a piece of paper, and you write it down. That way you let go of the loop.
- 5.7 You can use an actual anchor. It's always good to have an external anchor. In my previous courses, I've been talking about if you have schizophrenia, let's say, and you're driven by internal, not only thoughts, but hallucinations that are internally driven, what do you do? If you have your internal landscape, remember that whole loop of cognitive, higher-order thinking overriding all the sensory areas going from top down, and then you hallucinate. Remember that?

- 5.8 So I had this case study of a patient who was hallucinating. And what did I tell her to do? What was the antidote to the hallucinations and the inner imagery? It was taking an ice cube, remember?
- 5.9 Yes.
- 5.10 Holding it, right? Externally driven, Debbie, right? You do something that engages—you bring an anchor in to compete with that inner hallucination, that inner thought, that inner rumination. And this could be just focusing on that paragraph or the problem that's in front of you. It could be your breath, anything that's actually concrete that can temporarily compete with the words. You do that.
- 5.11 Fifth, you accept the mental noise. This will be the final point. You accept that you will have mental noise, that you will have distraction, you will have things come at you all the time, and you just accept it. Once you've written down the worst culprits, the most pernicious thoughts, you've written that down, then you accept that you have mental noise and you just learn to focus on your anchor and then keep going.
- 5.12 There are a few experiments—you have to know this; I want you to know this— a few experiments that are very interesting. In my early days, when I was in the laboratory, in that attention laboratory, they showed me this annoying—I found it pretty boring, annoying, but it's actually cool. There's this paradigm called the Stroop task. It goes something like this. The Stroop task is: you see the word "blue," but you see it in red ink. What happens then? You are told—okay, I want to be complete here. You see the word "blue," you see it in red ink, and you're told to read out that [ink color]. If you see the same word "blue" in blue ink, which one is faster? The first or the second?
- 5.13 The second.
- 5.14 The second is faster. Why is it faster? No conflict. What part of the brain is involved in conflict detection and attention? ACC, right? So the ACC, in the latter case, is not involved in any conflict detection. But in the first case, it is. It says, "My God, blue, and it's red," for example. It's like, this is conflict, right? This is me sitting on the computer and getting that email, that ding, ding notification. That's the conflict. And then when you have these people do it in the brain scan, you actually see the ACC light up in the first case, and you see the dorsolateral prefrontal cortex suppress the [word], and you go for the [color]. Is that clear? The Stroop task.
- 5.15 Then, of course, whatever pulls our attention—what is more likely to pull my attention? Josiah walking right here, going from A to B back in the backstages, or if I see a tiger over here, let's say a tiger running? Which one?
- 5.16 Tiger.
- 5.17 Tiger.
- 5.18 Tiger. Why?

- 5.19 Danger.
- 5.20 Danger? Okay, forget tiger. Tiger is—I can understand tiger. Let's say a more benign thing. Let's say a camel. Okay, there's a camel over there.
- 5.21 It's just unexpected.
- 5.22 Unexpected, odd, right?
- 5.23 Yes.
- 5.24 So it turns out that, of course, dopamine likes novelty. Attention and dopamine are cousins. Dopamine is driving attention all the time. Of course, the attention molecule is mainly acetylcholine, the nucleus basalis in the center of the brain—I've gone into that before—but it's dopamine-driven as well.
- 5.25 Now, there's this paradigm called the oddball paradigm. It works something like this. Each time you listen to sounds, and then there's an odd sound—this is called the oddball paradigm. And what do you have? What happens to your brain? Which chemical becomes excited? [inaudible] Probably, yes. But what else? What centers and what?
- 5.26 The acetylcholine?
- 5.27 The acetylcholine is unfocused. But then you have the oddball. You have the tiger, you have the camel, you have the giraffe, you have the space alien over there. What lights up?
- 5.28 Dopamine.
- 5.29 Dopamine. This is the oddball paradigm. You have to know this. These are some of the seminal, very important critical studies in this field of cognitive neuroscience.
- 5.30 Why early focus feels very hard—why does it feel very hard? We talked about the whole task switching, and it's costly. It's like turning the engine on and off and on and on. And that is itself very costly, right?
- 5.31 But there is this point that's also very important in your journey to more attentional focus, which is that early on, focus will feel heavy, not only when you start to train it in the initial days and weeks, but even in the first 10 minutes. And this is key because people give up, right, Joe? In the first 10 minutes, because of the pull of various things, various notifications in the environment, various cues, you feel it's—you just want to give up. You just want to go and watch that bloody movie or do that, check that email, whatever. But if you can hold it for 10 minutes, 15 minutes, that's when the charge goes away; it becomes easier. And so effort will feel more difficult in the beginning.
- 5.32 If you are sleep-deprived, especially if you are sleep-deprived, any hacks you can do? Anything you can do to circumvent the system?
- 5.33 You can take a little nap.

- 5.34** Power nap. How about a power nap? Yeah, I didn't think of it, but it's good. Power nap in there. Let's put power nap. That's one. See, I wanted to make this course very helpful for you. Practical, isn't that right? Not only theoretical, but practical. What's that? Good, good. I do power naps, too. It's very efficient, very efficient. Do a power nap or power meditation type nap. So it's half napping, half meditation.
- 5.35** Creatine. Do you like creatine? You take creatine. It's very good. Above 10 grams, good for the brain. It goes to the brain. I take about 10 grams daily. Honestly, I've been taking mostly 5 grams. But if I want to affect my brain, on those days, I'll do 10. I'll probably start taking 10 consistently. I've seen the literature, and it seems like it's good. It's safe, it's not going to damage your kidneys and all that. So I'll do 10 a day. Then at least you have enough to go to the brain. But definitely always when I'm lecturing, I'm doing anything mentally heavy load, 25 grams. And it's so cheap. It's not an expensive molecule. So we put creatine in there.
- 5.36** Okay, 40-hertz binaural beats. It's called binaural beats. It's like an auditory thing where you listen to some beats, some tones at about 40 hertz. This will be helpful for attention. It can be. So there are apps out there. Probably there are some free ones. You can download this, and this will put you in the gamma oscillation mode. So we have different alpha, beta, gamma, delta. There are various ways. So delta could be very, very slow waves, alpha semi-slow, beta fast, all that. Gamma, you can get into the gamma frequency range of brain waves and oscillations. It improves attention, working memory, executive functioning, and perception overall. And there are some studies suggesting you can do this.
- 5.37** So if you have struggles with attention, personally, would I go get these apps and start using this? No, I wouldn't. I mean, I would do some of the other tricks first, right? But if you're into that stuff, I mean, some people are, you can get those apps and you can actually use it for about maybe 10, 15 minutes before you start the deep focus work. And then you listen to these beats; you have this auditory entrainment. We talk about entrainment with music. You have auditory entrainment. And then you listen to that, maybe the first 20, 30 minutes of whatever you're focusing on. And I think those would be my final thoughts.
- 5.38** Oh, yeah, I probably should mention this. I'll mention this anyway. Should you do focused work while fasting? Glucose—should you be in the high-glucose state? Should you fast? Should you take ketones? What do you think, Juan? What's best? Should you take a good big meal before you go, like I do sometimes?
- 5.39** I had a lot of food at the restaurant yesterday. I regret it. You see? I got a little bit too much. I told them, "Don't give me that much. Don't give me that much food. I don't need that much fish." I told them, "Just give me a little bit of fish, a little bit of this, a little bit of that, a little bit of mushrooms." They gave me this big pile. And then I became, "Okay, my God. Have a bit more," and ended up with a big stomach. It feels so good. I said, "How can I lecture tomorrow?" It's not going to be easy, right?

- 5.40 But is it a good thing? What happens to your metabolism and your circulation, metabolism, brain metabolism? What happens to it? Does it go to your stomach? Where does circulation go? Where does all this blood flow go? To the brain or the stomach?
- 5.41 The stomach.
- 5.42 So is it a good thing or not a good thing?
- 5.43 No.
- 5.44 Should you eat a lot of sugar or no? Do you know what ketones are? Alternative brain fuel. Should you do this? What do you think, Sharon? No? You could. Maybe you could take ketones. Maybe you could do fasting. For some people—I skip breakfast. It just works for me. Just take some water, maybe some creatine, and that's it. And I think there's no real consensus. You have to follow your own path in this, but definitely avoid sugar and glucose spikes. This will be very bad for you.
- 5.45 And if you want to try ketones, I've tried ketones. I was on a podcast once, and this bodybuilder, he said, "You wanted some ketones?" Yeah, this sounds pretty cool. Let me try this. And I was like a crackhead, like cocaine for me. They don't work. They don't work, okay? I became like—I talked too much. I started rambling, you see? So some people may be more sensitized to these things, right? But it could work for some.
- 5.46 And on the whole dopamine business, let's actually stay on that. So what is dopamine? And what is dopamine all about? It is what? It is the reward chemical in the brain, and you have dopamine for two important functions. So plasticity is an outcome of that, definitely. But even more proximally, even more at a basic level, anticipation of goal-directed behavior and anticipation of reward.
- 5.47 Dopamine is also important for movement. Is there a link between movement and anticipation of reward? Just logically, think about it. Is there a link?
- 5.48 It gets you from point A to point B.
- 5.49 Yes. In what way?
- 5.50 Because you think about the reward when you start moving—
- 5.51 Towards the reward. Absolutely. So, you see that big meal over there, that gazelle. First, you see the gazelle, novelty, behind the bushes, and then you move towards the gazelle. What do you have there? Dopamine, right? So dopamine is involved in anticipation of reward and also movement.
- 5.52 Do we know of any conditions out there in psychiatry that illustrate this or neurology that illustrate these two basic functions of dopamine?
- 5.53 Parkinson's [inaudible]

- 5.54** Parkinson's, where you can't move. So what's happening in Parkinson's, since you want to bring up neurology, you want to be the neurologist, come on. Nikola, you've taken many of my courses now, many of my lectures. Lowering of dopamine in what circuit, my friend?
- 5.55** Basal ganglia.
- 5.56** Absolutely.
- 5.57** Nucleus accumbens.
- 5.58** This guy, he knows his business. Forget nucleus accumbens. Now you're going too far, okay? You have basal ganglia. Just keep it simple, but you're right. Basal ganglia. So you have dopamine in various circuits. You have mesolimbic cortical circuits. You have various circuits in the brain. I will get into that in a second. But within the habit circuit, the basal ganglia is the habit circuit, the automatic movement circuit. When I do habitual repetitive behaviors, you have a circuit for that called the basal ganglia, right? The putamen, the basal ganglia, those families of structures. All right.
- 5.59** Now, the neurons in Parkinson's, the dopaminergic neurons in Parkinson's die in that area, in that loop, in that whole structure. Is that clear? They die. And the outcome of that death of neurons is what? What happens to—
- 5.60** You can't move fluidly.
- 5.61** You become rigid in your movement. You've seen Muhammad Ali, Michael J. Fox. How do they move? Now, mind you, these guys, they do take dopaminergic boosters like L-dopa that will lead to boosting of dopamine. So it can be hard to say because they sometimes have movements like this. Have you noticed this? And this is because they had the death of the dopaminergic centers initially in the basal ganglia—and I've seen this. People who don't take the medication, they walk like this. They can't move, and they can't smile. They can't smile when you smile. They can't. Everything is flat, everything is rigid. And they even have depression. You sit with them, and you talk to them, and they're sad. They have anhedonia. No motivation. It's flat.
- 5.62** It's interesting with Parkinson's because we have—we talked about masking and unmasking circuits. Was that yesterday? Yeah, that was with the piano, right? We always have extra circuits behind the scenes for emergency. We call them emergency strings you can pull.
- 5.63** Somebody with Parkinson's, they can't move. It's very sad and all that. And then suddenly, if there's a fire or they see—and this is in many cases— they walk by, like, there is a sea over here, there's some water, and there's a drowning child, they will jump into the water and swim like an Olympic athlete, swim, swim, get the child and come out and then come back to this, like that. Or if there's a fire, they might start sprinting, and then they become rigid again.

- 5.64** Why? Because in the brain, we have these extra circuits we can pull on, extra strings. Very interesting. It's called unmasking, unmasking of circuitry. But there's the death of dopaminergic neurons, and that illustrates clearly that dopamine is involved in movement and motivation as well.
- 5.65** But motivation is more the motivational, called the mesocortical mesolimbic circuit. So the other would be the substantia nigra circuit. So it goes from the substantia nigra and those regions, basal ganglia and all that. But the other is the motivational, primarily motivational circuit.
- 5.66** And what disorder illustrates that? Which, Giovanni, which disorder is mostly emotionally flat, no motivation, just want to stay at home, I don't want to do anything?
- 5.67** Depression.
- 5.68** Depression. Exactly, depression. So depression would be the other example, in this case, a psychiatric disorder, where the dopaminergic centers—the dopaminergic neurons in the nucleus accumbens and the striatum—have become desensitized.
- 5.69** Okay, some technical stuff. So please write down: the mesolimbic pathway. The mesolimbic pathway regulates motivation, reward, anticipation, and effort allocation. This is the one implicated in? Depression. This is the depression circuit, the mesolimbic pathway. Guess what happens if I cut this circuit? If I cut this circuit, what happens? Anticipation of reward goes completely away. In fact, even in dreams, you will not dream anymore. Dreams are very similar. You see, in dreams, you have a lot of anticipation of reward, bliss. All of it goes away.
- 5.70** Then you have another circuit, circuit two, the mesocortical pathway. So, just to be clear, the first one is the nucleus accumbens. It goes from the VTA, the ventral tegmental area, to the nucleus accumbens. Nucleus accumbens, I keep mentioning this. When you think of anticipation of reward, think depression, think joy, me having a next meal, going for that mate, going for that whatever it be, always think nucleus accumbens. Nucleus accumbens. Over time, you'll get it. It comes. Over time, you'll have it.
- 5.71** But when you're talking about planning, decision-making, goal-directed behavior, not so much reward per se, but just goal-directed planning and future orientation, then we're talking about the mesocortical pathway, the ventral tegmental area, and the circuit goes to the prefrontal cortex.
- 5.72** And then you have the third pathway, the nigrostriatal pathway. It goes from the substantia nigra, piggy-black cells, to the striatum. The striatum is very much linked to the whole basal ganglia putamen. That's a whole family of neurons that have to do with movement and also some reward centers over there. It's very simple. Over time, you'll have this. No, it's so easy. Easy peasy, not very difficult. Anyway, essential for

movement, the last one is a movement circuit. This is the habit circuit. This is the substantia nigra basal ganglia circuit. This is the third one. Okay.

- 5.73** This is, again, technical. Dopamine, we talked about receptors, right? Do you know receptors? All about receptors. Joe, I know you know; you took several courses of mine. Receptors, you know what receptors are? I didn't know for a very long time. I just learned. I mean, it's not—It's technical.
- 5.74** Receptors are on the surface of cells. Cells, you have like a little parking spot. On this parking spot, this cell over here wants to talk to that cell. You see, it wants to say, "I want to boost you up, make you more excited. I want to send reward signals, whatever." So what it does, it will send this little signal, chemical, and it latches onto that parking spot.
- 5.75** Now, the parking spot can have different types of parking spots. So one parking spot, let's say, might be—let's call it this serotonin 2A parking spot—it means this is more when you tickle this parking spot, you get a sense of doom and gloom. The other one, you feel joy. The other one, you feel more happy. So it has nuances within the circuitry.
- 5.76** So the brain has all these layers. It's very sophisticated, right? Not only different neurochemicals and hormones, but also even within those hormones, you have receptors. Receptor C: it's isolated; you feel panic. A, I feel anxiety. C, I feel happy. So it has this fine-tuning and these nuances.
- 5.77** For dopamine, you have different receptors. One receptor would be D1. D1 receptor. The D1 receptor acts like the go signal. That's the one that supports motivation, drive, getting started, getting ahead. But you also have the D2 receptor. It's more caution, flexibility, and control. If the first one is the go, let's go. Let's have that meal. Let's have that mate. Let's go to that party. Let's go to that restaurant. Let's go to the gym. Let's have fun. The other one adjusts. You see, it works with the prefrontal cortex and adjusts behavior; it makes it flexible so it's not out of balance. That's the D2 one.
- 5.78** And now I go even more technical on you right now, but then we'll have fun after, okay? Don't worry. You have two types of firing rates in the brain: phasic and tonic. Tonic and phasic. Phasic and tonic.
- 5.79** Tonic is what? Overall background noise. That's tonic. Phasic is what? Baland sitting there on his computer, he gets an email, and it's somebody he likes, okay? He gets what? A phasic spike. He gets, "Wow, yeah, yes, that's great. That feels good." Phasic. And afterwards, dopamine has this pernicious nature to it, and I want to have more of that. I feel more excited. "So let me go for the chocolate now" and have a little bit of chocolate and eat chocolate and then do something else exciting. "Forget the work. I'm going to watch—" And then another spike. That is phasic.
- 5.80** Phasic is the spikes of dopamine, and tonic is the baseline background. In a moment, you'll see why this is all important because you can have different personality types.

You have people that are highly phasic, constantly—first of all, their tone might be high, but then they're riding on these phasic bursts all the time, so they need to have high levels of dopamine all the time; otherwise, they'll get depressed. They can't calm down, for example. So you have these two modes, phasic and tonic.

- 5.81** Now, you can have different types of personalities, fitting with the whole— your dopaminergic default wiring can vary from person to person. Have you seen some people, they're always excited? They seem to have a high baseline dopamine. Do you know these folks? They can't seem to relax. They're always happy; they're always this.
- 5.82** There seem to be two categories of people, actually. There are people with generally very low baseline, let's say. This is one category, very low baseline. These are boring people. I'm going to be frank. They're boring to be around. There's nothing happening. No, I'm sorry. Too much. I was going to say Germans, but we'd get in trouble. We're going to be seen by Germans, maybe. What's that? They know they're boring. Is that what you're saying? What about Italian? High dopamine baseline, right? Effusive, exactly. So put that down.
- 5.83** So you have basically a type of personality that's low baseline dopamine. In fact, if it's too constantly low, the baseline dopamine, you will have a danger of vulnerability to be more likely to fall into addictions. Because think about it, right, Alejandra? If the world is always boring, if things don't stimulate you, you get that text message, but you don't really get aroused by it, what happens then? You turn to what?
- 5.84** Drugs.
- 5.85** You turn to drugs. You need some dopamine, and then you really need high doses to get you aroused. So that's one version.
- 5.86** In fact, when it comes to addiction and addiction personality, there's another version. There's another personality type. And this personality, one, is the personality of the guy that has high phasic bursts. So the phasic firing tends to be very high. In fact, I tend to fall in this category. I have a problem with addiction. I have to be careful. I haven't had sugar since I was 25 years old. Believe me. Why? Because if I have that candy, it turns to two, three, four.
- 5.87** It started, actually, my dad had a pizza shop. Maybe I have mentioned this before. He had a pizza shop, like a pizzeria pizza shop, in Copenhagen. And I used to work there all the time. I wanted to read books, but then the customer would come, and I'd leave the book and go do a pizza and then do some burgers over here, some French fries. It was a very boring job. But anyway, sometimes interesting customers would come in, and we made some fun out of it. But I noticed whenever I had a Coca-Cola, we had the Coca-Cola machine—press it and it comes down. I had one; I couldn't stop. I had another one, I had a third one, and a fourth one. So I stopped drinking Coca-Cola altogether.

- 5.88** It's very important. You have to assess, what am I when it comes to dopamine? Am I low dopamine? Am I extremely low dopamine? Am I the one who is generally normal—this is my personality style, which is a very common one—it's generally normal, but then you have these peaks and salience peaks and phasic bursts when that girl comes or when this comes, and I just get very excited very easily. And this can also lead to addiction. This is also an addictive personality. So you have to know.
- 5.89** But generally, let's say if you have very high tone, and dopamine is very up all the time, there are ways to—this could be an artificial construct, meaning the following. Our society pushes us towards a more high-tonic state where we cannot tolerate boredom; we cannot tolerate sitting down, reading a book. We need the dopamine spikes all the time, right? We need that all the time, being fed with dopamine, which is a shame.
- 5.90** This is really where two personality types come in. This is a different type of personality. We'll get to that in a second. Giovanni, yesterday we talked about personality types and how that relates to dopamine. Remember that? Was it the gym? Remember we talked about type A and type B? Exactly.
- 5.91** And specifically, I was talking about how society today not only trains our dopamine centers to always want short-term reward. This is the social media. It is looking at this post, that post. Look, scroll your Insta feed, your TikTok, and you get rewards, short videos. "Oh, this is interesting. Oh, this is boring. This is interesting. This is boring." You keep scrolling.
- 5.92** Now, by the way, why is social media so damn addictive? What's interesting? What makes the TikTok and Instagram-type video so—the whole looking at your social media, why is it so addictive?
- 5.93** Constant reward.
- 5.94** What's that? Constant reward. Is that true? Is it constant? Okay, if I—hold on. I'm getting hallucinations over here. Getting all these sounds. If I was to look at my Instagram every single day and get 100 likes each time, I knew for sure. What happens over a week? One hundred likes each post. Would I get more addicted or less addicted to my social media? I'd get less. Why would I get less, Kelly? Why? I'll tell you. Because dopamine works best with intermittent reinforcement. Okay.
- 5.95** So if I get 10 first day—I had zero the first day. On day zero, I had zero. And then I make my social account, get 10 likes. Oh, my God, I feel happy. Next day, I come back, I look at it, I'm hoping to get 12. I get three. I feel a bit sad. It's annoying. Dopamine goes down. Literally, it goes down. Then the next day, I come back, I have 50. And then dopamine goes up. Then I have 50 again, maybe. Then I have 56, and then I have four again. And that keeps your dopamine addicted to the system. You want to have more all the time. You want to keep going with it.

- 5.96** So intermittent reinforcement is what gambling houses use. That keeps you hooked. You always want to go back. Or the way you text with a girl or whatever you do, if you do it intermittently, it'll keep the person hooked. You shouldn't do this, of course. You should be authentic yourself and all that. Of course, you may know some of these tricks as well, but it never hurts to know a little bit about human nature. But an intermittent schedule of reinforcement is very powerful.
- 5.97** But then I talked about, okay, so society is pulling our direction towards immediate gratification, immediate reward, dopaminergic spikes. Okay, recently I had this—remember the guy I told you about yesterday who did the interview? He interviewed me, the Kurdish guy, remember? He said, "Let's do an interview." He had this big list of all these questions, right? Now that guy, he told me—and by the way, I'm not publicly exposing him. This is a public interview, so you could also go back and listen to this Danish interview if you want.
- 5.98** He said to me, "Each morning I wake up and I check my bloody phone. I can't help it, but I have to. And then I check it and I scroll Instagram and TikTok for half—10 minutes, let's say, and then I have this crash afterwards. I feel bad." What is that about?
- 5.99** That was kind of more what I was thinking about originally. Other people's posts are the reward.
- 5.100** Yeah, that's something interesting there as well. But in his brain, what's happening in his brain? If there's a point of this lecture that you must remember, it's the following. Dopamine has this annoying arch. This is an annoying arch, but it's real. Whenever I have something good happen, let's say I have a great day. It was so fun. I had fun with my friends. I enjoyed it. I went to that restaurant. Everything was exciting. Everything was new. I loved it. I loved it. Dopamine on that day will be phasically very high. A lot of phasic bursts all over.
- 5.101** But then the next day, how do I feel? Tired? You don't want to move as much? Dopamine works in the following way—and this doesn't have to be from day to day. It could be a week of intensity and then a week of feeling tired and bored and less motivated. Unfortunately, dopamine, the way it works is that we have a limited capacity of dopamine. In other words, dopamine has to recharge. Dopaminergic centers will run out. We have a limited currency. Think of it as coins in your pocket. You only have so much of dopamine.
- 5.102** So if I run one week on holiday in Ibiza and just go freaking crazy, I'll come back and some people will say, "I felt down for a whole week. I didn't feel very motivated." That's because dopamine—you have a spike, spike, spike, spike, spike. It will lead to a tonic elevation, meaning the whole baseline is up now. You've used up all this dopamine. Now, you don't come back to baseline when you're back. You crash.
- 5.103** How much did you go up? Let's say on a scale, you went up this much. You must go down as much when you come back. So the more craziness you had at your party, the

less you will feel good when you come back. Isn't that annoying? That [can contribute to] postpartum depression and things like that. Once you have a child, you feel happy, excited. Then for a while, you can't really seem to find motivation.

- 5.104** And this tells us something crucially important. The way we use dopamine is we have to be careful about this. I don't want you to become too clinical and too scientific. I want to enjoy life, too, and be smooth and fluid about it, right? But at some point, knowing that, look, maybe I shouldn't go to parties three days in a row and just go crazy. We call it "in the sauce." Do you call it "in the sauce," too? It's like going crazy. You have rapper friends, they do stuff like that. They tell me. Yes, this means you don't want to do that. You want to know, okay, I did one, two days, and then maybe I should stay home. Maybe I should just go for a walk in the forest, leave my phone and just have a calm, restful dopamine recharge. What do you think? Dopamine recharge day, we call it in our world. This is good to know. It's actually very good to know.
- 5.105** But then Max will tell me, he says, "When you're scrolling all these bloody posts, you see all the Ferraris, see all the Lamborghinis, the big chains, you say, 'My God, I want one of those. It would be so cool to have that.'" It changes your personality almost. Does it change little girls' personalities? They get depressed, some of these girls. They see all this—not even girls, forget girls.
- 5.106** All of us want a beautiful car. There's something about a beautiful car. I think the brain looks at it and goes, "My God, not only is the car a chick magnet." You might go, you might say that. But it also likes the beauty of it. You've seen a Rolls-Royce, for God's sake? The interior, the leather, the steering wheel, everything's so bloody—they've spent time on it. You can see that these guys spent so much time on that car, and it's beautiful.
- 5.107** And I think we all like these things and we want to own them. It becomes almost—and this is where the danger comes in, and this is where, really, I want to go back to Giovanni's point, which is that this can become an extension of our egos. Society teaches us that owning these tools—this is very implicit, of course—but owning these things will make you greater. You'll have more girls; you'll be more successful in society, more societal status, more rank, and all that.
- 5.108** And it pushes people towards a personality type, which I really want to get into. This is what I call the compulsive personality type, the compulsive personality type. What is the compulsive personality type? This is the guy—and we all fall in this category; I don't want to be self-righteous over here. Many of us fall in this. And let me know if you know this; you can be honest and share. It's like group therapy. It's like, as I was saying yesterday, you can't stop. You just want to have more. You just want to strive and you want to go for it. What do you call it? There's a word for this. It's like—I forgot the word, but you just want to keep going and you just want to keep hustling. You want to keep hustling and you just want to—you can't stop.

- 5.109** And then we take a moment of rest. It's just a second of rest where you just want to have all your brain systems calm down, but you can't, right, Nikola? Because if you do that, then all the competitors, and then what happens to my business, then I won't be the greatest. I won't be the best, right?
- 5.110** Almost narcissistic in its nature. It's so self-absorbing. We talked about narcissism earlier. It has that almost narcissistic character to it. It's what I call a compulsive drive, very dopaminergic, Wolf of Wall Street drive. So it goes from one spike to the other. One spike up, one spike up. And then you go down and then, "My God, I can't deal with feeling bad, so I do even more cocaine, even more work, whatever I need to do that's got to get me to my goals." This personality type, we can all fall into it because society teaches us this is the way to be. You want to be like this because then you'll be great. Then you'll be the next CEO and the next great person.
- 5.111** But I much rather like the other personality type, which I call the purpose-driven personality type. This is your Maximus. This is the real Maximus because this guy has meaning and purpose. He has that balance Mr. Miyagi was talking about when he said, "Balance. Have balance." This balance is all about having a greater purpose in the way you're using your dopaminergic system and the way you're living life.
- 5.112** Being able to, first of all, tie your dopamine to effort versus external goals—you all write this down because this is very important: have a greater prefrontal contextualization of your life vis-à-vis dopamine. What is my life? What is my life all about? How will I feel 20 years from now when my mom is dead and I was spending my entire life chasing money? How did I fulfill my purpose? Why am I here on Earth? What is my bloody meaning? Religion is one thing. I don't want to go there because that's complex. But think about, what is your purpose on Earth? Why are you here?
- 5.113** If I met a person on the street and I said, "Hello, how are you doing?" "I'm doing fine. I'm doing well." I've met people like this. I said, "Where have you been?" They say, "I don't know." I say, "Where are you going?" They say, "I don't know." "What are you doing right here, right now?" They say, "I don't know." What would you call a person like this? Insane, right? That's what I say. I met a few people like this in my career.
- 5.114** Now, many of us, including myself sometimes, are just stuck in life, and we have no idea where we really were. We've forgotten the fact that we were kids. I was a kid in the playground in the hood. I didn't even have money to have bloody toys. I had no money, had nothing.
- 5.115** Now I'm here, feel like I'm all this and that, and I'm compulsive-driven, and I don't have time. My mom used to take care of me, sit with me for hours, teaching me cartoons and teaching me how to play and do things and about all these things. And now I'm a big shot, and I don't have time to sit with her and five minutes to show her an app, how you install this app? What is that about? You know this. You know what I'm talking

about? This is where the personality gets completely lost and dopamine circuits are completely attributed to the wrong thing.

- 5.116** And then future direction: what is my life about? And I think we should all—this is just me putting my own two cents in this whole lecture— service. Service is always good when you have service as the main driver, because service was given to you when you were in your diapers, you see. When you couldn't do anything, and you were sick, you had a fever in the hospital. You were like three, four years old. Your mom carried you. You had a fever. Who was serving you? You were being served.
- 5.117** So having some degree of service and having your dopaminergic circuits with that in mind in terms of future planning, and then also have a future plan in terms of how I'm going to live life, how I will distribute my dopamine. Will it be fully focused on my business only, or would I give dopamine to other things as well?
- 5.118** Dopamine is involved in learning, and each time you learn a new skill, you have dopamine involved in wiring that new skill. So let's think about a concrete example. If I was to learn, let's say, karate. I do karate. In the beginning, how am I doing? Poorly. And then I keep going at it, keep going at it, training my legs, up and kicking and all that. And then suddenly I do a great kick. What happens then? A little bit of a dopamine spike saying, "Wow, this is important." And then that will send signals from the nucleus accumbens and reward, goal-directed reward, down to the basal ganglia that has to do with repetition, rhythmic, habitual behavior, and wire that down. That's how it works, you see?
- 5.119** And then over time, that's where perfection comes in, in the movement. Could be dancing also, like I'm learning how to dance. Initially, very clumsy. You learn, you become better. And then slowly, you become more rhythmic, you become more smooth. All dopamine.
- 5.120** What movie do we think about when I mention this example of karate? What movie can you think about? A movie that illustrates this beautifully, how you can go from clumsiness to dopaminergic wiring, going from the nucleus accumbens, from goal-directed to habitual basal ganglia smoothness and repetitive rhythmic karate, let's say.
- 5.121** Have you seen The Matrix? It's in The Matrix. How is—what is his name? Neo, right? How is he in the beginning? He's clumsy. He falls. He can't do his karate. He jumps—Morpheus is training him. He jumps from roof to roof. Very bad. There's no sense of—he's very bad at it. He's very clumsy.
- 5.122** But then inside one of the simulations, he does one of the kicks. In fact, when he does one of the kicks, both of them, both Morpheus and Neo, can look at each other, and that's actually the moment perhaps he realizes this is the One. Or maybe he gets self-belief at that point, you see? Like, "I know I am the One at this point." This is the dopaminergic spike that comes.

- 5.123** And then when he enters the Matrix later on, he's now smooth. He's much more elegant. The agents come at him. How does he move? Smooth, right? The bullets, even. What about the bullets? How does he—what is that movement? What is that movement? Is that nucleus accumbens or is it basal ganglia? Is that the whole loop that goes from the nucleus accumbens and goal-directed behavior to the dorsal striatum, basal ganglia, automatic. Is that clear?
- 5.124** Yeah.
- 5.125** Very good.
- 5.126** Now we have a few minutes left, so let's go over to how you can improve your dopamine. The first point would just be to do a recontextualization of your life. Where have I been? Where am I going? And how do I want to plan my dopamine in the future? How do I want to live? Who am I as a person? What are my goals and what are my values? What are my values really like? Am I a superficial guy that's just going to go for the cars and the chicks? Or do I have something much deeper inside me, a more deep, meaningful purpose? And I want to live my life like this and have my dopamine drive based on this, based on this model. So you remodel your view of the world.
- 5.127** And then we said, attach dopamine to efforts, not spikes. Don't attach it to TikTok scrolling, external rewards like emails, praise, money, romantic uncertainty, let's say, but to actual effort you're doing in your life. You're moving towards your goal.
- 5.128** Then you restore the rhythm. Now, given the fact that you have a contextualization of your dopamine, you have meaning, values, you have it all written down, Juan. You know now, I'm not going to fall in the trap of becoming compulsive. See, the compulsive drive, it's not like it comes; you fall in the trap. You know what I mean? It's like a trap you fall into. And then a year, six months has gone by, you say, "My God, I've become compulsive." But in the beginning, you don't even know you're in the compulsive trap, for God's sake. It's like it is a trap. So you just become more aware of it, more cognizant.
- 5.129** Now, once you've become more cognizant, you restore the rhythm. Make sure you get sleep, exercise, you can relax more, lower cortisol, and so forth. And also you have the knowledge now in social media about intermittent reinforcement, gambling environments. You can deal with it better. You have that knowledge of dopamine and how it works.
- 5.130** Spirituality and religion can help; I think it goes with the first territory. So I'm not going to go too much into that.
- 5.131** Are there any substances we can take? Any drugs to boost dopamine?
- 5.132** Cocaine.
- 5.133** It all comes down to cocaine.

- 5.134** You're going too far now, okay? You're going too far now. We don't want to boost it so much. Here we just boost it a little bit, just to have an elevation in our baseline. What about caffeine? Who likes coffee? Caffeine, okay. You all like it. Good. It can, and the reason it can is that coffee will—
- 5.135** Now, okay, I remember that I mentioned the 90-minute bursts, right? The 90-minute cycles, ultradian cycles, where you work best. What happens? Why do you become tired after 90 minutes? It's because you have adenosine in the system. Adenosine is a chemical that builds up in your brain, and it's going to tickle all the places in the brain that make you tired and titillate them so you become more tired. That's adenosine.
- 5.136** Now, coffee has this function whereby it will latch onto the receptors for adenosine, the parking spots for adenosine, and block them so adenosine will not go inside the cells. So you don't become tired. That's why it also has that effect. It doesn't make you tired, and it sensitizes also the dopaminergic neurons. That's coffee. Coffee. So you may take it, of course. If you have too much high baseline of dopamine in your system, coffee can be agitating, make you frenzy-like. You don't want to do that either.
- 5.137** You can do cold exposure. This is interesting. So jump in a cold shower and boom, that could help. You get a lot of dopamine from that, especially since it can last for a few hours after that, so you elevate the whole state.

## Lecture 6: Recovery

At a glance — Recovery and sleep.

- 6.1** All right, so in this lecture, we'll be talking about recovery, sleep, how that is important for you. In recovery settings, let's say you have an addiction now. You developed, Rohana, an addiction; let's say—what could it be? You know, addictions are interesting. It doesn't have to be like, we see a shopping addiction. It could be. That's what I'm saying. It could be things like that.
- 6.2** I mean, we think about addiction as cocaine or social media, TikTok five hours a day. No, literally, you have people that are addicted to literary novels, or it could be addiction to movies, or it could be addiction to shopping, or it could be addiction to a man, or it could be addiction to a woman. It can be very strange.
- 6.3** Addiction merely means that your attention in your daily life goes toward that almost exclusively at the detriment of other things. You think about shopping 24 hours, and then I say, "Go to the gym or have fun with your family." You don't want to go. You feel like everything is flat. It's only that source that's interesting. You see?
- 6.4** When we go to treatment for addiction, let's say we have addiction, or forget treatment, you guys are all healthy, right? But let's say you don't have any addictions, but you have addiction tendencies. How many days, Kelly, would be enough to rewire your brain and make you feel okay again, less addicted?
- 6.5** Thirty days.
- 6.6** You're good. How do you know?
- 6.7** It's just a guess.
- 6.8** Really? Can be. Is that a guess? Exactly 30 days. Put that down: 30 days. This can't be randomness. There must be something going on. It's about 30 days.
- 6.9** So initially, you have a lot of—so let's say you have an addictive habit. It could be going to a shopping center. You just want to go and run amok and buy stuff. Okay, now you want to give up that habit over here. Okay, Rohana, what do you do? You have to, initially, find alternative sources of pleasure and enjoyment. This could be listening to music you liked in the past, okay? It could be going for a walk, meeting your family. Initially, it may not feel good, but you'll have to do it. You develop these alternative sources of dopamine, right?
- 6.10** In the first 10 to 14 days, the first couple of weeks, Giovanni, you will feel a bit of agitation, anxiety. Your amygdala will be activated. So what do you do about that? You abstain. You still abstain. You do not go to your drug of choice. You do not go to the mall now. You do not go to the horse racing or whatever you like. You stay away from that and do your other activities. Is that clear?

- 6.11** You want to have sunlight, exercise, sleep, and in approximately 30 days, your gaming addiction, Alexander—not you personally—it will be better. Is that clear? The craving and urge to go and play that PlayStation game will go away. All right. So you wanted to have that.
- 6.12** And of course, when you are stressed, whenever you feel stressed, whenever you feel like your life is hectic, stressed, deadlines—let's see, 30 days went by, you're good now. You don't want to go to the shopping mall. But then two months later, three months later, everything is chaotic in your life. Everything is hectic. Deadlines. Everything is just—emails are piling up. You can't just keep on track. Then what? Are you more or less likely to fall into that addiction? More likely. Why? Because the prefrontal cortex shuts down, right? Amygdala becomes more highly activated. And then anything in the reward circuitry of the nucleus accumbens is more free to reign and make you more impulsive. So that's when you have to be careful because stress can lead you to fall into these addictive traps. So reduce stress, and then that will help.
- 6.13** Vitamin D. If you are deficient in vitamin D, in Arizona, that is difficult to be. But sometimes if you stay indoors a lot, you could still be deficient. Do you know that? I was in Kuwait recently. That was before the whole war, Iran and all that. I was in Kuwait. It was actually about a month before that war. Can you believe it? I went on a podcast; they invited me. I went over there, and I was in two minds about going there. It's far away. How would I do in Kuwait? Anyway, they persuaded me. I went over there. I had fun.
- 6.14** And then they said, "We'll have you back for a second podcast in two months." That's exactly how I was about to go. And then the war happened. See, life sometimes—it's good. I was safe. I didn't go. But they told me, "Look, a lot of us Kuwaitian people, we have vitamin D deficiency." Can you believe this? I mean, Kuwait.
- 6.15** Have you been in Kuwait before? Have you been in the Middle East before? Have you been in Dubai, Abu Dhabi, all these countries in the Gulf? It's hot as you can imagine. The sun is shining. It's very hard to be vitamin D deficient over there. But yet you can be if you're indoors, especially if you have dark skin like me, Juan. It's problematic because we are less sensitive. So we need to have more sun, in fact, to have vitamin D synthesis in the skin. So do check your vitamin D levels. And yes, if they're low, they can actually affect dopaminergic sensitivity in the nucleus accumbens. They can affect that whole cascade.
- 6.16** Vitamin B. Can you believe it? It's involved in synthesis of dopamine. Check your B vitamins, especially if you're vegetarians. Make sure you have B6, B9, B12. I've been having a lot of oysters at my hotel. Do you like oysters?
- 6.17** Quite.
- 6.18** Do you like them?

- 6.19 Yeah.
- 6.20 They're bloody nasty, aren't they? They're not very good. They taste like a salty thing. I looked at nutrient density, though. It's very dense in nutrients, micronutrients. So I thought, okay, I'm not having a lot of meat over here and stuff. I eat a lot of fish. So I thought, let me have something very dense. So I tried these, one or two of them. I didn't like it. Not very good. Anyway, then I went back to the hotel room, and I read more about this, Rachel, all the benefits like zinc for testosterone and you're strong and have stamina. It's good. It's good for you, right? And then, B12, B9, all these vitamins and my dopamine. I thought, "Okay, look, the hell with it. I'm just going to have some of these bloody oysters." I went back to the hotel restaurant and ordered six more.
- 6.21 And then it's funny. You know acquired taste? What acquired taste is? Acquired taste is that initially you find something nasty and disgusting, and then you have more of it, and then you're tied to dopamine, and it tastes better and better and better. Now it tastes okay. I'm not saying it's great, but it's okay now. I can go with it, like four days in. Okay, vitamin B.
- 6.22 Omega-3 fatty acids. And I keep mentioning omega-3s and the fish oil, but it's so important. I've mentioned in my previous courses, perhaps in this course even, that it can compete with serotonergic agents for depression even. I mean, this is something that's not been talked about a lot, and the studies are so-so, but it's interesting. Fish oil, omega-3 fatty acids can be almost as good as SSRIs, perhaps. I'm not saying conclusively, but definitely, it's much less side effects for sure.
- 6.23 I don't need to mention this, but high cortisol directly affects dopamine synthesis. So be careful with that. And that would be my final thoughts on dopamine for now. Okay.
- 6.24 Let's go over to recovery. What happens when you're over, let's say you do something like me giving a lecture right now. Okay, one hour of me talking left and right. What happens in my brain? Cortisol goes up or down, Max?
- 6.25 Up.
- 6.26 Skyrockets. Cortisol is not a bad thing. It's a good thing. If I was in lecture mode 24 hours, we'd be having a problem, right? But for one hour, it's important.
- 6.27 What about dopamine? I'm paying attention. I'm learning new things along with you. I have goal-directed behavior. It goes up or down?
- 6.28 Up.
- 6.29 Noradrenaline goes up or down? Prefrontal cortex, highly activated, right? My point is that there's a lot of neural computation. It's very costly to be in a high state like I am right now. It's very costly. And so you have to have a recalibration of the system. It's very important to go up, but then also go down, understanding the importance of taking breaks. In the breaks, you meditate, you relax. And in the days between the

lectures, you relax. In the days between your, like, Alexander's board meeting, you rest. You have this restful state.

- 6.30** Why is that important? Because you have to replenish all those neurotransmitters, those chemicals that prefrontal fatigue will accumulate over time and over time, and you will just not be able to compete, not be able to—your speech will be affected, your cognition will be affected, your flexibility will be affected. So that is very important. And I think that will be a key point of this.
- 6.31** You see this with elite performers. Soccer players—have you seen FC Barcelona? Do you like FC Barcelona, my friend? Not really? I love them. I love them so much that I don't even watch their matches only. I also watch them on their YouTube channel. You can watch them when they're exercising in the weekdays. And you see them in their weekdays when they're exercising, they're just throwing the ball around. They're just walking around, laughing a little bit. It's like tiki-taka. You see tiki-taka when they play it? They have one in the middle, and then they have the ball and play it around. Very calm, very—that is the whole point that they have the match on the weekend, but then on the weekdays, it's relaxation. It's not relaxation completely, and that's also another important point, right, Juan? Let's go into that now.
- 6.32** If you get to a burnout stage, if you are burned out, let's say you had an exam. It was hectic. It was important for your life, so you were very stressed about it. For one month, you had to prepare, and everything was hectic. And then the moment that was done, the next day, you got the news that—give me some news you got. What news did you get, Nikola? Stressful news. Your flight got canceled for another thing you had, and now you have to completely change your schedule for the next three weeks as well. Then you're hyper-stressed. And then comes another thing. And then because of all this stress, your wife is angry at you, let's say. She gets angry at you, and she doesn't want to be around you anymore. She's getting a separation. I know, this is an extreme example, right? "I want to get separated, Juan. You're a workaholic. I don't want to be around you." So she leaves.
- 6.33** Now you get even more stressed. And then you get into the mindset, "Oh, my God, what happens? My wife is leaving me." And then you get into the compulsive mode. I have to get myself together. I have to get more money. I have to show her that I'm the man. I'm the cool guy. Then you get even more money, more motivated and compulsive. And this drive keeps going month after month, and you're in the cycle.
- 6.34** And I tell you, Juan—what do you call this under your eyes? These patches, they look very dark. I'm not you personally. You look good, but I'm saying the Juan we're imagining now, this is very dark. His skin is very dry. Why does he have dry skin? Because when you have elevated cortisol, you don't produce much—all this digestion, what do you call it, in the skin? The moisture in the skin goes down. You start to feel like you got a few wrinkles, my friend. What's happening, right? And you're just seeing all the fatigue, all the fatigue.

- 6.35** And I tell you, Juan, you look maybe stressed, you're acting a bit agitated, and you will not believe me. You keep going. You keep going in that cycle of stress and so forth. What was the point of all this, though?
- 6.36** Burnout.
- 6.37** Burn out, I know, but before then, there was another deep point. Oh, yes, yes, my friend. That's true. Okay. Sorry. The last lecture will always be a bit—
- 6.38** The question is now, finally, you come to me, and you say, "Well, you're right. I feel the exhaustion. I feel that I have come to the burnout stage. I'm burned out. I feel tired. What do I do?" And that's when I will give you—now you can listen to my advice, now you're open. And you're asking me, should I just go home on the couch now and just lay on the couch like a potato and watch soccer all day?
- 6.39** Is that the way to do it? Why not? Because you're actually saying we shouldn't do that. It could be a good idea, couldn't it? But it's a bold claim. You're saying we shouldn't. It's a bold claim, so you have to back it up. Because intuitively, if you have six months of just going at it, you want to replenish the system, you want to rest the system, right? So why don't you do the couch thing, Rachel?
- 6.40** For a day.
- 6.41** For a day, right? What about you over here? What about you? Rohana, Alejandro, do the couch. How many days would you do the couch now? About how long?
- 6.42** Just a weekend.
- 6.43** A weekend? But you're going at it for six months; you're burned out. You're maybe depressed even. See, what happens over time when I told you about the whole flip of dopamine. By the way, whenever you have cortisol, you also have dopamine. Cortisol and dopamine are friends. I know it sounds odd, but dopamine has many functions, right? And so you also have tons. In fact, cortisol has this function of depleting dopamine. It depletes it completely. The prefrontal now, my friend, is thin. Your amygdala is hyperactivated. It's become larger in size. So you have a lot of anxiety and fear, perhaps—I'm not picking on you, I'm saying, right?
- 6.44** And then your hippocampus, your memory part of the brain is also off. Your memory is a little bit—whenever you're stressed, have you noticed your memory gets really poor? Yeah. That's also because the receptors of cortisol can latch onto the hippocampus and really shrink it over time. It's interesting how tissue can grow larger and smaller. Yeah, it's interesting.
- 6.45** Okay, my friend, what do we do? Should you lay on the couch maybe one day, two days? I would say, okay, for a few days, it's okay. But then I wouldn't continue to do that because you are not merely somebody who's recovering after a couple of weeks of stress. He's had six months of intense stress, Kelly. We've got to do something more strategically optimal to get him back on track. So I would say, give it a couple of days.

Maybe give it a few days of rest on the couch, but then we have to get you back on track. Do things, but do things in a gradual way.

- 6.46** I would recommend that you do a daily walking thing, get exercise in. I would perhaps—if it's really bad—I would maybe leave my job for a couple of weeks, take a couple of weeks off, maybe two or three weeks, depending on how severe it is.
- 6.47** I would check you for mental health now. I would. I would do a systematic assessment, clinical assessment. Do you have anxiety, depression as a result of this? Was that triggered? Let's say it wasn't. I'll just give you that.
- 6.48** Then I would say, "Keep things moderate. Don't slouch on the couch." If you do so, you can end up in depression. That's the whole point of all this. If you do have a long period of burnout, and then you say, "Oh, my God, I'm going to crash," you can end up in a very depressive situation. Yeah, you can. I actually had something similar happen to me. It was about a few years ago. I went intense for six months, and then I thought, "Oh, my God, I'm just going to crash for one month. I'm just going to do nothing." So look at the wall. Don't look at the wall. It's not very good. It gets very boring very easily. And you feel like, "I should do a little bit more." And you feel a little bit—and then you start doing a bit more and you feel great. So really, I recommend—this is what the science says—have moderate exposure. Very important.
- 6.49** This is something that I've been thinking. Did I mention this before, the importance of walking? I think I mentioned Einstein, right? Why is walking so important? I think walking might be one of the most underrated things we can do as human beings. First of all, when you walk, you have something called anticipatory proliferation, which is a fancy term for the fact that human beings, whenever they walk, especially very long distances, it's a sign that they're moving to new territory.
- 6.50** Otherwise, why would you walk, Debbie? What's the whole purpose of walking? To walk a whole hour, an hour of walking, that must mean that I'm exploring new land, new landscapes. And so your brain is clever. It says, "Oh, let's ramp up the dopamine. Let's get the dopaminergic circuits fired up." Because you want to have new learning to go with that. So your dopaminergic centers go up in the good sense, in the tonic baseline sense, which is very good.
- 6.51** And then when you walk, it seems like you have, first of all, the prefrontal cortex is not completely shut down, but it's not highly activated either. So it's that mixed mode that's very healthy for you. You have increased mind wandering and imagination. So you do have the default mode network, but the default mode network is not so triggered by the amygdala. What is the amygdala?
- 6.52** Fear centers.
- 6.53** Fear centers of the brain. The amygdala will not change the default mode network. See, the default mode network is self-rumination and all that. But it's also mind wandering

and daydreaming. It doesn't have to be self-critical. I mean, it's only depending on your predisposition and how your amygdala is biasing the default mode network.

- 6.54** But whenever you're walking and you look at the trees, it's greenery; maybe there's a sea, your dopaminergic centers go down. In fact, whenever you are walking like this, you're engaging in approach behavior. And approach behavior is what? What's approach behavior?
- 6.55** You have two types of behavior: approach and?
- 6.56** Retreat.
- 6.57** Right. Approach and retreat. In the approach behavior, whenever you are moving directly towards an object or just walking, your brain thinks, "I am confidently moving towards my prey." In other words, you are not being hunted. Does that make sense? This is a hardwired algorithm. It's not like you're consciously thinking about this, Max. You're not going, "I'm chasing my prey out here in the forest, the desert of Arizona," right? You're not doing that, right? You're just walking. But your brain knows you're moving forward in space. That means I'm not being chased right now. I am consciously moving. I have agency. And that calms down the amygdala, and it allows the whole default mode network to work in a way where you are mind-wandering, but in a healthy way. You have creative output. You have lots of—of course, you have blood flow when you're walking. Another key point.
- 6.58** All right, so one of the key aspects of the whole recharge cycle—I'm telling Juan here, I say, "Now you have to take care of your sleep." Sleep is so important, right? But why is it important exactly? We're going to that now. And I will summarize—I have been talking about this in my sleep lectures—but you have two types of sleep: REM and non-REM. Have you all watched my sleep lecture? Okay, good. So this will be a summary. In science, we say repetition is always the key to learning. In fact, you want to repeat something about six times in a lecture to have optimal learning. I do it about 16 times, but still better than nothing.
- 6.59** Now, REM and non-REM. Non-REM is the initial stage. As you enter sleep, you go to bed. First of all, Alexander here, he will say, "What person am I? Am I somebody who needs to sleep?" What do you call it? Night owl? What is it? Night owl or?
- 6.60** Early bird.
- 6.61** Early bird? What do you call the middle person? Middle bird. Okay. Which one are you? This is genetically determined. Did you know that? There's a genetic component to that. Why? I'll tell you. Because when you used to live as humans in caves, it would be really good to have one that could be awake a little bit later to look out for the monsters and all that. Sorry, not monsters, but predators: tigers, saber-toothed tigers, lions. You have one that's awake, the other group is sleeping, and then you have that

throughout the night, ideally. So there is that. So it's a genetic component. So you, Alexander, want to know, which one am I?

**6.62** Night, I think.

**6.63** You're a night?

**6.64** Yeah.

**6.65** Good. And it can be tricky because for a long time, I thought I was a night owl. For years. I want to say for 30 years, I thought I was a night owl. I always slept at 2:00, 3:00 a.m. This is how all my friends slept. But then I turned 30. I thought, "Let me try to sleep early." And it felt so good. It felt so good. And then I kept doing it. So it's interesting. We can be tricked. It felt good before, too, I thought, but obviously, it didn't feel that good. Anyway, so you want to know how you fall in that group, which group you fall into.

**6.66** Then you enter sleep, non-REM first. Stage one, two, and three, heart rate, blood pressure, all that falls. Lower the temperature of your room. Who mentioned this today, by the way? Somebody mentioned that you have to have a low temperature. Oh, I know who mentioned this. It was a makeup artist. She mentioned this. It's very important. You want to have a low temperature when you go to sleep.

**6.67** I noticed this last night in the hotel. I went to bed, right? Usually when I go to bed here at the hotel, I especially notice this. The temperature, I make sure that the room is cold. Tonight, I went to bed and I was like, "Oh, I can't sleep. There's got to be something up. That's weird. It's been five, 10 minutes, 15 minutes." And I couldn't figure out what it was.

**6.68** Am I nervous? Oh, my God. Am I back to nervousness again? I can't be nervous. This can't be. What is it? I couldn't figure out what it was. And then I knew: it was the heat. Press the thing, 18.5 Celsius. Write that down—18.5, exactly. My God, I went to bed. Two minutes, I was sleeping like a rock. Can you believe this?

**6.69** Anyway, I went to deep sleep. My body temperature fell. I felt rested. My cardiovascular system, everything just went down. And then I entered the stage of deep sleep. I cycle through these stages. I have deep sleep, sleep. And in deep sleep—that's so important, deep sleep, and you know this. Cardiovascular function. Why is cardiovascular function so important? It's healed during deep sleep, my friend. But why is it so important? Why is it so important to have deep sleep so you can recharge your cardiovascular function? Do you want to have high blood pressure? Do you want to have strokes?

**6.70** No.

**6.71** That's all in deep sleep. It's literally like if somebody is in deep sleep and you wake him up, he'd be extremely groggy, and he'd be extremely tired. And that is because deep sleep takes you into that deep chamber of sleep, and so you have cardiovascular

healing. Skin. Do you want to look pretty for the lecture? You want to look good, right? Good. Healing of hair, skin, all that. Deep sleep.

- 6.72** What else do we have? Do you want to get sick all the time? Do you want to feel like you always have a cold, all these illnesses? [Immune function] and repair in deep sleep. Deep sleep. Do you want to get Alzheimer's? Presumably not. Also deep sleep. In deep sleep, you have something called glial cells. They shrink and they allow cerebrospinal fluid to bathe your brain and clean up toxins like beta-amyloid and plaques and tau proteins and all that. These are misfolded proteins that accumulate over your life and potentially lead to dementia.
- 6.73** Now, there's a genetic component to dementia, too. And I would actually—this is a personal recommendation. You don't have to do it. But you can look, does anybody in your family have dementia? For Alzheimer's, there's a genetic component that's actually a big genetic component. Big one. I have in my family, my grandmother, she had Alzheimer's. She got it after she was in Iraq. One day—she was this person. She used to walk every day for two hours, three hours. Beautiful woman. Everybody loved her. Social, beautiful, kind. Everybody loved her.
- 6.74** And one day she was walking. Boom, a car hit her. And the car was Saddam Hussein's—you know Saddam Hussein? Have you heard of him? His army, one of the cars hit her. Now, people say it was on purpose, probably on purpose. They didn't like the Kurds that much. Anyway, hit her, bumped her head. All the neural inflammation probably triggered her Alzheimer's. Within a few years, she had severe Alzheimer's.
- 6.75** Now, this tells me that in my genetics, there's a problem. She had severe—it's very severe the Alzheimer's. It probably runs in my genes. There's different genetic variants you can have for Alzheimer's, but it's very strong. You can get your genetics checked or maybe just look in your family, whether you have a family line of Alzheimer's. Deep sleep—that's all in deep sleep, where you do that cleansing for that.
- 6.76** What else do we have in deep sleep?
- 6.77** Is that synaptic pruning?
- 6.78** That's right, my friend. How do you know? Thank you. Exactly. Memory consolidation is also important in this stage. In other words, deep sleep is like the librarian that takes the information during the day. So Joe had all this information that he hopefully learned from this course, from this lecture. He's going home now. There's this librarian in deep sleep that takes that information and puts it in various spots in your brain.
- 6.79** And that happens with these—there's something called sleep spindle. So if I look at the waves of deep sleep, they go like this, like this, this, this, this. And then you have these spindles that take information from the hippocampus, the memory part of the brain, and places it in the vault, the outside vault, which is called the cortex. It happens [especially] in deep sleep.

- 6.80 And then you have synaptic pruning. Excess connections that you want to get rid of get chopped away. So the lecture was good, but the portion on dopamine over here wasn't that good. Chop it away. This portion wasn't that good. This was important. This is key for my life. I want to keep it. That happens. Synaptic pruning: pruning away whatever you didn't need, the unnecessary stuff and clutter. Get rid of that. That's deep sleep.
- 6.81 Then you enter?
- 6.82 REM.
- 6.83 REM sleep. REM sleep. During REM sleep, the neurons go crazy. They become agitated. If you were to look inside the skull of somebody in REM sleep and look at the neurons, they would be indistinguishable from a waking person. This is called beta waves. It's like these frantic, excited neurons that show you that so many places of the brain are activated at the same time. This is REM sleep. Your heart rate, respiration goes up, blood pressure, everything goes up, and you have vivid and lifelike dreams. This is REM sleep. Crucially important, that's why you want to have REM sleep. It's so important for neuroplasticity. And also having—
- 6.84 You know that knowledge I talked about before with Joe? We had that knowledge that the librarian put in all these boxes and all these different cabinets on these different shelves. That's all fine and good. But you want to have some wisdom extracted. How can I take all this information and weave it into the tapestry of my life? How can I make this meaningful? What does it actually mean in relation to me and my place in the world? This is where wisdom comes about, and that is in REM sleep. And you also have neuroplasticity as a—it's an extension of that. So the neurons are saying, "Okay, I want to connect more with these neurons over here, and I want to connect more with these," and then you build plasticity. So you have REM.
- 6.85 And I'll get into the dreaming aspect and what that is important for, mental health and all that. But before that, think about, roughly speaking, of deep sleep as important for physical health and REM sleep important for cognitive and emotional health. So you have things like working memory. We talked about that, how that is important for attention and all that and focus. And that's all REM sleep. So you have people do cognitive tasks, IQ tests. They have very little REM. They perform very poorly. That's REM. Very good.
- 6.86 Now, Alexander here is being skeptical and says, "How does this matter to me? How can I use this information in my life?" And I would say, do you know that REM sleep comes later in the night and deep, predominantly more early in the night? Meaning, in the first couple of hours of sleep—so you go through these 90-minute cycles. In the first cycle, let's say, you have 90 minutes and you have about half an hour, an hour of deep sleep, and you have a couple of minutes of REM. Meaning you have a lot of physical restoration and recharging, but very little emotional and cognitive recharging. Does that make sense?

- 6.87** REM, as we continue throughout the night, REM gets more and more and more and more. It expands. You get more and more REM. So, in the morning, Alejandra, you have tons of REM sleep. You have maybe an hour. Before, you had a couple of minutes in the first block. In the final block, at 7:00 a.m., you have tons of REM sleep.
- 6.88** Now, this is where the actual practical implications are useful. What happens if you—like me, I have to catch a plane and come here to Arizona to give this lecture. I wake up at 5:00 a.m. This is good because then I had a lot of deep sleep and I also had REM. But how much REM did I lose right now by waking up early?
- 6.89** A disproportionate amount.
- 6.90** A disproportionate amount, right? I didn't merely lose 25%. If I had 25% shaved off my total sleep time, I didn't lose 25% of my REM. I lost 75% of my REM now. Will I be able to give a good lecture now? Will I be sharp? Will I have cognition? Emotion?
- 6.91** Probably not.
- 6.92** Right? So it's very important to know the difference. But it also is important to know in my life, let's say, who likes sports? Who's an athlete over here? It's got to be an athlete. Giovanni, you're an athlete. What sport?
- 6.93** Golf and basketball.
- 6.94** Golf is a bit slow. Let's do basketball, okay? Now, as a basketball player, I'd say, "Look, you got your deep sleep in. Maybe you're okay." I'm not saying you should always do this, but at least now you can sprint. Now you can do your physical exhaustion exercises and you'll be fine, right?
- 6.95** So knowing how REM and deep sleep are distributed is crucial because you also know when—so if I'm giving a lecture, I would be very critical and get REM. But sometimes if there is leeway, if there's some way to get out of it, it's for physical exercise, you'd probably be okay if you have at least good deep sleep.
- 6.96** Here's another note. REM is critical for emotional well-being. Emotional well-being is REM, is REM sleep. So much so, did you know people with depression? People with depression will cheat sleep itself in order to have emotional healing. So I mentioned in the first block of sleep, REM sleep is about a couple of minutes in most of us, right? But if you have depression, you will skip some of that deep sleep and expand the REM stage and have more of it early at the cost of deep sleep. Can you believe this? So you push away deep sleep and say, "The hell with it, I need REM sleep," and you go for that. This is one of the symptoms.
- 6.97** I learned this early on in my career. I was intrigued. I saw all these graphs and I saw all these papers, and I said, "My God, when you have depression, you're just prioritizing REM," so much so that you don't care about deep sleep and all the cardiovascular healing and all the metabolic effects of deep sleep.

- 6.98** And what's the tragedy of that when you think about it? Somebody with depression or somebody who's semi-depressed, are they physically worn out? Are they tired? They can. So there's a cost to it, even though they get a little bit of benefit, right? But there's a huge cost, which is?
- 6.99** Physical.
- 6.100** Physical well-being. Absolutely.
- 6.101** Dreams. What are dreams about? And what are dreams good for? I'll tell you now. Dreams are the stage of sleep that happens in REM. You are paralyzed from head to toe during the REM stage. You know this. I know because you've been in my lectures. Glycine and GABA are completely paralyzing you during REM sleep so you can't move or you can't speak. And this is when you have REM sleep and you have REM dreams, vivid and lifelike, crisp dreams, and you find yourself on the moon having tea with the queen. Everything is strange. People, times, places, everything is warped.
- 6.102** Now, Giovanni is asking, I think you're asking, why is this so strange? Why is this so bizarre, so emotionally arousing, titillating, exciting? And I'll tell you why. In REM sleep, it seems like, first of all, our emotions are being recalibrated and reconfigured such that whatever we experience in our daily lives, let's say you have a dealing with a boss that was hectic and hard on you, let's say. What else could be? Give me a personal example.
- 6.103** For a stressful situation?
- 6.104** Yes.
- 6.105** Can't get to sleep.
- 6.106** Chad, you can't get to sleep, right? Annoying. Even more severe. Give me something more severe than that. Let's say you were in a car, like a semi-car crash, not traumatic, but a semi-car crash. Now you want to take that information and dream about it. Because when you dream about it, in the REM stage of sleep, first of all, you symbolically represent the actual event, so you won't see yourself in the car crashing. That only happens in PTSD. That's a completely separate lecture. I have that in a separate lecture.
- 6.107** But if you're just a non-PTSD-like person and you have a car crash and you want to process that, you bring it up as a symbol in REM, something that reminds you and evokes that emotion. So you have that emotion, and that experience is symbolically represented. You are in this scenario on the moon, and things are being unstable. Let's say gravity is off. I mean, dreams are so strange that anything could be happening. The point is, you know that it's that event that's anchoring the whole dream. You know that emotionally. You know what I'm saying? You know that, but you have these symbols to represent that.
- 6.108** But here's the key. That event is being processed in the absence of noradrenaline, the chemical cousin of adrenaline in the body that makes you aroused. Now, when that is

gone, and you experience that emotion without chemical agitation, what happens then? You take out the sting and the charge of that event.

- 6.109** It's like me going to an arousing event. Let's say I have this—what's really arousing that's going to make you agitated? I'm trying to think of an analogy. Okay, I'm going to just say it. You're going to go to see that girl that makes you very anxious, okay? You go over to her, but before you go, you take this shot over here. I block your noradrenaline completely. So you will go completely calm. You'll become like a psychopath, completely calm. You go over to her and you talk to her and you have a chat with her, and you find yourself, "Oh, my heart is not racing. I'm not agitated and out of it. I'm calm."
- 6.110** What happens then? Your brain learns that the next time I'll be calm again. I'll learn plastically—REM is involved in plasticity. I learned now that this whole thing is not that dangerous. It won't make me agitated and anxious in the future. That's the role of REM sleep. You see? You put that down. That is the role of REM. And so it's so critical for us.
- 6.111** Creativity and REM, we talked about creativity. I just want to mention quickly, because of the absence of noradrenaline and the activation of the creativity network, the default mode network is not only the self-rumination network, it's also the creativity network. If it's used and not biased by the amygdala in dreams, for example, if you don't have nightmares, it's actually a creativity machine because it leads you to link all these concepts in the world in a novel and associative way. Unrelated patterns are connected more easily, and that happens during REM sleep.
- 6.112** In fact, if I was to wake Juan up from REM sleep, I say, Juan, "sun" and "moon," I pair these two words, and I say "sun" and "chair," you'd be faster to detect "sun" and "chair." But if I do that during wakefulness, which one are you more fast at detecting? What pair, "sun" and "moon" or "sun" and "chair"?
- 6.113** Sun and moon.
- 6.114** Yes. You're following me? I just want to make sure that I'm not rambling. Okay.
- 6.115** Rachel, you're sleeping in REM sleep. Now, if I wake you up from REM, in the seconds after REM sleep, you will have remnants of REM chemicals in your brain present. So you'll be a semi-REM person, even when you're woken, seconds after REM. So you'll be a semi-REM person. Now I say, Rachel, I show you two words. I show you two pairs of words. First pair, "sun" and "moon," and I detect how fast you are at pairing these, "sun" and "moon." I say, "Okay, you're relatively fast, but not that fast." And then I say, "sun" and "floor." You'll be much faster at the latter. But you should be much faster at "sun" and "moon."
- 6.116** Why? Because these are semantically related. If I were to do the experiment two hours later and ask you the same question, you'd be much faster at "sun" and "moon." The reason is because you lack noradrenaline. Noradrenaline is the chemical that makes us

focused. You become less focused. You become more associative, more creative. The default mode network makes you more creative. And that's the whole point.

- 6.117** See, if I was to wake you up from REM and show you random anagrams like "oseog"—it's just five [letters], but actually it's "goose"—and ask you to solve this, you're much faster when you wake up from REM. You're more creative. You have these gist ideas and creativity. All happens during REM. So REM, if you want to be more creative, have more wisdom, have more emotional focus, and just be more overall, less rigid, you want to have a lot of REM sleep.
- 6.118** And dreams. Dreams are good. In fact, dreaming about the events is so good that if I put Max, if I have you, Max—there's this experiment. You have rats running around this maze. Forget rats. Human beings, okay? Video game. They play a video game where they are finding themselves going out and in, out of these labyrinths, these mazes, just like rats would do, okay? Now, it's a human playing a video game. How good are you at this? How good are you? Depends on how much REM you've had.
- 6.119** If you do the game and then two hours later ask you, "Do it again," you haven't improved much. If you have REM sleep, you'll be much better at this. You solve the exit much faster. It's not about fatigue. You've had plenty of sleep and rest. You'd be better at solving the maze. But here's the punchline: if you dream about the maze itself, you'd be 10 times better at solving it. Can you believe this? This is very interesting.
- 6.120** In fact, Giovanni, on that point, I don't want to go too far. I have to cover a lot. We may go over time for this one, but people that are depressed, that have depression—excuse me, they don't have depression; they go through a divorce. They go through a divorce, which is a highly traumatizing, adverse event, they will be much less likely to develop depression if they dream about their ex in the months after. Can you believe this?
- 6.121** It shows you how much coping is done during REM sleep. So you have a divorce, and for some reason, you do not dream about your ex, what happens? Likely, depression a year later. But the fortunate of us who dream about the ex, what happens? Less depression.
- 6.122** Anyway, let's go to the sleep tips. I have a lot of tips I have to cover. I can go on and on about sleep. So how can we improve our sleep? We mentioned a few things. What did we mention already? Remind me. Temperature of the room. What else?
- 6.123** Light.
- 6.124** Did we? Yes. Light is important. The lighting of the room and overall exposure to light in the evening. But why is that important?
- 6.125** Circadian rhythm.
- 6.126** Circadian rhythm. What is a circadian rhythm? You want to go there? Let's go there, my friend. I wasn't planning to.

- 6.127** It has to do with light and being exposed to—something to do with melatonin?
- 6.128** Yes, it inhibits melatonin production. So in the suprachiasmatic nucleus of the hypothalamus. Sounds so geeky. Anyway, in the suprachiasmatic nucleus of the hypothalamus, you have a little clock there that times whenever you want to sleep and when you want to be awake. Is that clear?
- 6.129** And it's interesting. Literally, it will tell melatonin, "Now it's evening. You feel tired, Kelly. You want to feel tired, you want to go to sleep." And anything you do that's overwhelming in the system can mess with that clock and make you tired at different times and mess up the rhythm of whenever you want to feel awake and whenever you want to feel tired. Is that clear?
- 6.130** So that could be light. So light hits the optic nerve, goes into the [suprachiasmatic nucleus], and then hits the pineal gland, and then that will affect, ultimately, melatonin production. So light is one cue. But these are called zeitgebers, "time-givers" in German. And it's not only light. Can you believe it?
- 6.131** Having a big meal like I had at the restaurant yesterday can also affect your circadian rhythm. It can. If you don't have big meals, usually like me, I don't have big meals at night. I usually eat four hours before I sleep. Do you know that? I give it four hours. So I usually sleep around 10:00 p.m. I sleep. From 6:00 p.m., I do not eat. Yesterday, I had the big meal at what? At 7:00, 8:00? Slept at 9:00, 10:00? Not good. Be careful.
- 6.132** Zeitgebers. What else? Social interactions. If you suddenly start calling your friends and going out and having, not even parties, but just engaging in social interactions in the evening, that will affect your circadian rhythm and your sleep will be off. So that's another crucial aspect of this circadian clock and timing of sleep.
- 6.133** How can we reset it in the morning? Is there anything we can do? So cortisol—forget cortisol for now. How can you practically reset your circadian clock in the morning, make sure that it's set such that it is ready for evening?
- 6.134** Sleep and wake up every day with direct sunlight?
- 6.135** Okay. Sleeping the same time and waking up the same time. Studies have been done on this in Boston, showing that this is better than having 7 hours one night, 11 the other, 8, 5, 7, 9, 12. Having 6, 7 each night is much better than having this other schedule. So yes, sleep and wake up at the same time every single day and get the same amount. Okay?
- 6.136** What else? You want to tell your brain that it's now daylight and you want to have—what do you do to convince it of daylight? To convince it of morning time that it is now morning. Exposure to light, right? Go outside, right? Go outside, have sun. About five minutes I would do every single day, or at least stick my head out of the window. I'd do that. This is good for resetting the clock and ensuring that you know it's morning time. You can do some exercise in the morning as well. Do a cold shower. All these things will

boost noradrenaline and all the chemicals involved in wakefulness. And yeah, they're very healthy. Very good for the whole sleep cycle.

- 6.137** Should you drink lots of wine before you go to bed? You know when I was in Iraq, as a kid, you know what they used to do? Can you believe this, Joe? This is crazy when you think about it. This is just crazy ideas. But when the babies couldn't sleep, they would give them whiskey.
- 6.138** I mean, my time in Iraq as a kid, I was about—I think the first time, I'm six or seven or something, I saw a Kalashnikov, you know these big ones? These big rifles on the tables, some bazookas, tanks. We'll have dinner and there'll just be this AK-47 on the table. Normal. How is this? How do you like this? It's completely normal. From Copenhagen, Denmark to that, I was just visiting. Imagine. Anyway, they gave the kids whiskey, and I understand the culture was a bit—a lot of war. What can you do?
- 6.139** Is alcohol good for you? Did you take a lot of—not saying you personally—but is it good to have a lot of alcohol? Why not? What does alcohol do to your brain? So it makes you tired, right? You can make the baby collapse, almost faint. But there's actually a derivative of alcohol that will fragment REM sleep and make REM sleep less solid and less stable. So it's a good tranquilizer, but it's not a very good sleep aid.
- 6.140** I think that's all. I think that's about all when it comes to tips on sleep. I'd say have a good night routine. Your brain is very cue-dependent. It just lacks cues and everything that reminds us of "this is sleep time." I have a certain tea that I drink in the evening. I know the tone of the lights is a certain way. I have certain behaviors that are repetitive and that cue my brain to sleep. And I think that's very good to have.
- 6.141** Do you have this, too? It's wind down, right? Wind down. This is not some mumbo jumbo. There is actual science to this. So I would have that, too.
- 6.142** And I think, darkness of the room. Do you use a sleep mask? I'll use a sleep mask. For me, I find them—I don't like them. So I just make sure I have curtains on that are covering everything. But that's important too for light. I would have that.
- 6.143** Oh, for body temperature, this is just a little extra bonus for body temperature. If you want to go to bed and you want to sleep faster, take a hot shower. That would put your body temperature, core temperature down. It has to fall about one Celsius or so. So I take a hot shower. Maybe you can even wash your face and head. That's good too, or face, in this case. Head is a bit difficult. Exercise during the day, that would be my other point. I think regarding sleep, this would be it.
- 6.144** Now, I did have some other tips for you, and these are left over from the other lectures, but I just feel like they are very important. The question is when to put it in. Should we talk about it now or maybe tomorrow? But I also have tons for tomorrow, you see. So I just want you to go home from this course and just really feel like I had so many things I can use in my life now, so many tools in my toolbox.



## Lecture 7: Stress

At a glance — Stress and its management.

- 7.1** So today we'll be talking about stress. Stress is so important. I mean, you can't do a course on optimal performance and well-being without talking a lot about stress. Stress is so important. So we'll talk a bit more about stress. Who mentioned burnout yesterday? They'd like to talk about burnout. Was it here? It was here. So we'll go into burnout a bit more, talk about that, what burnout is, a bit more. And then I guess we'll just go over to practical tips, what we can do to improve our well-being and performance. So we'll make this last day very practical. Does that sound good, Kelly? It will be super practical.
- 7.2** The inverted U-shape of stress. What's the inverted U-curve of stress? What is that? So you have a U, right? How does a U look like? Like this. Then we invert it, so it looks like this. To be able to perform optimally, you have to think about stress in the following way. If you have a little bit of stress, is that a good thing? A little bit is good, right? Moderate stress, in fact, you perform best when you have moderate stress. When stress becomes too—the U is the performance, right? So the more stress you have, it goes like this: stress, stress, stress, stress, stress, you perform well. But if stress continues, it goes down. So just draw that U. This is called the Yerkes-Dodson curve. So write, Yerkes-Dodson curve.
- 7.3** When you have little stress, you have boredom, you have drifting of attention, you have reduced motivation, you have difficulty initiating effort. Does that make sense? So you have to have that moderate level. Now, when you do reach that moderate level, what happens? Attention sharpens. Working memory capacity improves.
- 7.4** Where is working memory, Debbie, in the brain? What part of the brain is involved in working memory? You know this. I know you know this. Working memory, that's when you're juggling numbers in your head, right? Or you're doing some simple subtractions. It's in the prefrontal cortex, right? Is it in the medial or is it in the lateral? It's in the medial?
- 7.5** The lateral.
- 7.6** The lateral? What's the lateral thing?
- 7.7** Cold calculation.
- 7.8** Cold calculation and working memory as well. So medial is the emotional part. The medial is when you have emotion and cognition blended. Is that right?
- 7.9** Yes.

- 7.10** So when you experience embarrassment, you're giving a lecture and you say something embarrassing, what part of the brain lights up like a pinball machine? Medial. Yes. Self-referential thinking, the medial or lateral?
- 7.11** The medial.
- 7.12** Okay, good. Now, we said—this is all Joe's insights yesterday: attention is sharpened during moderate stress. Working memory capacity is maximized. Decision making remains strategic. You're flexible in your output, so you have cognitive flexibility, being able to flexibly adjust your behavior. Again, the DLPFC, the lateral side of the prefrontal cortex, is heightened during moderate stress. Reaction time improves. What's the reaction time? Max, what's the reaction time?
- 7.13** How quickly you are able to act.
- 7.14** How quickly you are to, for example, I say, identify the smiling face. The time it takes for you to identify it is what? The reaction time. That improves, too. So that improves during moderate stress.
- 7.15** Okay. What happens during high arousal? Racing thoughts, tunnel vision, emotional flooding, rigid thinking, blanking out, basically, going to survival mode, right? Noradrenaline floods the brain. Cortisol floods the brain. Dopamine floods the brain. The amygdala, the emotional part of the brain, goes on fire. The prefrontal cortex loses its grip on reality. The very front part goes down. The medial goes down. Lateral goes down. And that has to do with what? This neurochemical completely flooding the system.
- 7.16** Is there any situation, Nikola, where it's good to have this reaction?
- 7.17** Too much stress.
- 7.18** I'm saying—yes, extreme stress, high stress. Can you think of any situation?
- 7.19** Being chased by a tiger?
- 7.20** [Inaudible]
- 7.21** What's that?
- 7.22** If you freeze. It's like you—
- 7.23** Yeah, but you're going to the extreme far now. You know why? No, it's not extreme in what you're saying. The input is good. So there is boredom. Then there's moderate stress—the optimal zone for a soccer match when you're playing on the soccer field; that's the moderate. And then you have high stress—this is where you're running away from a tiger, right? You're being chased and you're running away, right? This is immediate survival and all that.

- 7.24** But then Alexander is saying there's another stage where you completely freeze and you detach completely. And what is that about? This is also good to know. When do you detach completely? Sharon?
- 7.25** When you see something horrible?
- 7.26** Yeah, you do. And I say this is important because if this is zone one, two, and three, this would be zone—let's call it zone X. It's even like when your brain is completely overwhelmed and you're no longer even running away and trying to escape the lion, that's no longer an option. You can't even escape the lion. You're saying, "My God, I will die now." This is when rape scenarios happen, actually during the rape or when somebody's actually trying to kill you or something.
- 7.27** What happens, your brain says, "The hell with stress, I'm going to shut the whole system down." So the hypothalamus, you have the hypothalamus, the hormone factory in the brain, and it communicates with this ACC that's involved in conflict monitoring. It says, "Look, sending out hormones now makes no sense. So we shut down the whole system. Even the hormones, the stress hormone that we shut down now and we detach completely. And even behavioral outputs, don't know movement, play dead."
- 7.28** And then it will send additional signals to the parietal lobes. Now, in the parietal lobes, you have a multimedia studio of your sense of self and me and my body in space. And it says, "From now on, picture yourself from the outside, looking at yourself as an external observer." That's why they will tell you during rape scenarios—whenever they were violated, let's say, when somebody jumped you and they kicked you and all that—they'll say, "I saw myself from the outside. I was a third-person observer. It was me, but it wasn't really me. I was more like a Netflix movie observing the whole thing." This is called "detachment," "dissociation."
- 7.29** In fact, if I was to zap that part of the brain using an electrical current, what would happen?
- 7.30** Neuroplasticity.
- 7.31** No, not really. That's true, too. You can get neuroplasticity, of course, if you zap certain regions. But I'm saying, temporarily, if I zap the parietal lobe—
- 7.32** You have an out-of-body experience.
- 7.33** You have an out-of-body experience, and you see yourself from the outside. So I temporarily construct that tiger scenario, that rape scenario. I do an artificial replica of that.
- 7.34** But we never really answered my question. I think maybe we did. We said running away from a tiger or a lion. Let's take a modern example. Running away from tigers, it's not so common these days. So what is more common, Joe? What could be a really stressful scenario that would actually benefit from?

- 7.35 Someone cutting me off in traffic, and me getting out of the way.
- 7.36 It could be, right? For short snippets. And I actually don't know the answer fully because I'm thinking, let's say you're in a house and the house is on fire. The whole building is on fire. In the hood, you have these big building blocks and the whole building is on fire. You're on the top floor, you want to get down. You want to save your family, you want to save your laptop with—you want to save your computer and all that, the information. No, seriously, you have six months of work there. You have all these things you want to save, let's say.
- 7.37 If you have extreme stress, that is good for survival. But I just wonder whether it's still optimal because you lack flexibility. That's why you're going to panic. So that's when you might take your laptop instead of your child, right? I'm being extreme here, but I'm saying. I think it's really rare, the high stress, I think it's really rarely you would actually have that as a beneficial survival mechanism. But I suppose it does. I think in some scenarios, it could be beneficial. You want to say something?
- 7.38 What if you're an athlete and there's five seconds on the clock and it's high stress, you want to shoot the shot and it actually helps you focus in on?
- 7.39 No, because we said during high stress, though, the focus is not very—it's more racing. It's more—
- 7.40 Even above.
- 7.41 Yes, it's even above that. I would say it could be—in extreme short-term scenarios, it could be beneficial, perhaps. And that's also when you turn on the endorphin systems and you don't feel pain. That's when the zebra is running away from the lion, Giovanni, and it's intense and everything is just spilling out and it doesn't feel anything. It could be in a few seconds like that, 10 seconds, 20 seconds, a few minutes, it could be beneficial. But not more than that because you lose flexibility. So strategic thinking, how do I get out of here and all that, it becomes much more rigid.
- 7.42 I think those would be my main thoughts on stress. Let's look more on—we covered most of it. So let's take a look at some more practical things. Let's go into the practical world a bit more. Have you noticed—so this happens to me, and I actually wondered what was going on. Have you ever had—I guess, who's lifting heavy weights in the gym once in a while? Like, heavy. Anybody? It's a few of you, right? A little bit? Okay, Giovanni, good.
- 7.43 Have you noticed the next day, you feel a bit off sometimes, especially if you go really hard at it, right? If you go really hard at it, and I sometimes go a little bit too hard at it, especially if I'm—there's some girl, there's something going on, and you get your frustration out, and you take a little bit more extra weight. You know this feeling?
- 7.44 Absolutely.

- 7.45** You know that feeling, right? A little bit too much and too many sets, right? God damn it, I have to have those muscles, right? But then the next day, you feel a little off, something off. You may even feel hyper excited. You have what might mimic mild anxiety, not necessarily anxiety, but it just feels you're a bit more agitated, easily stressed, maybe even more sometimes fatigued, you're tired. Have you noticed? Do you notice this feeling?
- 7.46** Yeah.
- 7.47** Sometimes. So what's going on there? What might be going on? Rachel, do you experience this? Not much? Not so much anymore? All right, for those who lift heavy weights or plan to, sometimes when you do heavy lifting like this, especially if it's many sets and you really fatigue the body like this, you will have excess noradrenaline, adrenaline, cortisol, and these hormones will spill into the next day.
- 7.48** I often talk about my mentor in California, Ramachandran. We often had this conversation. If there were some troubles, especially if there's something anxiety-provoking. Again, it could be with a girl or something, right? Something you're anxious for a week, given some situation, right? Let's say. Have you noticed when the anxiety goes away, the actual trigger of the anxiety goes away, everything is fine now, you resolved your issues, you can still be a little bit anxious. You feel a little bit on the edge for another five days, one week. Have you noticed this?
- 7.49** This is because your brain is not like—it can't just shut things off instantaneously, right? There's this timeline. So even though for you, cognitively, the anxiety trigger is gone—mind you, did you know that when you are lifting heavy weights, very high weights, this is cortisol inducing, high cortisol inducing.
- 7.50** But your brain doesn't necessarily know the difference between psychological cortisol release and physical cortisol release. For the brain, it's similar. Did you know that? It's not like it says, "Oh, this is psychological." In fact, people who have depression, let's say you're depressed and you have high levels of cortisol in the system being spilled out. And if you go to the gym and lift heavy weights that can compound and make you even more. So this is stuff to think about, right?
- 7.51** But my point was all of this was when you do heavy lifting or anything of this sort, it can spill over to the next day. Or if you have a period of, let's say, trouble, anxiety-provoking things, and that goes away, you will have a time where that anxiety stays in the system. Again, I think it's always about reinterpreting that excitement and saying, what does this feeling mean? Is this anxiety or is it excess noradrenaline?
- 7.52** Having a neuroscientific lens, Nikola, like saying, "Okay, this is just noradrenaline. This is this cortisol that's high in the system," helps you reinterpret all of it, and it also takes away the fear of it. Anxiety in itself can be a little bit fear-triggering. It's annoying, but what is it? Why do I feel like this? And if you don't have the answer, you don't know that this is because of hormones in your body surging, and you're trying to understand,

"Oh, something must be up with my life. Maybe my life is not right. Maybe I have this inner thing that I'm inhibiting." You're trying to get into all these Freudian interpretation, but really, it's just your physical body having all this excess hormone surging and surfing through your bloodstream and you're trying to make sense of it.

- 7.53** And of course, when you do work out like that, you have tons of dopamine. I mean, so much dopamine. And dopamine, what happens when dopamine goes up? It must go down. Does it go down to baseline? Or does it go—
- 7.54** Lower.
- 7.55** Lower than baseline. Exactly.
- 7.56** And then you have microdamage and inflammation. And then think about inflammation. Inflammation is serious because we don't take inflammation very seriously often. What is inflammation?
- 7.57** When you have high cortisol in the system, your body also releases certain cytokines, inflammatory molecules that will also activate your immune system. And they're again running around your system and make you feel tired. Why do they make you feel tired? Well, your body says, "If I have inflammatory molecules running through my system right now, that means I must maybe be sick. Maybe I'm sick right now. I'm activating my immune system. And if I'm doing that, that's a sign that I need to rest."
- 7.58** You know what I mean? When you have a fever, you also feel tired. Kind of like that, your body says, "Well, if I have this, it must mean I must rest so I can overcome the infection." And it also messes with dopamine. So you don't feel motivation as well.
- 7.59** So these cytokines, they're very tricky. These molecules will cross the blood-brain barrier. With the brain and the body, you have a barrier, so molecules usually can't go in and mess up your brain. But these cytokines can do that and they can mess up your brain and cause all kinds of issues with dopamine as well.
- 7.60** Max, one of the interesting studies, you can put this here. It was hepatitis C. They had them, I think it was interferon-alpha or something. It's one of those cytokines they injected into them just for [hepatitis C treatment] reasons. And they found that within weeks, low mood, fatigue, anhedonia, lack of motivation, all those symptoms you associate with depression, they had those symptoms. Then they scanned their brains. They looked inside their brains to see what was going on in the ventral tegmental area and the nucleus accumbens, which are? Ventral tegmental area and nucleus accumbens, what kind of—
- 7.61** Dopamine production.
- 7.62** Dopamine production. In my last course I taught this. Dopamine production in this part of the brain goes down, goes down just from injecting you with these inflammatory markers, the same inflammatory markers that you have when you're chronically stressed. Dopamine synthesis and generation of these neurons went completely down.

- 7.63** What else is important? What's important to know about—what induces these cytokines and inflammatory markers besides psychological stress? So we said they go to the brain and mess up the brain, but what else can produce them?
- 7.64** Illness.
- 7.65** Illness, of course, yes, if you have illness. In fact, I talked about before, if you have a tooth infection, the inflammation from that can actually cause depressive-like symptoms. So depression doesn't have to be a psychological issue. You don't have to have great adverse effects. Let's say you lose a loved one or something. I mean, you can have a tooth problem, and you get depression from that. Think about this. It's crazy.
- 7.66** But even something like visceral fat, like fat around your stomach and the organs, can be a release of cytokines. Okay? So this is very important to think about. Try to exercise and reduce—the fat around the stomach, especially, is very, very important.
- 7.67** What about vitamin deficiencies?
- 7.68** Vitamin deficiencies and inflammation? Yes, I would think so. Yeah, I would think so. There'd be a link there, too.
- 7.69** There's a special body [type] called the stress body. And we'll get into that. There's actually two very interesting cases in pathology. So this is not psychology, but it's pathology. What do you call this? Endocrinology, I think. It's the area of hormone issues and hormone problems. You know the high cortisol person? Skinny arms, big belly. This is how the body turns out. So everything, the rest of the body is completely skinny, and they just have this belly that points out. And this is really where it's unhealthy. But we'll look at those symptoms later as well.
- 7.70** Okay, so we haven't really talked about testosterone. You can't do a course on well-being and performance without talking about testosterone, can you? So what is testosterone? We all know it's a hormone. But what does it do to you? Nikola, is it good to have high testosterone?
- 7.71** Generally yes, but not too high.
- 7.72** Why is that?
- 7.73** You could become overly aggressive.
- 7.74** Is that true? Or is it a mix with cortisol? Perhaps it is—cortisol and testosterone. These two have to be in balance. First of all, they're always competing. Always competing. So testosterone can make you more assertive. It can give you energy. You feel good. You feel—you're risk-taking. Do you know those times in your life when you don't want to take risks? You just don't want to take any risks. You know you should do this thing over here. You should probably do it. But something is holding you back. You just feel like, I don't want to do it. You're just withdrawn. You have a lot of withdrawal in your

behavior. What is that about? Testosterone as well. Your social risk, you don't take any of that.

- 7.75** And the more cortisol you have and the longer you have chronic elevation of cortisol in the system, what happens to testosterone? It goes down. It goes down. So risk tolerance, social assertiveness, approach motivation, competitive drive, all of this goes down.
- 7.76** So there is high testosterone and low cortisol. That's what we should aim for. This is where you have focused assertiveness, motivation. You just have energy and you feel good. It's not impulsive and aggressive. It's just socially balanced.
- 7.77** When you have high testosterone and high cortisol, that's not a good combination. This is volatility and impulsivity. I see these guys in the gym all the time. These are the anabolic steroids. They have big muscles. They're very aggressive. They go around. You can really sense. If you look at them the wrong way, what happens? Do you have those guys in Arizona? We have them in Copenhagen in the hood. We have them. They're everywhere. This is high cortisol and high testosterone. You don't want to have that.
- 7.78** And then—just flip it around—there are various permutations. Then you have low testosterone and high cortisol. Low testosterone, high cortisol. What is that about? What personality is this, Max?
- 7.79** Stressed.
- 7.80** Stressed, withdrawn, no risk-taking, scared, anxious, perhaps depressed in their behavior, not clinically necessarily, but in that category. And it's interesting. We can all fall in these—I mean, obviously, we can't test our testosterone from day to day. We don't go around having blood tests, but there's such a psychological component to testosterone, right?
- 7.81** I mean, one day, okay, let's say, this is actually for my life. This is for my life. I've had recently very stressful time, let's say. I have a very stressful time. I have so much going on. Getting all these emails. You have to do this, you have to do this and that, promote this and that. And I say, "I don't want to do this." "You have to. You spend all this time writing this, so you have to promote this and that." And I say, "I don't want to" and all that. This is where assertiveness comes in.
- 7.82** I thought about this last night as I was sleeping. I thought—last night, I couldn't sleep much. That's why I can ramble sometimes. When I don't sleep, I ramble. Forgive me. I couldn't sleep, and I thought, "Okay, how can I give them practical tips tomorrow?" And this kept me even more awake. What practical tips can I give them? The whole thing of turning down, making the room colder, it didn't work at all. It didn't work. So what practical tips can I give them? And something that came to mind was a very simple thing, which is you have to be more assertive in your life.

- 7.83** Assertive is not a bad thing. It's just you having boundaries, and that helps you perform better and just live better, right? This is so critically important. I mean, especially somebody like me who always—I love people—I love being around people. Like, for example, the guy who drove me here to the studio today, he wanted to chat with me. He was from Iraq. I was from Iraq, and he kept going, in Arabic, right? Great. I wanted to talk to him, too. I'd love that. But I thought, "If I talk to him, I won't be energized." This is the sixth day of lecturing. So I had to say, "Look, buddy, I have to meditate a little bit. Forgive me. I love—thank you, but I have to." You have to do that. I hate it, but you have to do that.
- 7.84** I think assertiveness is so important. It can be trained. The more you do it, the more it can be trained. And then each time you train it by just doing a little bit, you actually have more testosterone. Why? You're being less stressed and you're being more socially bold. Whenever you're more socially bold and you feel like you're winning in a social situation, or at least you did not suppress your emotions, you actually have more testosterone.
- 7.85** So there's a huge social component to testosterone. Let's say you watch soccer—FC Barcelona fans? Madrid? Hopefully not Madrid.
- 7.86** Barcelona.
- 7.87** Barcelona, right? Okay. I told you not to people please, my friend. So you like FC Barcelona. You watch them win. You watch them win. And Messi scores. He doesn't play there anymore, but he scores back in the day. Does testosterone go up or down?
- 7.88** Probably down.
- 7.89** It actually goes up. It actually goes up. And if your favorite team loses, it goes down. Can you believe it? So there's a huge psychological and hierarchical aspect to testosterone as well. And by having wins in life and being more assertive and having wins, almost like I'm acting out being a high-testosterone person, and it gives me more testosterone. I'm tricking the system.
- 7.90** Now, coming back to my own stressful period. So I was going through all this stress in my life, having all these pulls—what do you call it? Drawing me; they were pulling me, right? And then I was like, "I'm very stressed because I'm not sleeping very well." And when you don't sleep, you have what? You become very tired. Actually, one night of sleep deprivation, 60% more activation in the amygdala, the fear part of the brain. The amygdala goes down and chit-chats with the hypothalamus. The hormone factory says, "Look, buddy, send down commands to the adrenals and release some of these hormones, cortisol and adrenaline," especially cortisol. What does it do to testosterone? It lowers it. Over time, definitely chronic decline of testosterone.

- 7.91** Now, I have too many of these sleepless nights; it's not very good for me. As time goes, testosterone goes maybe more down and down and down. I can feel it in the gym, less strength as well. Now my sleep is really messed up.
- 7.92** But then, this is where you can actually flip the system. This has happened to me, and it worked. You have this thing you have to do that scares you. Let's say you have to talk to a certain person. It could be a girl—sorry, I have to keep mentioning this, because this is my own life. I have to be honest, right? So you have to go talk to her, but you have low testosterone now. What do you do? So now you're a neuroscientist, so you know what a low-testosterone person looks like, and you don't want to look like that because you know she's not going to be attracted to that. So you fool the system. You can all do this. I said to myself, "I'm going to act like I'm a high-testosterone person, even though I know the last month, my testosterone must have gone down because of the stress."
- 7.93** So I did something extremely socially bold. Mind you, I was so anxious, so stressed out: heart racing, everything's just completely in panic, but I did it anyway. And it worked. It worked, and my testosterone did what? Jumped. Right away. In fact, afterwards, I felt much more bold. And I felt bold in other situations as well. So I had boldness spread out throughout my life. I got this more bold in other scenarios. Did you know, Juan, you can train boldness? You can train your boldness muscle, which can lead then to more testosterone. And this is how you can engineer it yourself, manufacture these things and trick the system.
- 7.94** And I think this is a huge part of this course: that knowing that life sometimes throws you curveballs, you become more stressed, testosterone will go down. It could be that you're a new parent or something like that. But doing socially bold things, doing risk-taking anyway, being courageous, all of these things will boost your testosterone and make you feel better.
- 7.95** Some steps for testosterone. Some steps for testosterone: how we can improve it. Now I want you to check out your nice notebooks and jot down your notes. Consistent sleep, controlled workload, recovery after high activation, and limit chronic rumination—that would be the pillars.
- 7.96** Anyway, so the first one is reducing—yes, reducing stress would be the first one. Reducing stress. And you have the tools for that now. Then you have sleep, stabilizing sleep.
- 7.97** And I want you all, anybody in here that doesn't do resistance training—I talked about the importance of cardiovascular training, didn't I? For BDNF and GDNF neuroplasticity and changing your brain. What about resistance training? Do I recommend this? Highly. I do. Anybody who doesn't do it in here should now start. This is my recommendation: twice a week at least, lift some weights. This is very good for testosterone. I would focus on the large muscle groups, like the thighs. You need to have adequate recovery for that. Very important. So you don't overwhelm the system.

- 7.98** Think about your body composition and metabolic health overall. Do I lack protein? Am I good in iron? Do I have enough zinc? Zinc is good, the oysters. They're very good for testosterone. So you want to have adequate protein.
- 7.99** I have periods of my life where I feel like my protein is—I have less protein. I should eat more protein. I'm exercising, I'm doing all these things, but I'm not getting enough protein. I have a few protein shakes, and I just feel better. So protein is very important to think about.
- 7.100** Who mentioned sunlight and vitamin D? Rachel, you did, right? You did. Was that in relation to testosterone? Maybe not. It could have been, but it's very important for it. It's very important for that. So get your vitamin D.
- 7.101** Here's the more interesting one, I think, for testosterone. If you can engage in any competitive sport, that'd be good. This could be anything: negotiation, challenge, anything that's competitive, competitive in nature. What would be a good example that's practical for all of you guys?
- 7.102** Basketball.
- 7.103** Basketball. Giovanni?
- 7.104** I agree with basketball.
- 7.105** Basketball?
- 7.106** Any sport.
- 7.107** Video games could help—
- 7.108** What's that?
- 7.109** Playing video games.
- 7.110** Yeah, if it's very competitive and against a competitor, sure. Max, you have one.
- 7.111** Taking my job interviews.
- 7.112** Yeah, sure.
- 7.113** Negotiation.
- 7.114** Negotiation. Absolutely.
- 7.115** Psychological posture and behavior. There's such a huge psychological component to testosterone. Is it good to be all like this? That's not a good thing. What about like this? Is it better? Good posture? Absolutely. There's so much there in psychology and psychological terms of posture and having a good self-body, like a good image of yourself.
- 7.116** Let's focus a little bit on burnout and recovery. I mean, we covered a bit yesterday. There seems to be two types of burnout patterns. Two types. Juan—sorry, can I use you again? Okay, thank you, brother. So you have pattern A, the first type. This burnout

pattern is the following. We've all tried this, I think, many of us. You wake up in the morning; you're supposed to wake up at seven, you wake up at five. You feel you have energy, racing thoughts. You can't go back to sleep. And then you just feel a little bit too excited, and you know it's not real energy. There's something up. It's not the real type of energy where you feel calm and you feel energized. It's more like a racing energy. But then you hit noon and you start crashing a little bit and you start feeling tired. And overall, motivation goes down and you're not feeling very well, right, Alexander? You know this feeling? This is version A. Starts energized, starts with racing thoughts, starts with the illusion of "I'm on top of the world right now."

- 7.117** Version B, the second version. You wake up tired, sluggish. You can barely make it out of bed. Supposed to wake up at seven, but Juan sleeps till 11:00 and 12:00. Okay? And then he wakes up, can barely make breakfast. He wanted to comb his hair, he wanted to get dressed. He just doesn't have any motivation to do anything. Hours pass by, you're just going around the house. And then by seven or eight, you suddenly feel energized again. That's when you want to play your video games, Max. You're going on video games and you play to 4:00 a.m. And then you feel sluggish and tired again.
- 7.118** Two patterns. How do we address these two patterns? How do we reset these two patterns that are very common? Pattern A. Sharon, when you wake up with these racing thoughts and energized and all that, what do we do? We have to calm down the system, not get up right away. I do this all the time. I have to blame myself because I do it when I get this. I wake up at 5:00 or 6:00, I say, "Oh, I have energy. Great." And I go out and do the gym and I do all these things. But really, what you should do is give it one more hour of rest in bed, do meditation, focus on your breath, and get your cortisol down. Get your cortisol and noradrenaline down.
- 7.119** In fact, you have a peak of cortisol as you rise from sleep. And you have noradrenaline also flooding the brain when you rise from sleep. So you want to get the whole system down and calm it down. And then slowly, once you have the system down, you want to gauge your daily activities and slowly do your exercise and do your various activities. We'll get into what a perfect day could look like, generally, for all of us in the next lecture. But the idea is really calming, calming the system, especially in the morning, because when you do that, you don't end up with the crash later.
- 7.120** Now, the other person, what do you do with him? The one who wakes up at 12 and all that.
- 7.121** Enforcing a regular schedule, like waking up at the same time every day.
- 7.122** Yes, along those lines, Max, along those lines, in fact, what you would do is the following. You have to activate a system in the morning, and the way you do that is by, first of all, when he wakes up, you have to say, "Okay, this guy has to wake up at seven." And then you have to force yourself out of bed, get sun exposure, maybe take an ice bath. You have to stimulate the system, the opposite.

- 7.123** Mind you, I also have to make it clear here that version B of burnout usually comes later in the stage of burnout. This is not after six months; this is after one year. It's closer to depression. You can see it's very close to depression, even. It's almost like two steps before depression.
- 7.124** Get the system activated, get sunlight, ideally do some exercise in the morning, maybe drink a cup of coffee, and get the cortisol spike, which, before the cortisol spike, this person wasn't spiking as it should. Person A, too much cortisol. Person B, too little, so you get him up.
- 7.125** And then, critically for version number two, you have to have a very suitable wind-down night routine. So at night, when the energy comes, the urge comes to go and watch Netflix, do all these crazy things on the internet, what do you do? You rest, relax, drink a cup of tea, go for a little walk, these things. You wind down the system. This helps you reset the cortisol rhythm.
- 7.126** So to be clear, cortisol works like this. It's a spike in the morning, and then it goes up, and then it falls by the evening; it goes like this. This is how cortisol is supposed to look. It's interesting in people with mental illness, like depression, you find a different pattern. Usually it's down. Instead of going like this and then down, it goes like this, and then it peaks in the evening. You see, you're not supposed to peak in the evening, right? So what you can do is, whether you have late-phase burnout, version B burnout, or depression, or mental illness, you flip the system by raising cortisol in the morning, by doing exercise, by drinking coffee, by stimulating yourself. I think we covered most of this.
- 7.127** What should you look out for? What's important when you're dealing with a burnout situation? What do you look for? Okay, so Max wants to know whether his brother is going through burnout. How do you know? How do you identify this? Any thoughts?
- 7.128** It might look like depression.
- 7.129** It might look like depression. Could be, but you look for early warning signs, even for yourself, right? You look for fatigue, you look for disruptive sleep. You look for anything that might signal that you're going through some burnout right now.
- 7.130** And I think also to add with burnout, and I think this is a crucial point, I'm not sure if I really got into this, but the whole notion of burnout is that the longer you get in burnout, in the phases of burnout, the longer it takes to recover. And I think this is key.
- 7.131** So initially, basically, if the stress has been for one week, two weeks, three weeks, it's easier to recoup the system. The longer the period has gone for, you keep going, the longer you will feel like—the longer it will take to actually recoup. Okay?
- 7.132** What else can you take, early measures? And I wish I had taken some of these steps myself early in the days. How about, is a therapist a good thing? What do you think? Okay, who says that therapy is only for those who are clinically sick, depressed,

anxious? I think one of the best steps one could take, forget burnout. Just overall in life, it's getting a therapist. Just put that down. I would say that if there's any advice, find a good therapist. Find somebody who's good with interpersonal relations, skills.

**7.133** I mean, I wish the whole education system was based around skills you learn in therapy, like how to deal with people, how to calm the system, how to be less stressed, how to look out for burnout, how to sleep better, how to reduce anxiety, all these things, how to boost testosterone. We should have a system for this, but having a therapist is, I think, very important. They can find your flaws. They can see it much better than you and tell you, do this and that. And then in the burnout phase, they can definitely help you not reach the burnout. So I say therapist is very important.

**7.134** It is not embarrassing. In my world, it was very embarrassing to get a therapist. My God, you can't get a therapist. If my friends knew, so embarrassing. It would be so embarrassing. My scientist colleagues would be okay. I'm talking about the other friends. It wouldn't be very cool, not very macho to get a therapist.

**7.135** But it's so good. It's so good to talk to somebody about your life, your problems. You don't have to do it like twice a week, three times a week, just like once a month. Give them a recap of your life. And they say, "Oh, you have to do this more, do that more. Be careful here." It's worth the investment. Definitely.

**7.136** And this I have to add: there are so many studies showing that your brain will structurally rewire after therapy. So the therapist doing all this cognitive restructuring exposure therapy, saying, "Go do this and that that you're afraid of." You do it, and your brain starts changing and you're starting to reinterpret your identity in a positive manner. And you will have neuroplasticity, change of your neural circuits before and after. Psychotherapy, merely talking to a therapist. Actual structural brain changes.

**7.137** Early signs: cognitive changes; concentration may be down. Lack of—you just can't think right. You just can't solve the problem. And this is where it becomes paradoxical because then what happens? You try harder. Exactly. You've tried this before. I've so many times. You can't concentrate, you can't focus, you can't stay on the task, but you keep trying even more. The next day, you do even more. Really, it just requires you to step away from the task and do something completely different, relaxing, resetting the system.

**7.138** Okay, I mentioned these disorders because it helps us really understand what extreme low [cortisol] looks like and extreme high cortisol looks like. So one is called Cushing syndrome. This is when you have too much cortisol leading to excessive stress activation. This is really an endocrinology disorder. So hormonal imbalances, these guys are not psychologically stressed. They have an issue in their brain, the hypothalamus and the pituitary and all that, the stress parts of the brain.

**7.139** So what does Cushing look like? Too much cortisol. You have that round belly. You have a moon face, like a rounded face, but thin arms and legs. Fatigued, but a restlessness as

well that goes with it. You have high blood pressure, high blood sugar, fragile skin. Skin doesn't look very good. You don't look very good, okay? You're irritable, you're anxious, you're depressed, have difficulty sleeping. And this is high cortisol in its extreme form. But it's not that far from somebody who has chronic psychological stress. And that's my point. I just want to give you the actual extreme example. But there's definitely—that frame also fits somebody who has psychological stress.

- 7.140** And then you have Addison's disease. Too little cortisol. Which condition in psychiatry is characterized by too little cortisol? Alexander knows, so don't say it. I want to test somebody else. Which condition in psychiatry is characterized by too little cortisol? Baseline.
- 7.141** I was going to say narcissism.
- 7.142** No, no, no. Alexander, say it.
- 7.143** I actually don't know.
- 7.144** You don't know? Come on, man. I mentioned it in my previous courses. Come on. In post-traumatic stress disorder, after the trauma, the person develops too low cortisol baseline. They can't regulate their cortisol baseline, their stress and cortisol, and so their baseline is basically flat. But having low cortisol as your main baseline is also bad because now you can't regulate cortisol very well. And cortisol is a very healthy hormone. It's very good for us. It's just when it's highly elevated, chronically, it's a problem.
- 7.145** Okay, so what are the symptoms of too much—too low cortisol? This is called Addison's, and the symptoms are constant fatigue, no energy, weight loss, muscle weakness, low blood pressure, salt cravings, skin pigmentation, difficulty coping with stress. Okay? So that would be the opposite scenario of high cortisol. It's the opposite scenario.
- 7.146** Max, my behaviors, what I do from day to day, how I interact with people. Let's say I do stress-reducing activities and behaviors. Will that affect me? Three months, six months from now?
- 7.147** Do that stuff today.
- 7.148** Yes. Let's see. I do heavily in and now and do psychotherapy. I do yoga, meditation. Take care of my sleep. I go for a daily walk. Take vitamin D as needed. Do all these things. Take care of my stress, psychological stress. Don't overload. Don't make myself a meaning-driven person, all that. Will it affect me six months from now if I do this? It will. How?
- 7.149** You form new habits.
- 7.150** New habits, new circuitry in the brain, new synapses, expanding the prefrontal cortex, all that.

- 7.151** But Kelly, I have a question. Will it affect my children now?
- 7.152** Yes.
- 7.153** It will? How? How will you make a change to that? Your children are not yet born, let's say. Will it affect your unborn children? Of course, if they're born, they will be affected because they see you stressed and agitated, and they become stressed as well. They may be modeling your behavior, and they learn that stressful patterns are the way to cope with the world. And so they copy that, and they start shouting in kindergarten and all that. We don't want that.
- 7.154** Now, will they affect your not yet, future kids? They will? How? This is a bold claim now. You're making a pretty bold claim. How does it affect your children?
- 7.155** I think it's in the cells.
- 7.156** In the DNA, exactly. Okay, so this is important. This is very important. Within the DNA, there are what? Switches, methyl groups that sit on the DNA and turn genes on and off, right, Alexander?
- 7.157** Right.
- 7.158** So if my parents—and this was actually the case; my parents were Kurdish from Iraq. They experienced a lot of stress, trauma, explosion, death. Everything was just—that was just everyday life for them. My mom was always stressed, always anxious. Within her DNA and genetics for stress—and it's polygenetic, of course, anxiety and stress. So she turns on these genes and voilà, I come to the world, and I will have stress, too, and anxiety, too, because those genes were carried over into me.
- 7.159** So you have a mouse, Joe, you have a mouse, right, you expose it to a smell, and you give it an electric shock, right, Giovanni? It now fears that smell. It fears it. But it turns out that that mouse, the kid of that mouse, its baby, who was never exposed to that smell, will also fear the same smell. But it doesn't know why. It doesn't know why I fear the smell, but I just do. For some reason, I fear this; I don't want to be near that smell at all. And that kid—sorry, that mouse—when that mouse has a baby, so now we're two generations down, even that mouse, the mouse's baby's baby, will fear that smell, having had no electric shock ever.
- 7.160** It's like me, for example, as a kid, I used to always be scared of, I don't know, my friend's parents. Always scared, hiding from them in the house. So I see them, I see this big guy, I got scared. I got scared. He may have some tattoos. Maybe bald, tattoos and a big chain and all that. It was in the ghetto. So I got really sad. I would just hide. I wouldn't be near them. And I think it's because my mom in Iraq, she saw so much killing and blood and all that, and it just passed down my genetics. So we have to be careful, not only for ourselves, ourselves six months from now, a year from now, five years from now, but also our future generation and our children.

- 7.161** I wanted to end on that note. Let's go to Q&A, and then the next lecture will be almost all practical. Okay?
- 7.162** All right. I was wondering if we could talk a little bit about the potential link to rising anxiety disorders and this societal push that you talked about a few courses ago, and its push for more spontaneous decision-making. What was the word you used, particularly—impulsive behaviors.
- 7.163** Impulsive behaviors, yes.
- 7.164** A link to those impulsive behaviors and the rise of anxiety disorders. In particular, I've noticed quite a few of my anxious friends who maybe even exhibit something like burnout A, where they feel guilty, relaxing, and high stress just trying to relax and such. Then potentially, if we could try to really nail home some of the best ways for us to maybe try to deal with that rising push and stress in society.
- 7.165** How do we push stress in society?
- 7.166** Deal with the push of stress in society, I suppose.
- 7.167** Yeah, it's a good question. I think it has to do with a whole change in society. And it will always be difficult because what we are seeing now is the rise of social media over the last 10, 15 years. And nobody's dictating, really; nobody's dictating what should be on these social media channels in terms of—people are just putting their lives out there. But they notice that when they put that Ferrari on there, they put themselves next to that chick, they get more likes. People admire them more. So it's driven by survival of the fittest. I don't know. It's just a jungle, and people are trying to use that.
- 7.168** And so how do we change it? I think we have to have, as a society, I think I want to say the media channels, but the media channels are so bloody corrupt. Yeah, they seem to be bought, really, many of them. So it's very hard to—
- 7.169** But you have to have people that are just courageous, that are not selfish, right? The people that, let's say, they have a certain profession, they get exposure for whatever they're doing. These people have to take more responsibility and be less selfish, less greedy, less greedy for instant recognition and fame, and think about the responsibility that they have for their children to pass on a good message to people and just be human and be ethical, and think of it, it goes down to, why are you doing what you're doing? Is it because of yourself and your self-interest, or do you have a greater responsibility? I think that would be more responsibility on their part.
- 7.170** But how do you get them to do that? How do you get people—it's a deep question. I don't really have a clear answer on how we push society. It would be training in the social—It would be in school, the school system, teaching kids good values, counteracting.
- 7.171** But to be honest with you, I've been so disappointed. The school system is so disappointing. So disappointing, I'm telling you. In my entire school years, I've always

been told since I was a kid, "You're not going to amount to anything." I swear to God, there are these guys in the hood; they go around and mentor people. And me, they would come to me, and they would mentor me as well, and they would talk to me and, "Don't go away. Stay away from gangs, stay away from drugs, stay away from all that."

- 7.172** But they were so hypocritical. When I graduated high school and got through high school, they wouldn't believe me. "You're going to get through high school? No, you're never going to make it in high school. You're never going to pass through high school." And also class teachers. I always got the impression that I would never amount to anything. I had one teacher, Jørgen, this tall, blonde-haired, blue-eyed Danish person. He was so empathetic. Held our hands, always talked to us, very kind, very warm. But most were just never any good values of you can make it, you can be socially responsible. Always just, "Here's the next math. Here's this."
- 7.173** And I think, how do you revive that? How do you change that system? I don't have the answer, Max. I can't answer you. I don't know. That's my answer. It's a complex—I think you and I and all of us can make a change in our lives and be more responsible. How about that? I think that's the only thing.
- 7.174** So there are two parts to this question, and it piggybacks off the first where you can turn on stress genes and pass it off to your kids. The first part is, if you recover from your stress, let's say you had a bad past, can you maybe face it and turn off those genes so that you don't pass them on? Then the second part of that is, I heard that if you encounter hardship, you can also turn on genes by facing issues. You can flip on—
- 7.175** Flip on the good genes?
- 7.176** Yeah, the good parts of your genes.
- 7.177** Or at least turn off the bad ones.
- 7.178** Yeah, or turn off the bad ones.
- 7.179** Because usually in brain science, there's a fine line. What is a good feeling? Is a good feeling the absence of a bad feeling? And so this gets into a blurry neuroscientific area, but I see what you're saying. Can we switch on these genes, right? On and off? Definitely, yeah. Definitely, you can. That's the whole point.
- 7.180** So let's say you have—I said before that I had bad genes switched on from my mom, probably my dad, too, both orphans in Iraq. But my mother's love was so overwhelming. She was always kissing me, hugging me, taking care of me, taking me to the zoo, taking me here and there. That, I think, turned on all the good genes. That's why I never had any issues, really. I mean, if you want to call it, I still have some issues, but you know what I mean. I didn't end up behind bars, right? So I think definitely all that.

- 7.181** And then we can do it ourselves, too, by engaging in social interaction, playing with our nieces, nephews, having fun, being social, all that will turn off some of those bad effects. Good question.
- 7.182** Love is the answer to everything. And if God is love, we have love inside of us, so we can activate that. And it will counter darkness and evil.
- 7.183** It will. It will. So you're going through so much darkness in your life and then being able to have something greater for some people. For you, it could be God. For me, it's definitely also God. You evoke that light in the darkness gives you meaning. I mean, human beings have this ability to contemplate ourselves and our place in the universe and future beyond this life. And having that is, from a scientific perspective, extremely powerful.
- 7.184** Yes, because we're not only this body, we're the spirit inside the body.
- 7.185** Definitely, yeah. From a scientific—again, science can't prove that, but I personally believe it, so why not? Good question.

## Lecture 8: Decisions & Direction

**At a glance** — Decisions and direction, the final lecture.

- 8.1** I think it's very important for me that this final lecture, any questions you may have had about well-being, about performance, is really answered. I don't want you to leave Arizona and feel like you had something that you wanted me to talk about, and I didn't answer your question. So we'll make sure of that, okay? That's very important to me.
- 8.2** I'm going to make this final portion very practical, and I'll look at practical things we can do to improve our lives. And interestingly, many of these tips I have are issues that I've been struggling with myself, like things that I face day to day. I feel like knowing them has just enriched my life and made things easier. I can just run more smoothly.
- 8.3** So let's look at one of them. There's something called decision fatigue. This is where your prefrontal cortex is overwhelmed. The DLPFC, let's say if it's a bunch of strategic decisions in your day, you have to do A, B, C, and D in your life, you have all these things you have to address, it overwhelms the prefrontal cortex, the lateral.
- 8.4** If it's emotional decisions all day, do I have to—what's an emotional decision? That's a bit extreme, Max, but let's say. You don't have that on a daily basis, but you have to make emotional decisions. What am I having for dinner? Is that emotional? Semi-emotional, right? Too many of these, they accumulate and create what's called decision fatigue. So you want to find a way to reduce decision fatigue in your life.
- 8.5** I mean, this is something—have you seen the CEOs that always wear the same T-shirt? Or they wear the same hat all the time, or they wear the same—I wasn't thinking about me. We have another hat. You guys gave me a hat, too, right? So I have another extra hat now I can use.
- 8.6** But really, it has cognitive load. Load is something. Having too much load on the brain and having the prefrontal cortex always make decisions and always go, "Oh, A, B, C, D." My mom, whenever I'm with my mom, visiting my mom, "What are you having for dinner? What are you having for lunch? What are you having?" I just want to—like, "Mom, whatever you make, it's fine. Whatever it is, I'm going to love it. It's okay." But anyway, she still keeps asking and asking.
- 8.7** And this is taxing. You're going, "Okay, I should make these decisions about these very important decisions in my life. I have all this, but then I have to also decide what to eat." And then I go, "Oh, I can see the importance of, like, do I want chicken?" It's important, too, right? I mean, your tummy certainly feels it's important, and it competes with other very important tasks. So I'm saying, try as much as possible to flatten that whole thing and reduce cognitive noise in that way. Reduce that load. Reduce decisions. If you can have the same clothes you're wearing every single day, try to do something like that.

- 8.8** I mean, for years, I had the same dinner every single day. I know it sounds really boring, pathologically boring. Giovanni knows. You know it? Same thing every single day. Can you believe? Every single day. Just a pizza slice. Really, it's true. It's actually very true. This is what—back in Ramachandran's lab, I was having the same—and the people in the laboratory knew that I was having the same thing over and over.
- 8.9** One day, this guy came to the laboratory. He was a little bit chubby, but he was so warm. He was so kind. He was so nice. And he told me this guy is a writer. He writes books. And he also eats the same meal, not one meal a day, but the same meal every single day for dinner. Same thing. And I was so happy to know that I had something in common with this guy.
- 8.10** His name was Oliver Sacks, one of the great neurologists in science. You know Oliver Sacks? Look him up. He's very famous. Look him up. Oliver Sacks, okay? I was so happy to know that I had something in common with Oliver Sacks himself. And I think it has to do with reducing cognitive load, having habits you do every single day. And yes, maybe not as extreme as having the same dinner, but definitely for clothing, decision-making, anything you can think of, try to make these less.
- 8.11** What else can we think of, Nikola? How can we reduce our cognitive load and make things run more smoothly? Okay, here's a question. Task switching, is that costly for the brain? It's costly. Why is it costly? Start over the engine, turn it on, off, on.
- 8.12** What about bulking tasks? So similar tasks, let's say emails. I do emails, let's say, twice a week. At a certain time, I do all my emails. Instead of going like one hour you work and then you do email. Is that what you do now, Rosanne? That's what I do sometimes.
- 8.13** I have two major things that I write on when I publish today.
- 8.14** Yeah.
- 8.15** And I can go from there.
- 8.16** You go from there. But really, bulking is the key. If you can bulk similar tasks, that's really good. So I have deep work. Let's say I do preparation for my class here. And then I have a similar task. It has to do preparation for, let's say, an article that I'm writing. Same. And then I have other types of tasks that are similar. I bulk them together. That could be play. Think of play as a duty in your life. You have to have play in your life, too. Play with your family, with your family members. So you have time for that. Similar tasks.
- 8.17** What else, similar task, Rachel?
- 8.18** Can you plan things out?
- 8.19** Absolutely. See? She's much better than I am. Absolutely. You plan it all out. Reduce the cognitive noise. The entire week, you plan out the things you want to do. In fact, having cues like I'm going to sit at 7:00 a.m., I'm going to wake up at 6:00, let's say, but

at 7:00, I'm going to start working, and I'm going to work for one hour, and I'm going to write two pages. Having specificity like that is extremely good for the brain. It likes cues and specific tasks. The dopaminergic system, the goal-directed system in the ventral tegmental area and the nucleus accumbens, loves that. Specificity and writing it all down. Now, after you wear the same T-shirt every single day, wear the same clothes, same meals, everything is written out. You do emails every day at 4:00.

- 8.20** Have you heard of decision paralysis? This used to hit me really badly in the past. This is when you have five options. Let's say you have—what can be decision paralysis about? I want to say five T-shirts I want to buy. Five T-shirts I want to buy. I have red, blue, green. Is that a good example? Kind of, right? Any decision in your life, you may hit paralysis. You start analyzing. "Oh, this one is better. The brand is better. The fabric is better. This one, it's like the sleeves are a little bit like this." And you start overanalyzing and you can't make a decision. And you wait for weeks and weeks sometimes in analyzing this.
- 8.21** So a T-shirt may be a bad example, but let's say it's a job. You have four job opportunities, right? And you just can't make a decision. So how do you make this? How do you avoid this decision paralysis where you get stuck and you just can't choose the right—any of them, in fact, you can't choose anything. What do you do? Are you faced with this sometimes, Rosanne? Not so much?
- 8.22** Not with work.
- 8.23** Not so much? Other things?
- 8.24** Yeah.
- 8.25** Yeah. Sharon, you nodded.
- 8.26** When I have too much to do, I freeze.
- 8.27** You freeze. Too many decisions, you get paralyzed. Analysis, paralysis, right? You just can't choose and you overanalyze. So what do you do? How do you get over this? I mean, if you can have—if you're much faster at decision making, your life would be much easier. You would just have it much smoother. For me, I overanalyze, too, sometimes. I just get, like, "Should I do this or that? Should I say this or that?"
- 8.28** And by the way, I just want to add before I go into decision paralysis and how you cure it—if there is one lesson I have not mentioned that will destroy your well-being, you know what that is? Perfectionism. It's so bad. Who's a perfectionist here? Perfectionism, huh? It's so bad. You know what I'm saying, Alexander? It's not fun. It's so bad. Perfectionism is not a good thing.
- 8.29** Excellence is a good thing. Excellence. You want to prepare for this lecture. You want to prepare for that job interview. You want to prepare for that date. You want to prepare for that soccer match. You want to prepare for that board meeting. You want to prepare for that family occasion, mentally, whatever.

- 8.30** But perfectionism is the enemy. Enemy. It makes you rigid. It makes you over-focused on details. It makes you—it's time-consuming. It's inefficient. It leads to nothing good. "How do I craft this email to my boss? If I put this here, you may think I'm trying to suck up to him. If I remove this, he thinks I'm—" You know this feeling, right? You don't know what to do. So bad. Perfectionism is the killer. You have to do what instead of perfectionism? Excellence. But how do you deal with, let's say, that email you want to craft perfectly or you want to have the perfect dress for that date you're having or whatever it is? How do you—what's that? You let go, right? Let go.
- 8.31** You make a choice.
- 8.32** You make a choice and let go.
- 8.33** I think for perfectionism, it is knowing that these details will not matter. This is very important. For perfectionism, details such as, "Oh, did I wear—did I have this perfect outfit? Did I look all—" It's not going to matter at the end. There's room for flexibility. If you put the words like this in the email, it's not going to get you fired.
- 8.34** I think letting go and then exposure is so important in the sense that, let's say, you expose yourself to what you fear. Let's say you fear sending an email in two minutes instead of five minutes, let's say, to your boss in this high-stakes situation, sending that SMS, right? Whoever it is, you're going to send that SMS. It needs to be perfect; every word is so carefully crafted, Alexander? So carefully crafted. Every single word you think about.
- 8.35** What if one day you say, "The hell with it? I'm just going to just do a fast SMS. Whatever I intuitively feel is the right thing, I'm just going to write that and send that off." Try that once. I know it's a little bit scary because it's high stakes. You don't want to sound clumsy and stupid. But you send it anyway, and lo and behold, you see, "Oh, nothing bad came out of it." And your brain now learns that lack of perfectionism is okay. It's boldness as well. It's boldness in the face of perfectionism. Perfectionism is cowardish in a sense. So put that down.
- 8.36** Now, decision paralysis—before I went off on my tangent here over here—is, how do you deal with that? I say the decision paralysis is the following. You want to eliminate the ones that certainly are not good options. So you have 10 job candidates, five potential wives, five potential women you want to marry. Sorry. You have these options, and then you say, okay, this one is definitely not the right one. This job is not good for me. This is definitely too far. And you look at the pros and the cons, quickly analyze. So you remove; now there are three left.
- 8.37** At this point, you go, okay, instead of going, "I'm going to think about these next three for the next month, for the next three weeks," you give yourself five days, especially if it's not too important. If it's not too important, give yourself—if it's very important, let's say career, job stuff, give yourself one week: "I'm going to just let it rest. I have three

options." But you give it a time limit. This is very important. Elimination and then a time limit.

- 8.38** Now your brain has a specific anchor of time. It knows I have only this amount of time. I would say within that week, rest your prefrontal cortex. Go for walks, relax, and feel what your gut feeling is telling you. Intuition. And say, "You know what? No matter which one I choose, if I can't choose between them, it must mean that they're similar and it's good enough." So this is where you go against perfectionism, and then you just choose one. You just choose one.
- 8.39** So this is where you realize that going with your intuition is actually a good thing. Intuition is really—what is intuition? It is when you let go of the prefrontal cortex and high top-down control of your mind. You let things flow and you take in memories from the past and everything that worked in the past, maybe emotional memories, stored in the emotional part behind your ears, the limbic centers. All these different experiences you've had, you integrate all that in a loose associative way. And you say, "Oh, based on all my data"—like AI; it's machine learning. You say, "Okay, probably this is the best one." And this would require time and letting go of prefrontal control, restriction, narrowness, right? And so this is how you'd make that decision.
- 8.40** Oh, and this is an important one. There's something called the [two-way] door decision, the [two-way] door decisions. And this is—I think once I really got a grasp on this one, it saved me so much time. Should I go and propose to that girl? Let's say. Should I go and — okay, that's an extreme example. Should I go and buy this thing over here that is, let's say, a new suit?
- 8.41** What are the decisions? Let's take a practical example from your life, Juan.
- 8.42** Yeah, I think what kind of clothes I use here in the lecture.
- 8.43** Yeah, that could be, but I would say an even more high-stakes decision. So this, of course, is also a decision. But let's say—Nikola, you want to say something?
- 8.44** Do I move to this country or that country? Is that too extreme?
- 8.45** Not too extreme, not too extreme, but a little bit extreme. It's a big decision. That's what I mean by extreme. So like marriage, buying a house, moving—these are big decisions.
- 8.46** But I want to say the following, even for that, the [two-way] door decision business works, which means that if I just take a smaller example, right now, I'm sitting in front of the computer, I'm going, "Okay, there are these very nice sunglasses on Amazon right now. Should I order them?" We all have these dilemmas, right? And then you think, calculating in your brain, "Okay, this is \$200, glasses, let's say. Can I really— my budget, is it vanity? Why am I doing this? Should I buy them? Maybe it's good to have sunglasses." You have all these calculations. And you go over and over for the whole

day, and then you wasted the whole day. You wasted \$500 of work for \$200 glasses. You see? Not good.

- 8.47** The [two-way] door decision is the following. You have the thing in front of you. Give it 10 minutes, five, 10 minutes, and you buy the damn thing because you know you can always return it. You can always go back on it. Even moving to a different country—after one month, what can you do? Can you come back? Yes, you can. Almost any decision is reversible. Almost any decision. I mean, there are a few decisions that are irreversible. But many, the ones that we face in our daily lives that slow us down, are reversible.
- 8.48** And therefore, you reduce this friction between the emotional part of the brain, the limbic and the prefrontal. You reduce the friction and you make it easier. You say, "Okay, I'm just going to go for those sunglasses. I'll buy that suit." What else? "I'll go to that event, and if I don't like it, I'll just come home after half an hour."
- 8.49** Texts the girl, huh? You want to know that? Okay. All right, Juan. Don Juan over here. Okay. I think that texting is reversible. It's very reversible. It's very reversible, I tell you. Hot-cold contrast. I mentioned this in a different lecture in a different course. But anyway, I'll give you a snippet, okay? Go watch the other one, and recommend it to your friends.
- 8.50** Okay.
- 8.51** If with a girl, if you are too hot—hot meaning too emotionally warm—you go over to her or you send that text message to her and you fear her rejection, fearing that she doesn't respond in a warm manner or whatever, it doesn't matter.
- 8.52** Your next step is what? Remove your affection for five days, six days. Be a little bit, not cold, detach a little bit. What does that do to the system? It recalibrates. Time is what resets perception. So your warmth on day A, on day E becomes an asset for you. It's a good thing now. Oh, the guy showed me warmth. Now the affection is gone, the attention is gone. You see what I'm saying?
- 8.53** It's the only problem when you double text Juan over here. Don't double text her. Just know that you can be yourself, be emotionally warm, be charming, be all this. But then if there's no reciprocation and it's not working as planned, that's when you just withdraw the affection a little bit. Hot-cold contrast: her dopaminergic system will start craving you, especially after two weeks. Then it's a peak. You wait two weeks. She'll crave you for sure if she likes you. If not, let her go. She doesn't like you anyway, right? All right. It's going too far. Different course. A very different course. But it's well-being, right? You want to have well-being; this is well-being, too. Yes.
- 8.54** What's the sunk cost fallacy? This is also very, very important. And I faced this—oh, my God, I faced this for—I wasted years, years on projects. I had this project. We were

trying to raise money from angel investors. I wasn't the business person involved in it; I was more of an ambassador, just hanging around.

- 8.55** But we wasted so much time on this project, years, because the more you go into the project, the more time you spend on the project, the more meetings you do, the more investors you meet, whatever you do, over time, what happens?
- 8.56** You feel like you can't go back. You can't go back anymore.
- 8.57** Because your amygdala flags this, the ACC flags this conflict. Oh, my God, I spent all this time on this project here. Now it's been months, it's been years. And what do I do? Letting go of this project that's not working, I'm not getting the investments. It's obviously not taking off. But you keep digging yourself into the hole deeper and deeper.
- 8.58** So I think letting go is so important. It's called the sunk cost fallacy. It goes against what your emotional brain, your ACC, your limbic, your amygdala would want. But at some point, you have to make this decision to leave it and say, "I'm going to just leave it and I'm going to focus on new stuff." And then you could do a cognitive reappraisal, meaning mentally say, "Well, it wasn't really a waste because I learned all these new tactics, these new strategies, all these new things I learned and I can use in my life now." So this is a way to cognitively make yourself happy, appease yourself.
- 8.59** I would say for the sunk cost fallacy, I think what's really crucial is that at some point you start mixing your investment or whatever you're doing with identity. It becomes too much like, if this thing is a part of me, maybe it's my baby. You've heard this expression? "Oh, it's my baby. I can't let it go." I think it's very important to have a separation and try to look at it objectively. And if you can't, that's when you have an outside perspective. So you have a third-person perspective. You bring in your friends, bring in people that are not emotionally invested and have them make a decision for you, help you.
- 8.60** I mean, this can be in other things. It doesn't have to be a business venture. It could be with a relationship that's obviously not working out. It could be with a job that you feel like you're digging your own grave, right?
- 8.61** And then I would say, in our brain, the ACC is important for what?
- 8.62** Focus.
- 8.63** Right. Is it important for motivation as well? If you have a trauma to the ACC or to the midcingulate cortex, what happens? Brain damage. I'll tell you what happens. You become a vegetable. You're lying in your bed, and I want you to have a slice of pizza. No movement. I say, "Come on, let's go to the restaurant. Let's go have fun. Let's go to the park." Nothing. Nothing happens. You become completely zombified. This is the midcingulate cortex. It's important, Alejandra, for a sense of motivation, wanting to do

stuff in your life and all that. You see? And so if you have damage to that part, you don't want to do anything. You're not motivated. You feel like life is just not worth it.

- 8.64** So obviously, all of us, we don't have damage to that part of the brain, right? Hopefully, all of us. Sometimes I—no. But if you have less activation in that part of the brain, what happens? Less motivation. So if you are a depressed person and you can't get out of bed, you don't want to go to the gym, you don't want to go out and go for a walk, you don't want to go to your job, you have less activation in the midcingulate cortex. The neurons there have shrunk. The neurons there have shrunk.
- 8.65** But here's the good news. The good news, Giovanni, is the more you actually go against that feeling of not wanting to do something that feels off, the more the midcingulate cortex grows, and it becomes larger. And then you're lying there in your bed the next day and—should I do that ice-cold ice bath or not? And within minutes, you decide to do so. It becomes much easier with time. You become better at going against friction. That's the midcingulate cortex. So it grows.
- 8.66** But I wanted to add here, have you heard about the concept called bed rotting? What's bed rotting, Rachel? I know you know. Come on. Doing what? Doing nothing? Just lie there, scroll. One more scrolling. One more scrolling. Bed rotting. Yeah, you're just rotting in bed. It's not fun, is it? I know the feeling. We all know it. We all know it.
- 8.67** Now, bed rotting. The antidote to bed rotting in any scenario where you're stuck and you don't want to do it is to do it within the first five minutes, Joe. First five minutes, that's when you want to take action; your brain is best. You can do it best, and that's when you stimulate your midcingulate cortex the best. Okay? First five minutes. The longer you wait, the less activation in that circuitry. Should I go to the gym? Within five minutes, decide to go. Should I go to this and that? Within five minutes, decide to do it.
- 8.68** Design your room optimally for stimulation or lack thereof. Meaning—and I've talked about this in some of the previous lectures—but definitely reduce the notifications and the emails coming in. Remove those tabs from your laptop.
- 8.69** What else can we do to improve our room? You can have nice posters. I would put posters that—in Ramachandran's laboratory, we used to have Einstein hanging here, Newton over here. There were all these great scientists, and this was inspiring. You feel like you're part of that whole history, and you're sitting there and doing your science, it's very inspiring. And he had skeletons of dinosaurs, too. And there's all these—one of them was named after him as well. Can you believe this? A great guy.
- 8.70** Now, how can we make our rooms more stimulating or less stimulating? How do we optimally design our rooms? As a scientist, I gave you how it would work for a scientist. Now you have to tell me as a non-scientist. Anyone? Giovanni?
- 8.71** I'd say to enhance the room if you want more motivation.

- 8.72** What's your area of interest, let's say, you want to optimize for? I think it may be field-specific, not entirely, but to some extent, maybe.
- 8.73** Creativity.
- 8.74** You want to optimize for creativity? Right, yeah. What colors would you use?
- 8.75** Bright. So I'd go with maybe a brighter green, maybe a brighter blue.
- 8.76** Yeah, that'd be soothing for sure. Yeah, I would do that. Would you have a painting there maybe? Like a beautiful Salvador Dalí picture, maybe, painting?
- 8.77** Nothing like Picasso.
- 8.78** Nothing like Picasso, right?
- 8.79** Something a little more enlightening.
- 8.80** More enlightening? Yeah. Would you have big windows?
- 8.81** Yeah, I would if I could.
- 8.82** If you lived on the freeway, would you have big windows? The freeway, you wouldn't, right? But if you live in a beautiful area, greenery, maybe a desert-like area, you would look out in the beautiful mountainous area, soothing and stimulating. That's what I do. As I said—I often mention, I look out the window, it's green where we live.
- 8.83** And then occasionally you have this person walk by, and it's like you're looking at him and you're looking at how he moves and everything. It's like you do an analysis of him, but in a good way. You wonder, what is he thinking right now? What is his life like? Does he have a wife waiting for him at home, waiting for a meal? Maybe he has a daughter. What kind of work does he do? You look at him, and you analyze him. It's fun. It's stimulation, but in a good way. So have good, soothing big windows. I would have removed distractions, right? So those are some of the things you could do.
- 8.84** In the morning, you have the glymphatic system clean your brain of toxins. You have a recharge of dopamine, serotonergic attention. Acetylcholine is top. Now, assuming you don't have burnout, you're not burned out, Sharon, over here. You're fine, right? What should you do when you wake up? What is the best?
- 8.85** Sunlight.
- 8.86** What's that?
- 8.87** Go out to see sunlight.
- 8.88** Yes, very good. So the first thing we do is we go out in sunlight. Go for a walk outside, right? Then what? Okay, now we have the gym option. Should we go to the gym now or should we go do focused work right now? What's optimal for the brain? Gym?
- 8.89** I'd say focused work.

- 8.90** Focused work. Good, Giovanni. Exactly. Let's put that down. Deep work. Deep work for about three hours. That's when the brain is optimally ready to just delve in very hard and do some deep work. This is when I sit down. And what is the example of deep work? Preparing for this lecture, writing that chapter, writing that article. What else? What's deep work? I want to say, this is my life, what I'm doing. What could be deep work for you, Rosanne? Preparing for a meeting, maybe?
- 8.91** Yeah.
- 8.92** Could be? Preparing for a meeting. Kelly?
- 8.93** Studying.
- 8.94** Studying? Could be one. Anything else? Deep work. Practically in your life, deep work.
- 8.95** Listening to your courses on Peterson Academy.
- 8.96** Absolutely, right? This is all deep work. So you do this for two hours now, two to three hours. Okay, once you've done that, you can relax. So you do that for 90 minutes. Is that the ultimate perfect best cycle? Ultradian cycle, isn't that 90 minutes? How much break?
- 8.97** Ten to 20 minutes.
- 8.98** Exactly, 10 to 20 minutes. Right. Okay, now you can bring in exercise. I would do exercise now. I would go to the gym, and you can choose between various types of midday exercise. It could be, depending on the day, resistance, cardiovascular, whatever you want to do.
- 8.99** Now, the afternoon comes. You're ready again for—by the way, I don't do this myself all the time. What I usually do—and this is the optimal one—because of my own life, I do feel like sometimes when I wake up in the morning, I like to wake up and rush to the gym. This is what I like to do. I know if I have a heavy-duty cognitive load and deep work, I may not go resistance training full in; I may just do cardiovascular and mixed with resistance. But I usually come back then, I may take a little nap, 30 minutes, and then start deep work. But the deep work is still within the early hours. It's before midday, before 1:00, 2:00 p.m. But you can mix this up.
- 8.100** Late afternoon, when your brain, your prefrontal cortex, is a little bit more fatigued and tired, you have another bout of work you can do. Here, I would engage in more creative endeavors, thinking outside the box, anything, meetings you can have now where you're thinking, you're thinking ahead, any creative. What could be a creative endeavor? For me, it could be thinking about a next project, maybe taking notes for a future book. Maybe it could be for an article, for a talk, something like that. That's just less intense. While in the morning, I was sitting down and writing that chapter that had to be written, like four pages.

- 8.101** I can't do the whole two days for emails a week. For me, it doesn't work. So I have to, every single day, do emails. Maybe I do it at four o'clock. So I do my emails, and I do all those things. Now, as we get later into the afternoon, I would do all the automatic things, all the basal ganglia stuff. Now, the dopaminergic centers of your nucleus accumbens are now tired, all the goal-directed dopamine. So I use the basal ganglia and the more automatic dopamine now. Replying to emails, I would not be perfectionistic. I would make decisions automatically; all that easy stuff I'd do now.
- 8.102** And then the evening comes in. What would I do with the evening? What's that? Wind down. I would wind down. I would spend time with my family, just resting, socializing. I would do anything social, really. That's when you have tea with your friends. I don't know. All those things I would do at that time.
- 8.103** So I think these are overall tips for improving performance, well-being. But I think there's one critical component of well-being that I haven't touched upon. And it may be the most important one for well-being and performance also, for both. And it has to do with the fact that—did you know that lying, lying, we all lie once in a while. Sometimes you call it a white lie, an innocent lie. Lying is not good for the brain. You know that. It's bad for the brain. It's bad for the brain because there's friction between the emotional part, the limbic, and the prefrontal part of the brain that's lying.
- 8.104** Somebody in there knows that you're lying, and your emotional brain doesn't like it. It can sense the inconsistency. It knows. The prefrontal is saying, on the one part, say this, but the other part knows it's a lie, sends signals down to the emotional core, and you have this lack of smoothness. Something is up.
- 8.105** Now, I want to go further a little bit and say most of us, when it comes to life in general, we seem to be lying to ourselves, like we are living a big lie. I'm not saying you specifically, okay? But we are, many of us, for large periods of our lives, living a lie.
- 8.106** As a kid, Nikola, where did you grow up? In Washington, in Seattle?
- 8.107** Montenegro.
- 8.108** Montenegro, okay? As a kid in Montenegro, what were your hopes, dreams, aspirations, hopes for the future? What did you want to do?
- 8.109** Play soccer.
- 8.110** You had certain goals. Maybe those goals have changed over time. But living in an existence where you are congruent with your actual desires and hopes brings smoothness. It brings ease, brings comfort. Things are running. But when you decide to live a life that's based on a lie—your parents want you to go to that university and study to become a lawyer. You don't want to become a lawyer. You know it's boring to be a lawyer. It's so bloody boring. You don't want to do that. But you're choosing that path because you feel like, "That's the only thing that I have. That's the only choice that's been given to me." You feel like you don't have options.

- 8.111** Or maybe you do have a dream inside of you, something that you really want to do, but you fear doing that because it will have all these consequences, all these outcomes. You're with me?
- 8.112** In a sense, you're living in a lie that's always based on limbic friction. And that's why your life is running not really smoothly. Something's up inside of you and you know what it is. It's you not pursuing that dream, that goal you have, that ambition you have. You're not pursuing it. You know you're not pursuing it. And you know you're not pursuing it because perhaps you fear rejection, you fear you won't be good enough. Whatever it might be, you're living a lie. And I think all of us can fall prey to this. And whenever we do live in alignment with what we actually want, what our dreams are, what our hopes are, what we dare dream about and actually go for it, our life is much smoother.
- 8.113** Now, let's say one example could be a girl that I'm interested in, and I want to propose to her, but I fear what?
- 8.114** Rejection.
- 8.115** Isn't that true? I fear rejection. Is that a [two-way] door decision? It is, right? She could say no, but a week later, one month later, she might what? She might say yes. If she likes you, she might say yes. Maybe she's just hesitant at the moment. Maybe she's just fearing the future, right? Anything like that is reversible, ultimately. Most things are.
- 8.116** And so I would say, to overcome that stagnation and the fear of following dreams—first of all, I don't want you to be reckless, Nikola. I don't want you to go and drop out of law school now, med school. Don't drop out of psychology. I know you study psychology, metaphor, blindness, and all that. Don't drop out of all that, okay? You have to do your science. You have to look into the future. Don't be reckless. Have a plan. Sit down and write down, "In the future, I want to have all these plans X, Y, Z, and I want to get to them by doing this and that. I want to have all these plans in order to get to that goal," Juan, right?
- 8.117** But the thing you have to ask yourself then is, am I in the right direction? Is this the right thing I'm doing? Am I going the wrong way now by doing this over here? Or is this the right thing? And life is constantly changing.
- 8.118** I'll give you an example in my own life. A couple of years ago—and this led me to sleepless nights. Harvard Medical School, I do research at Harvard. But Harvard Medical School, the branch over there, they gave me the option to say, "Look, you've been doing this research for a long time. We want to promote you. Within a few months, we'll give you this fellowship. And within a short time, you'll be a professor here at Harvard Medical School." So, my God, this is what I've been dreaming about for all these years. Fantastic.

- 8.119** But then I kept thinking about it. Something was up. I couldn't sleep right. Something was just not right. It didn't sit right with me. And then I pictured myself every single day just seeing patients. There was a very clinical job. You see, you had to see patients every single day. Patients, patients— zap his brain next. Patient, zap his brain next. All my creativity would go lost. Everything that I'm doing, where I'm lecturing, traveling, living the life that I want to live, having fun, all that would just be reduced to me sitting in a—
- 8.120** I'll get the diploma, Professor XYZ at the best medical school in the world, seeing patients. My parents might be happy, but I knew it wasn't the right choice. And this is when I, after many months, I decided to say no to this opportunity. When I said no, I said, "I'm going to regret this probably." But I didn't regret it. I took my own path. I still do work over there at the university, but I didn't take this particular clinical path.
- 8.121** I think all of us can be tempted by outside prestige, things that are telling us, "Oh, this is good society." You become a new lawyer, Juan. This is very good. It's good for the family. It looks good on paper and all that. But then what? Is that really what life is all about? You sitting and doing this boring thing?
- 8.122** You have to be congruent with yourself. It's part of the whole boldness process. Be bold enough. Train your prefrontal cortex to override your fears of social rejection. I mean, look at me. I'm wearing a bloody hat all the time, and I'm wearing it; I go on and do this thing here. There's a camera; it's me with a hat. Who the hell would do that, right? But I think the reason that I also do it is, like, there has to be an element in all of you of "F" the system a little bit. You have to follow your own path. Don't be afraid to take bold decisions, and don't be afraid to be yourself. Follow your own path and have that. And I think that links to that congruence. Does that make sense? Or is it too far? I'm going too far a little bit, right?
- 8.123** No.
- 8.124** Okay, good. Anyway, this is how I think, and I think how life should be lived. On that note, did you want to hear that poem or no?
- 8.125** Yeah.
- 8.126** Okay. This has a good story around that poem, you see? It's very important to me, very important to my heart.
- 8.127** On a tiny table, a tranquil candle inquires about a match so sad. "Dear match, why are you distressed? I sense in you angst and unrest. Is life for you that bad?" The match truly utters, "My soul is like a cold October day. That day you first notice the chills of the gloomy autumn breeze. That day you see the first leaves, or two, or so, dripping, tenderly down the trees. That day, the sullen voice of doom makes you drowsy, reminding you that you must soon leave. For how may I find comfort knowing my tiny head shall soon burn to ashes, and my body thrown away? You, glorious candle, are

much more fortunate than I. For you shall not soon decay—your life is much longer. Dear candle, why is life so short?" the match added in utter dismay.

- 8.128** The candle gently replies, "You are desirous of a long life, yet do not comprehend the insignificance of length. My life may be longer than yours, but I depart this life slowly and painfully, eroding bit by bit, little by little, second by second, anticipating the dreary hand of death to snatch my soul unwillingly. But I am joyful. My heart is filled with outmost delight, for I serve those around me by providing them with light. My life is full of meaning."
- 8.129** The conversation between the match and the candle continues until midnight. A wrinkled and grey hand suddenly reaches out for the match and strikes its head. The match barely stutters, "Life is..." before going out in a daze, light upon light, sparkling and majestic, filling the room with rays.
- 8.130** Excellent. Well done. I love that voice, man. It's good. It's very good. Did you understand the poem? So this poem is about a match and a candle. And what's the issue that's going on between them? The match is very sad about life and the shortness of life. And the candle comes in and says, "Look, my life is worse. I'm eroding bit by bit, melting away. But I provide those around me with light, right? I provide those around me. I give them life. So my life is full of meaning. So my pain becomes meaningful now." And then the match understands this message and eventually dies as well.
- 8.131** And I think this is a very dark poem, but back in the days when I was writing this, I was 16, I wrote this poem. I was in Copenhagen, and I was performing so bad in school. Everything—I had bad grades. Everything was dark. And they put me in a school outside high school, outside the ghetto, so I could perform a little bit better. But I had no friends over there because I think there's a social difference, a class difference, poor and rich. It just didn't match.
- 8.132** And just one day, I was just sitting in my room, and it was like an autumn-fall break, and I was thinking about, how can I— I want to do something. I have an essay, and for the first time in my life, I actually want to try to write something.
- 8.133** And I just sat down; I thought about my own life, about meaning, about all those things, and I wrote this poem. It took me two weeks, night and day; I was starting to think about poetry. I just came up with this poem, and I took it to class. And I was so excited, gave it to the teacher and everything.
- 8.134** Two weeks later, I looked at it, and it said with big letters, "Where have you stolen this poem?" My heart sank. The first time, imagine you've done anything in class, they didn't believe it was me. Now, it could be flattering, too, because maybe it was a good poem, but that's not how I felt. I felt really disappointed. Cut a long story short. I think this poem right here encapsulates what well-being is all about, what performance is all about. It is meaning and service to others.

- 8.135** If you can link any activity you have in life to meaning, to serving others, like I want to go get that job, and I want to have that career change or whatever I'm trying to do, or I want to win that soccer match because I want to provide light to people, I want to illuminate people, other people that were in my seat.
- 8.136** See, these days, when I get a DM from a young kid in the ghetto that says I inspired them, this moves me. It moves me so much. When the rich and the famous write to me, "come here to do this," I ignore them. I don't like it. But when the poor kid in the ghetto says to me, "I read your book, and I couldn't stop," this touches me because I see myself in that little 16-, 17-year-old kid in the ghetto, having no hope. Everybody telling you, "You won't become anything. Reading a book? Graduating high school? Never for you."
- 8.137** But now linking everything to meaning, serving others, especially your mother—I say that: your mother, your mother, your father, your family, this is what it is. When you do that, you have that meaning. God, of course, if you believe in that—I do—then well-being will be easy. You'll never lie to yourself, and congruence and cognitive friction between the prefrontal and limbic would be less. And so that's what I had. Thank you so much.